Spatial scale space from hemigroup axioms: rotation-invariant self-decomposable kernels in dimension d

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Abstract

Scale-space axioms usually require the cascade to be a semigroup, which with rotation and scale invariance leads to Gaussian and stable kernels. The axioms studied here require only that the stages compose, a hemigroup indexed by two scales, and that rescaling the signal relabels the scales. This paper derives the rotation-invariant families they allow in $\mathbb{R}^d$. The kernels that take the signal to each scale are the dilations of one rotation-invariant law that is self-decomposable: the convolution of each contracted copy of itself with another law. Every such law but the point mass occurs. Suitably labelled, the kernel at scale t has Fourier transform $e^{-F(t|\xi|)}$, for one function F. Its density is bounded near the centre exactly when $c > d$, where $c \leq \infty$ is the power at which the transform falls off. Otherwise a point source is recorded as a spike, whose growth and constant are given. For finite c a stage between positive scales passes a fixed fraction of the record unchanged. Since c depends on F alone, a kernel bounded on the line can have a spike in the plane: the Matérn kernels of the Bessel scale space, with transform $(1 + |\xi|^2/2)^{-\gamma}$, do for $\frac{1}{2} < \gamma \leq 1$. Each higher dimension allows strictly fewer F. An allowed $F(r) \sim c \log r$ determines a radial characteristic function, the Fourier transform of a law, and is allowed in $\mathbb{R}^{d+1}$ exactly when that function is a characteristic function there. The statements are machine-checked in Lean 4, assuming seven theorems of the literature, except those that §1.1 lists as proved in print.

1 Introduction

A scale-space representation measures a signal through an aperture whose size is the scale. A measurement is convolution with a probability kernel, and the record at a coarser scale is computed from the record at a finer one by a further convolution, a *stage*. The stages together are the *cascade*, and the *aperture* at scale t is the kernel that takes the signal itself to its record at scale t. Scale-space axioms usually require the cascade to be a semigroup in one scale parameter, as those of the extended class of Pauwels, Van Gool, Fiddelaers and Moons do [44]. A semigroup requires more than a cascade needs: every stage has to be itself one of the apertures, so that one parameter indexes the stages and the apertures alike. With rotation and scale invariance the semigroup axioms of the literature leave the probability kernels whose Fourier transform at the frequency $\xi \in \mathbb{R}^d$ is $e^{-t|\xi|^p}$, for one p with $0 < p \leq 2$. This paper calls them the *stable kernels* of index p. The Gaussian is the stable kernel of index 2. The axioms studied here require only that the stages compose. With $\mu_{s,t}$ the kernel of the stage from scale s to scale t, this is $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ for $r \leq s \leq t$, and how a stage depends on its two scales is left free. Such a family is a *convolution hemigroup* ([27], Def. 2.14.16, p. 380). What is gained is a larger class of apertures. The two-scale cascade has a precedent in scale-space theory, and "Relation to scale-space theory" below sets the present axioms beside it and beside the semigroup axioms.

The axioms are those of the author's paper on the line [15], the *line paper* below, with rotation invariance in place of the reflection symmetry (Definition 3.1): an aperture has a size, and no shape and no preferred direction. Read on the kernels, they say that the stages compose, that they depend continuously on the pair of scales, that each is rotation invariant, and that a dilation of the signal only relabels the scales (Definition 3.2). A family of kernels with these properties is an *isotropic cascade family*, an *isotropic family* for short. The axioms read the same in every dimension. In imaging d is 2 or 3.

On the line the families are classified in the line paper (Theorem 2.3), and a second paper [16], the *cone paper*, describes the set of them. What is used from the two is restated in §2. A kernel is a probability measure, and that classification, like everything after it in this paper, rests on the theory of such measures in probability. So from here on the paper uses the vocabulary of that theory: a kernel is a *law*, and its Fourier transform, the transfer function of the smoothing, is the characteristic function of the law. Stage, scale and cascade keep their meaning, and "kernel" stays in use for a law in its role as an aperture. A frequency is written ω on the line and ξ in $\mathbb{R}^d$, and $\widehat{\mu}(\omega) = \int e^{i\omega x}\,\mu(dx)$ is the Fourier transform of a kernel μ on the line, its *transform* for short. The axiom on dilations makes the apertures, the kernels $\mu_{0,t}$ *from scale zero*, the dilations of one law: up to a relabelling of the scales, $\mu_{0,t}$ has the transform $e^{-F(t|\omega|)}$ for one function F, the *exponent* of the family. The cascade makes that law *self-decomposable*: for every $0 < \lambda < 1$ it is the convolution of its own image under $x \mapsto \lambda x$ with another law ([47], Def. 15.1, p. 90; [56], Ch. V, (1.1)). In a cascade this is the relation between two scales. The kernel at the finer scale is a contracted copy of the kernel at the coarser one, and the other law is the stage between them. Every symmetric self-decomposable law other than the point mass occurs. Such a law is the convolution of a centred Gaussian, which may be absent, with the law of a sum of independent displacements. A law of the second kind alone has *no Gaussian part*. Its *Lévy measure* gives the expected number of displacements of each size. On the line the density of that measure at the size x, the two directions counted together and multiplied by x, is the *folded profile* of the law. In $\mathbb{R}^d$ the same quantity is taken in polar coordinates, the *polar profile* of §3.2. The Gaussian part and the profile determine the exponent, and the exponent determines them (§2.2; Proposition 3.11(1)).

This is what the weaker axiom admits. In probability the stable kernels are the rotation-invariant stable laws ([47], Thm. 14.14, p. 86). Among the kernels added are the laws on the line with transform $(1 + \omega^2/2)^{-\gamma}$, $\gamma > 0$, whose folded profile is $2\gamma e^{-\sqrt{2}\,x}$. This paper calls these laws and their dilations the *Matérn kernels* of index γ. Scale-space theory knows them as the kernels of the Bessel scale space ("Relation to scale-space theory" below). On the line the kernels of integer index are those computed by a cascade of identical first-order recursive filters, run forward and backward along the signal ([16, Prop. 7.2 and Cor. 5.5]). Read as functions of the length $|\xi|$ of a frequency $\xi \in \mathbb{R}^d$, the transforms of the stable and the Matérn laws are the transforms of self-decomposable laws on $\mathbb{R}^d$, for every d (Propositions 2.6(2) and 3.12). For $\gamma > d/2$ the density of the Matérn kernel on $\mathbb{R}^d$ is a Matérn covariance function, of smoothness $\gamma - d/2$. Spatial statistics uses that family to model random fields whose smoothness is a parameter, and Matérn obtained it as a gamma mixture of Gaussians ([42], (2.4.5)–(2.4.7), pp. 17–18; [55], §2.7, p. 31, and §2.10, pp. 48–49; [25]; [45]). In probability the Matérn kernels are the symmetric variance-gamma laws [20], and in potential theory their densities are dilations of the Bessel kernel [2] (Example 3.19 has the formulas).

The questions. Three questions arise.

1. Which kernel families do the axioms allow in $\mathbb{R}^d$? Their kernels from scale zero are the apertures that a scale space with these axioms can use in the plane and in space.

2. What does the dimension change in a kernel? Chiefly, is the kernel bounded at its *centre*, the point $x = 0$, or does it record an ideal point source as a spike?

3. Does the exponent of a family of the line, read as a function of $|\xi|$, give a family in the plane and in space? This is the rotation-invariant extension of the family, not the separable one. The Gaussian and the α-scale spaces of Duits et al. [13] pass. Is that general, or does the set of allowed exponents shrink as d grows?

The answers. The main result answers the first question. A probability law on $\mathbb{R}^d$ is a kernel from scale zero to a positive scale of an isotropic family, an aperture of a scale space with these axioms, exactly when it is rotation invariant, self-decomposable and not the point mass at the origin. In every isotropic family the kernels from scale zero are, up to a relabelling of the scales, the dilations of one such law, and every such law gives an isotropic family, for every relabelling of the scales (Corollary 3.9, with Theorem 3.6 and Lemma 3.8; on the operators the theorem is Corollary 3.7). The reason is a reduction to the line. A rotation-invariant kernel is determined by its projection on a line through the origin, and the projections of an isotropic family form a family on the line (Lemmas 3.4 and 3.5). Each aperture has a radially nonincreasing density, so it weights a nearer point at least as much as a farther one (Proposition 3.15).

Second, the dimension is a threshold for the kernel at its centre. A kernel with a Gaussian part is bounded and smooth in every dimension: its transform falls off faster than every power (§3.3). For a kernel from scale zero with no Gaussian part, let c be the power at which its transform falls off at high frequencies: $-\log \widehat{\mu}(\xi)/\log|\xi| \to c$ as $|\xi| \to \infty$, with $0 < c \leq \infty$ (Corollary 3.14). The kernel has a density that is bounded near its centre exactly when $c > d$, and for $c > d + N$ the density is sure to have N continuous derivatives (Theorem 3.13). The number c belongs to the exponent. It is the same in every dimension that admits the exponent, and only the threshold moves. So one exponent can give a kernel that is bounded on the line and has a spike in the plane, and the Matérn kernels, for which $c = 2\gamma$, do this for $\frac{1}{2} < \gamma \leq 1$. When $c < d$ the density grows at the centre like $|x|^{c-d}$ times a *slowly varying* factor, one that changes more slowly than every power, as a constant or a logarithm does, and when $c = d$ it grows at least like $\log(1/|x|)$. The factor and the constant are computed from the Lévy measure (Proposition 3.18). Its small displacements decide whether the kernel is bounded at its centre and how many derivatives it is sure to have, and for a spike its power and its slowly varying factor. The large ones enter the constant of the spike.

A stage of the cascade need not be an aperture. When the kernels have no Gaussian part and c is finite, the stage from scale s to scale $t > s > 0$ passes a fixed fraction of the finer record through unchanged and blurs the rest. The fraction is $(s/t)^c$ when the scales are labelled so that the kernel at scale t is the dilation by t. With a Gaussian part, or with $c = \infty$, as for the stable kernels, a stage is a blur by a density (Proposition 3.20).

Third, each dimension admits strictly fewer exponents than the one before. The sets $\mathcal{A}_d$ of the exponents admitted in dimension d form a strictly decreasing sequence, the *dimension filtration* (Propositions 2.6 and 3.22). The exponents that remain in every dimension are those of the Gaussians whose variance is drawn at random from a kernel of a time-causal scale space with the same axioms, one that measures a signal in time from its past alone. These Gaussian variance mixtures include the stable and the Matérn kernels (§2.3; Propositions 2.6(3) and 3.12). Every exponent of the line is a superposition of a Gaussian term and of elementary exponents, each with a Lévy measure whose density is proportional to $1/|x|$ up to a cut-off and zero beyond it (Proposition 2.7). The exponents admitted in the plane are still such superpositions, the Matérn and the stable ones among them. What is lost first is each elementary exponent taken alone: none of them is admitted in the plane (Proposition 3.22). For the next dimension there is a test. The exponent of a kernel with no Gaussian part and a finite c determines a function of the radius $|\xi|$ that is the Fourier transform of a probability law on $\mathbb{R}^d$, and the exponent is admitted in $\mathbb{R}^{d+1}$ exactly when that function is still the transform of a law there (Proposition 3.21). And one explicit family of exponents is followed through the dimensions: the dimensions that admit a member are read off its parameter (Theorem 3.25).

Scope. The axioms and the classification are stated on operators and on their kernels, and the two forms describe the same families (Lemma 3.3). The other statements are on kernels. The paper is theory: it has no discretization, no implementation and no experiment, and §4 lists its limitations. Apertures with a shape are the subject of a companion paper on affine covariance, not yet released.

Relation to scale-space theory. The two-scale cascade is the causality axiom of Alvarez, Guichard, Lions and Morel, who state it as the weaker form of the semigroup, and their scale invariance, like axiom (A8), lets a dilation relabel the scales in a way that is not prescribed ([1], pp. 203, 206 and 226). They require in addition that the stages be regular and local. For linear operators that commute with the isometries the heat equation follows, up to a rescaling of the scale (Thm. 3, p. 214). The present axioms keep their comparison principle as positivity and do not require locality.

The axiom systems that keep a semigroup with scale invariance arrive at the transforms $e^{-a|\xi|^p}$: Pauwels et al. from recursivity and scale invariance ([44], p. 694 and §III, p. 695), Duits et al. as the semigroups generated by $-(-\Delta)^\alpha$, positive for $\alpha \in (0, 1]$, where $p = 2\alpha$ ([13], pp. 267 and 277), with the Poisson scale space of Felsberg and Sommer at $\alpha = \frac{1}{2}$ ([19], Thm. 1, p. 7). A requirement of another kind leaves the Gaussian alone, and each derivation has its own. Iijima's axioms, the first, give on the line the transforms $\exp(-\xi^{2m})$, $m = 1, 2, \ldots$, of which positivity leaves the Gaussian ([59], pp. 239–240). His derivation supposes a transform that is differentiable at the origin, which that of the Poisson kernel is not ([19], p. 8). Koenderink prohibits spurious resolution, his causality, and with homogeneity and isotropy obtains the diffusion equation ([34], p. 365). Florack et al. require separability ([22], (23)–(25), p. 380). Florack's book requires filters that are positive Schwartz functions and form a convolution algebra ([21], p. 57 and Result 3.2, p. 61). Lindeberg requires that local extrema not be enhanced ([39], §3.1 and §4.1.1).

Without scale invariance a semigroup of rotation-invariant kernels need not be of this kind. The Bessel scale space of Burgeth, Didas and Weickert is the semigroup in $t \geq 0$ of the kernels with transform $(1 + |\xi|^2)^{-t/2}$, which at $t = 2\gamma$ are dilations of the Matérn kernels of index γ ([8], pp. 86 and 89). It runs along the index and is not scale invariant. Its authors also dilate the kernels at a fixed index, with no rule for composing the dilations (p. 90). The hemigroup is that rule (Proposition 3.12). The relativistic scale spaces of the same authors are a second such semigroup [9].

Relation to probability. The laws that the axioms select are a known class. Read on laws, the classification is in print, for the self-similar processes with independent increments ([47], Thm. 16.1) and for hemigroups on groups ([27], Thm. 2.14.21). In both the action of a dilation on the parameter is prescribed. Here it is a relabelling of the scales that the axioms leave free. The statements on the kernels have counterparts in print as well, for a case or in another form: on the line, results of Zolotarev, of Wolfe and of Sato and Yamazato (§2.4); in $\mathbb{R}^d$, Tauberian theorems, which pass from the decay of a transform to the growth of a density. Table 3 in §A.1 says, statement by statement, where each is in print and for which case. The proofs given here are written out from the cited facts of §2.6 and the statements restated in §2.

Plan of the paper. §2 restates what is used from the line paper and the cone paper, and the theorems of the literature that the proofs cite, the *cited facts*. §3 is the isotropic theory: the axioms, then the three questions in their order. §4 has the limitations and what is left open. Appendix A has a table of the notation (Table 2), the sources (§A.1), the proofs of the classification, the statements that §3 names and does not state, and the proofs of the statements that are proved in print only. Appendix B says where the formal development is. The other proofs that Lean checked, the cited facts in full and the list of the formal declarations are in the supplement. A statement, a section, an equation or a table whose number begins with an

S is one of the supplement. What is machine-checked, and what is proved in print only, is said next.

1.1 What is machine-checked

The statements of §3, of Appendix A and of §S3 have formal counterparts in the Lean 4 proof assistant, on its mathematical library Mathlib, except those of the second list below. Lean checked that each formal statement follows from the formal definitions, its own logical foundations and seven theorems of the literature, the cited facts C1, C2, C3, C4, C5, C6 and C7, which are its only assumptions. §2.6 names the seven, each with the page of its source, and says with which statement each enters: Theorems 3.6 and 3.13, Propositions 3.11, 3.15 and 3.18, and Lemma S.13 of the supplement. Table 4 in Appendix B has the cited facts of every statement. The statements of the second list are proved in §A.4, and their proofs were refereed by AI agents that saw the statements and their proofs and no other part of the work. The formal statements were compared with the printed ones by AI agents. §S1 has the account of that review, of the form of the formal statements and of what the text computes, and the statement on the use of AI at the end of the paper says how the work was done.

- *Machine-checked*: every definition, lemma, proposition, theorem and corollary of these three parts that the second list does not name.
- *Not machine-checked*, proved in print: Proposition 3.21, the dictionary between the exponents of logarithmic growth and the radial characteristic functions; Corollary A.7, the two readings of the test in that dictionary; Proposition A.8, the separating family as laws; Lemma 3.17, the closed form of the constant $\varkappa_d$; and Proposition 3.23, the step exponent one and two dimensions up.

The statements of §2 are restated from the line paper and the cone paper. The formal development proves what it needs of them from the seven cited facts, and in their own papers part of them is proved in print. Proposition 3.12 is stated for the kernels of a time-causal family. That each of its three named delays has such a family is the classification of the time-causal paper, with Halgreen's theorem for the inverse-gamma delay, both cited after the proposition. The formal statement takes the family as given, so these two results are outside the formal development. The examples, Table 1, the list of c in §3.3 and the figures are computed, in the text or by scripts, and the statements are independent of the computed values. That a kernel with a Gaussian part is bounded is argued in §3.3 from C1, with no formal counterpart.

2 What is used from the line papers

This section restates, without proof, the statements of the line paper [15] and of the cone paper [16] that the later sections use, and names the facts of the literature that the proofs use and do not prove. The two papers work on the real line $\mathbb{R}$, with signals of one variable, and are cited in their released version 0.1, by the statement numbers of that version. With the restatements the statements of this paper can be read without the two papers, and its proofs rest on theirs: the classification rests on that of the line paper (Theorem 2.3). A restated statement carries its released number in its title, and its proof is at that number.

2.1 Notation

For a probability measure μ on $\mathbb{R}^d$ the characteristic function is $\widehat{\mu}(\xi) = \int e^{i\langle \xi, x\rangle}\, \mu(dx)$. When $\widehat{\mu} = e^{-F}$ for a real function F, F is the *exponent* of μ. For a measurable map T, $T_*\mu := \mu \circ T^{-1}$ is the image measure. On the line the dilation by $\lambda > 0$ is $D_\lambda f = \lambda^{-1} f(\cdot/\lambda)$, the translation by a is $\tau_a f = f(\cdot - a)$ and the reflection is $Rf = f(-\,\cdot)$. $\mathrm{O}(d)$ is the orthogonal group, S^{d-1} the unit

sphere, σ_{d-1} its surface measure and $|S^{d-1}| = \sigma_{d-1}(S^{d-1})$ its surface area. In §3 the frequency on $\mathbb{R}^d$ is written ξ and the frequency on the line ω. Table 2, at the head of Appendix A, lists the symbols that recur, each with its role, and names the letters that keep a second role. Inside a cited fact the letters are its source's.

2.2 The axioms and the classification on the line

The line paper's object is a family of operators on $L^1(\mathbb{R})$ indexed by pairs of scales $s \le t$. The operator $\Phi_{s,t}$ takes the record of a signal at scale s to its record at the coarser scale t. The axioms are these.

Definition 2.1 (symmetric cascade measurement family, [15, Def. 3.1])**.** A family $(\Phi_{s,t})_{0\le s\le t}$ of linear operators on $L^1(\mathbb{R})$ is a *symmetric cascade measurement family* if:

(A1) each $\Phi_{s,t}$ is a bounded operator on $L^1(\mathbb{R})$;

(A2) $\Phi_{s,t}\tau_a = \tau_a\Phi_{s,t}$ for every $a \in \mathbb{R}$;

(A3) $\Phi_{s,t}R = R\,\Phi_{s,t}$;

(A4) $\Phi_{s,t}$ maps nonnegative functions to nonnegative functions;

(A5) $\int \Phi_{s,t}f = \int f$ for every nonnegative $f \in L^1(\mathbb{R})$;

(A6) $\Phi_{t,t} = \mathrm{Id}$, and $\Phi_{s,t}\Phi_{r,s} = \Phi_{r,t}$ whenever $r \le s \le t$;

(A7) for every $f \in L^1(\mathbb{R})$ the map $(s,t) \mapsto \Phi_{s,t}f$ is continuous from $\{0 \le s \le t\}$ to $L^1(\mathbb{R})$;

(A8) there is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0,\infty)$ with $D_\lambda\Phi_{s,t} = \Phi_{S_\lambda s,\,S_\lambda t}D_\lambda$ for all $0 \le s \le t$ and all $\lambda > 0$;

(ND) $\Phi_{s,t} \ne \mathrm{Id}$ whenever $s < t$.

Axiom (A6), the *cascade*, requires that the stages compose, and leaves free how the stage from s to t depends on the two scales: the family need not be a semigroup in any scale parameter. Axiom (A8) is the scale invariance of the theory. Dilating the signal changes which scale a stage belongs to, and (A8) requires only that it relabel the scales by an increasing bijection S_λ, without fixing in advance how. Axiom (ND) excludes stages that do nothing. The first consequence of the axioms is that every stage is a convolution.

Lemma 2.2 (from operators to kernels on the line, [15, Lem. 4.1])**.** *Under (A1), (A2), (A4) and (A5), for each pair $0 \le s \le t$ there is a unique probability measure $\nu_{s,t}$ on $\mathbb{R}$ with $\Phi_{s,t}f = \nu_{s,t} * f$ for every $f \in L^1(\mathbb{R})$; under (A3) it is symmetric. Conversely every symmetric probability measure on $\mathbb{R}$ defines an operator satisfying (A1)–(A5).*

A *gauge* is an increasing bijection χ of $[0,\infty)$, hence continuous, with $\chi(0) = 0$. It labels the scales: the classification below determines a family up to its gauge, and in a family with gauge χ axiom (A8) holds with $S_\lambda = \chi^{-1}(\lambda\,\chi(\cdot))$, so that a dilation by λ multiplies χ by λ. A function $F : [0,\infty) \to [0,\infty)$ is *admissible* if it has the form [15, (2.3)]

$$F(r) = a_{\mathrm{V}}\, r^2 + \int_0^\infty \bigl(1 - \cos rx\bigr)\, k(x)\, \frac{dx}{x}, \tag{2.1}$$

with $a_{\mathrm{V}} \ge 0$ and $k : (0,\infty) \to [0,\infty)$ nonincreasing, $\int_0^1 x\,k(x)\,dx < \infty$ and $\int_1^\infty k(x)\,x^{-1}dx < \infty$. The number a_{V} is the *Gaussian coefficient* and k the *folded profile*. A subscript V marks an object of the line, and a subscript I, from §2.3 on, one of the time-causal theory. An admissible F is the exponent of a symmetric *infinitely divisible* law, one that is for every n the n-fold convolution of a law with itself, and the measure $k(x)\,dx/x$ on $(0,\infty)$ is that law's *Lévy measure* folded onto the half-line: displacements of size about x arrive at rate $k(x)\,dx/x$, the two directions counted together. The two integrability conditions are those of [15, Lem. 2.15].

Theorem 2.3 (the classification on the line, [15, Thm. 7.3])**.** *A family $(\Phi_{s,t})_{0\le s\le t}$ of operators on $L^1(\mathbb{R})$ satisfies (A1)–(A8) and (ND) if and only if there are a gauge χ and an admissible $F \not\equiv 0$ such that $\Phi_{s,t} f = \nu_{s,t} * f$ with symmetric probability measures $\nu_{s,t}$ on $\mathbb{R}$,*

$$\widehat{\nu}_{s,t}(\omega) = \exp\big[-\big(F(\chi(t)\,\omega) - F(\chi(s)\,\omega)\big)\big].$$

The pair (χ, F) is unique up to the normalization $\chi(1) = 1$.

In the *canonical gauge*, χ the identity, the kernel from scale zero to scale t has transform $e^{-F(t\omega)}$: it is the dilation by t of one law. In a formula, a law μ on $\mathbb{R}^d$ is self-decomposable (§1) if for every $\lambda \in (0,1)$ there is a law ρ_λ, the *residual*, with

$$\widehat{\mu}(\xi) = \widehat{\mu}(\lambda\xi)\,\widehat{\rho}_\lambda(\xi), \qquad \xi \in \mathbb{R}^d \tag{2.2}$$

(on the line [15, Def. 2.12]). The residuals are the stages of a cascade. The admissible functions are exactly the exponents of the symmetric self-decomposable laws on the line [15, Prop. 2.13].

2.3 Subordination and the dimension filtration

Subordination builds new smoothing kernels from the Gaussian. Blur the signal with a Gaussian whose variance is not fixed but drawn at random, and average over the draws. The result is a *Gaussian variance mixture*, the law of a Brownian motion at a random time (Bochner's subordination, [47], §30, p. 197). A familiar example is the kernel of the Poisson scale space of Felsberg and Sommer ([19], Thm. 1, p. 7), the $\alpha = \frac{1}{2}$ member of the α-scale spaces ([13], p. 267). With transform $e^{-t|\omega|/\sqrt{2}}$ it is the Gaussian blurred at a variance drawn from the one-sided stable law of index $\frac{1}{2}$ with Laplace transform $e^{-t\sqrt{\sigma}}$ (Proposition 3.12(3) with $\alpha = \frac{1}{2}$).

Subordination matters here because it is how the time-causal theory meets the spatial one. A time-causal scale space measures a signal in time from its past alone ([35]; [41]; [14]; [40]). In the time-causal theory of the same axioms [17] a kernel is a law on the half-line $[0,\infty)$ of delays: the record at a time uses the signal at earlier times only, and the delay is how far back it reaches. A *time-causal family* is a cascade of such laws that satisfies the axioms of the time-causal paper. That paper classifies these families: in its canonical gauge a family is given by an exponent of the form of Definition 2.4 below, and every such exponent other than 0 occurs ([17], version 1.1.0, Thm. 7.3). Using the delay as the variance turns a time-causal family into a family on the line (Proposition 2.5 below, for the exponents), and into an isotropic family in every $\mathbb{R}^d$, and no other exponents give an isotropic family in every dimension (Proposition 2.6(3) below).

Definition 2.4 (causally admissible exponents, [16, Def. 4.1])**.** A function $F_{\mathrm{I}} : [0,\infty) \to [0,\infty)$ is *causally admissible* if

$$F_{\mathrm{I}}(\sigma) = b_0\,\sigma + \int_0^\infty \big(1 - e^{-\sigma u}\big)\,k_{\mathrm{I}}(u)\,\frac{du}{u},$$

with $b_0 \ge 0$ and $k_{\mathrm{I}} : (0,\infty) \to [0,\infty)$ nonincreasing, $\int_0^1 k_{\mathrm{I}}(u)\,du < \infty$ and $\int_1^\infty k_{\mathrm{I}}(u)\,u^{-1}du < \infty$.

The law on $[0,\infty)$ with Laplace transform $e^{-F_{\mathrm{I}}}$ is the *delay law* of F_{I} [16, Rem. 4.3]. In probability F_{I} is the Laplace exponent of a subordinator, an increasing *Lévy process*, that is, a process with stationary independent increments ([47], Def. 21.4, p. 137), with the further condition that k_{I} is nonincreasing. The process advances at the constant rate b_0 and by jumps of size about u at rate $k_{\mathrm{I}}(u)\,du/u$, and the delay law is its law at time 1. In its canonical gauge the kernel of a time-causal family with exponent F_{I}, from scale x to scale $y \ge x$, is a law $\mu^{\mathrm{I}}_{x,y}$ on $[0,\infty)$ with Laplace transform $\exp[-(F_{\mathrm{I}}(y\sigma) - F_{\mathrm{I}}(x\sigma))]$, $\sigma \ge 0$, as in the proof of [16, Prop. 4.5], and $\mu^{\mathrm{I}}_{0,1}$ is the delay law.

Two further notions are restated for later use. A pair (b_0, k_{I}) with $b_0 \geq 0$, the two integrability conditions and a measurable k_{I}, monotone or not, is a *delay datum* [16, Def. 4.11], with *delay profile* k_{I} and the exponent F_{I} of the formula of Definition 2.4. Definition 3.10 and Theorem 3.13(3) use it with a nonincreasing profile, and §3.4 without, because the delay profiles of the family followed there are not monotone. The cited fact C6 of §2.6 is about *Bernstein functions*: nonnegative smooth functions on $(0,\infty)$ whose derivative is completely monotone, $(-1)^n (f')^{(n)} \geq 0$ for every $n \geq 0$. A causally admissible exponent is a Bernstein function, and it vanishes at the origin ([16, §4.1]).

The bridge to the line is the substitution $\sigma = \omega^2/2$. A centred Gaussian of variance u has the transform $e^{-u\omega^2/2}$, so its mixture against the delay law has the transform $e^{-F_{\mathrm{I}}(\omega^2/2)}$.

Proposition 2.5 (the bridge on exponents, [16, Lem. 4.2])**.** *Let F_{I} be causally admissible with data (b_0, k_{I}). Then $F(\omega) := F_{\mathrm{I}}(\omega^2/2)$ is admissible, with Gaussian coefficient $a_{\mathrm{V}} = \frac{1}{2} b_0$.*

The cone paper lifts this to families [16, Prop. 4.5]: mixing the centred Gaussian laws of variance u against the kernels $\mu^{\mathrm{I}}_{s^2,t^2}$ gives the kernels of a family on the line that satisfies the axioms, with exponent F in its canonical gauge. The scales enter squared because the variance of a Gaussian is the square of its width. Proposition 3.12 is the same construction in $\mathbb{R}^d$. The *subordinated slice* $\mathcal{S}$ is the set of the exponents $F_{\mathrm{I}}(\cdot^2/2)$, F_{I} causally admissible [16, Prop. 4.6]. The admissible exponents form a convex cone, and the next proposition is about the part of it that each dimension admits.

Proposition 2.6 (the dimension filtration, [16, Prop. 4.9(1), (2) and (4)])**.** *For an integer $d \geq 1$ let $\mathcal{A}_d$ be the set of admissible F for which $\xi \mapsto F(|\xi|)$, $\xi \in \mathbb{R}^d$, is the exponent of a self-decomposable law on $\mathbb{R}^d$, and let $\mathcal{A}_\infty$ be the intersection of all the $\mathcal{A}_d$.*

1. *$\mathcal{A}_1$ is the whole admissible cone, and $\mathcal{A}_{d+1} \subseteq \mathcal{A}_d$ for every $d \geq 1$.*
2. *$\mathcal{S} \subseteq \mathcal{A}_d$ for every $d \geq 1$.*
3. *$\mathcal{A}_\infty = \mathcal{S}$.*

So $\mathcal{S} = \mathcal{A}_\infty \subseteq \cdots \subseteq \mathcal{A}_2 \subseteq \mathcal{A}_1$.

Membership in $\mathcal{A}_d$ is what an isotropic family in $\mathbb{R}^d$ requires of its exponent (Theorem 3.6). The cone paper proves that the first two inclusions of the chain are strict [16, Lem. 4.12(3)], and reports numerical evidence for the later ones [16, Rem. 4.10]. That every inclusion is strict is Proposition 3.22(2) here.

The extreme rays of the cone of admissible exponents have an explicit form.

Proposition 2.7 (the extreme rays, [16, Prop. 3.3, Lem. 3.1])**.** *Write* $\mathrm{Cin}(z) := \int_0^z (1 - \cos v)\, v^{-1} dv$. *The extreme rays of the cone of admissible exponents are the Gaussian ray, through $\omega \mapsto \omega^2$, and the* Cin *rays, through the functions $\omega \mapsto \mathrm{Cin}(\tau|\omega|)$, $\tau > 0$, whose folded profile is the unit step $\mathbf{1}_{(0,\tau)}$. Every admissible exponent is a unique superposition of* Cin *rays and a Gaussian term:*

$$F(\omega) = a_{\mathrm{V}}\, \omega^2 + \int_{(0,\infty)} \mathrm{Cin}(\tau|\omega|)\, \varpi(d\tau),$$

with $\varpi = -dk$ the measure on $(0,\infty)$ given by the folded profile, $\varpi((x,\infty)) = k(x)$ for almost every $x > 0$.

A member of the Cin ray through $\mathrm{Cin}(\tau|\omega|)$ is $F(\omega) = b\, \mathrm{Cin}(\tau|\omega|)$ with $b > 0$, with folded profile $b\, \mathbf{1}_{(0,\tau)}$; the function $\mathrm{Cin}(\tau|\omega|)$ itself is the ray's *unit member*.

2.4 The kernel at its centre, on the line

On the line, whether a kernel from scale zero with no Gaussian part is bounded at its own centre is decided by one number, the folded profile at the origin, $c := k(0+)$. The next corollary says so when c is finite and positive: the kernel is bounded at its centre exactly when $c > 1$. In it ϕ_1 is the density of the kernel from scale zero to scale 1, in the canonical gauge. Besides the Cin rays, the corollary names the Matérn family. Its member of index $\gamma > 0$ and range $\theta > 0$ is $F(\omega) = \gamma \log(1 + \theta^2\omega^2)$, the admissible exponent with Gaussian coefficient 0 and folded profile $2\gamma e^{-x/\theta}$ ([16, §5.2]).

Corollary 2.8 (the boundedness threshold on the line, [16, Cor. 3.9])**.** *Let F be admissible with Gaussian coefficient* 0 *and folded profile k, let ϕ_1 be its kernel at canonical scale* 1*, and suppose $c := k(0+)$ is finite and positive.*

1. *If $c > 1$, then ϕ_1 has a representative continuous on the whole line. If $c \leq 1$, no representative of ϕ_1 is bounded near the origin.*
2. *The unit member of every* Cin *ray has $c = 1$ and sits at the threshold. The Matérn member of index γ and range θ has $c = 2\gamma$, so its kernel is bounded at the origin exactly for $\gamma > \frac{1}{2}$.*

The threshold is classical. A symmetric self-decomposable law on the line with no Gaussian part has a continuous density exactly when $c > 1$ ([47], Thm. 28.4, p. 191, where the two half-lines are counted together in c, as in the folded profile). Sato's Remark 28.6 (p. 193) credits this to Zolotarev and to Wolfe: [63], Thm. 3, p. 131, is on the derivatives of the density for $c > 1$, and [60], Thm. 4, pp. 2069–2070, the condition in both directions. That for $c \leq 1$ the density is unbounded, and not only discontinuous, is a theorem of Sato and Yamazato ([48], Thm. 1.5, p. 276). They also give the rate at which an unbounded density grows at the centre, with its constant ([48], Thm. 1.7, p. 277; [47], Thm. 53.8, pp. 410–411). The form of the corollary in $\mathbb{R}^d$ is Theorem 3.13, where the threshold is d, and the rate is Proposition 3.18.

2.5 Two facts of the line used in proofs

Two released statements of the line paper are used inside proofs of §3, and are restated here. Lévy's classical continuity theorem on the line, which [15, Prop. 2.8] restates, is used in Lemma 3.5: if the transforms of laws on $\mathbb{R}$ converge at every point to the transform of a law, the laws converge weakly to it. A function h continuous at 0 with $h(0) = 0$ and $h(\omega) = h(\varkappa\omega)$ for every ω and one fixed $\varkappa > 0$ with $\varkappa \neq 1$ vanishes identically [15, Lem. 6.3]. It is used in Theorem 3.6.

2.6 The cited facts

Seven theorems of the literature are used and not proved. They are named C1 to C7, C for cited, to keep them apart from the axioms (A1)–(A8) of the line. the supplement states each as it is used (§S2), and Table 4 says which of them each statement rests on. Three depart from the letter of their source, and §S2 has the account. C5 is Wolfe's statement. His printed proof covers the laws with no Gaussian part, and two results of Kanter's paper [32] cover the others. C6 composes two theorems of one book and leaves out the uniqueness that the book states. C7 is one direction of an equivalence, under narrower hypotheses.

C1 *Fourier integrability.* A probability measure μ on $\mathbb{R}^d$ with $\int |\widehat{\mu}(z)|\,|z|^n\,dz < \infty$ has a density of class C^n, whose partial derivatives of orders up to n tend to 0 at infinity ([47], Prop. 28.1, p. 190).

C2 *The polar criterion.* An infinitely divisible law on $\mathbb{R}^d$ is self-decomposable exactly when its Lévy measure is, in polar coordinates, $\lambda(d\eta)\,k_\eta(r)\,dr/r$ with λ a finite measure on the sphere and $k_\eta(r)$ nonincreasing in r ([47], Thm. 15.10, p. 95).

C3 *The Lévy–Khintchine representation.* (i) An infinitely divisible law on $\mathbb{R}^d$ has a *generating triplet*: a Gaussian covariance matrix, a Lévy measure and a drift. (ii) The triplet is unique. (iii) Every such triplet is that of an infinitely divisible law ([47], Thm. 8.1, pp. 37–38).

C4 A self-decomposable law on $\mathbb{R}^d$ is infinitely divisible ([47], Prop. 15.5, p. 93).

C5 *Wolfe's unimodality theorem.* A symmetric self-decomposable law on $\mathbb{R}^d$ is a mixture of uniform laws on compact convex symmetric sets ([61], Thm. 2, p. 144, with the definitions of [32], pp. 66 and 70).

C6 *Schoenberg's radial theorem, with the representation of a Bernstein function.* A real function ψ on $\mathbb{R}^d$ is *negative definite* when $\sum_{j,k=1}^{n}\big(\psi(\xi_j)+\psi(\xi_k)-\psi(\xi_j-\xi_k)\big)c_j\overline{c_k}\geq 0$ for every n, all points $\xi_1,\dots,\xi_n$ of $\mathbb{R}^d$ and all complex numbers $c_1,\dots,c_n$. A function $f\geq 0$ for which $\xi\mapsto f(|\xi|^2)$ is continuous and negative definite on $\mathbb{R}^d$ for every d is $f(\sigma)=a+b\sigma+\int_{(0,\infty)}(1-e^{-\sigma t})\,m(dt)$ for $\sigma>0$, with $a,b\geq 0$ and $\int(1\wedge t)\,m(dt)<\infty$ ([51], Thm. 13.14, p. 212, and Thm. 3.2, p. 21; the definition is its Def. 4.3, p. 36, in the form (4.8), pp. 45–46).

C7 *Karamata's Tauberian theorem at the origin.* If the Laplace transform of a finite measure on $[0,\infty)$ is asymptotic at infinity to $c\,s^{-\rho}\ell(s)$, with $c,\rho>0$ and ℓ slowly varying, then the mass that the measure gives to $[0,x]$ is asymptotic to $c\,x^{\rho}\ell(1/x)/\Gamma(1+\rho)$ as $x\downarrow 0$ ([3], Thm. 1.7.1′, p. 38).

Each enters with one statement. C3 enters with the classification (Theorem 3.6), C2 and C4 with the description of the classes (Proposition 3.11), C1 with the threshold (Theorem 3.13), C5 with the shape of the density (Proposition 3.15), C7 with the rate (Proposition 3.18) and C6 with the top of the dimension filtration, $\mathcal{A}_\infty=\mathcal{S}$ (Lemma S.13 of the supplement).

3 The isotropic theory

An isotropic family measures a signal on $\mathbb{R}^d$ through rotation-invariant kernels, and a dilation of the signal only relabels the scales. This section states the axioms (§3.1), classifies the families by reduction to the line (§3.2), describes a kernel at its centre (§3.3) and says how the dimension restricts the exponents (§3.4).

3.1 The axioms

The axioms are those of the line, Definition 2.1, with $\mathbb{R}^d$ in place of $\mathbb{R}$ and the orthogonal group in place of the reflection. They are stated on operators, which assumes neither a convolution nor a probability kernel, and on kernels, which is the form the proofs use. Lemma 3.3 joins the two. The operator $\Phi_{s,t}$ takes the record of an image at scale s to its record at the coarser scale t.

Definition 3.1 (isotropic cascade measurement family)**.** Fix an integer $d\geq 1$. On the real space $L^1(\mathbb{R}^d)$ write $T_af:=f(\cdot-a)$ for the translation by $a\in\mathbb{R}^d$, $R_Qf:=f(Q^{-1}\cdot)$ for the action of an orthogonal matrix $Q\in\mathrm{O}(d)$, and $D_\lambda f:=\lambda^{-d}f(\cdot/\lambda)$ for the dilation by $\lambda>0$. A family $(\Phi_{s,t})_{0\leq s\leq t}$ of linear operators on $L^1(\mathbb{R}^d)$ is an *isotropic cascade measurement family in dimension d* if:

(A1) each $\Phi_{s,t}$ is a bounded operator on $L^1(\mathbb{R}^d)$;

(A2) *(translation invariance)* $\Phi_{s,t}T_a=T_a\Phi_{s,t}$ for every $a\in\mathbb{R}^d$;

(A3) *(rotation invariance)* $\Phi_{s,t}R_Q=R_Q\Phi_{s,t}$ for every $Q\in\mathrm{O}(d)$;

(A4) *(positivity)* $\Phi_{s,t}$ maps nonnegative functions to nonnegative functions;

(A5) *(mass)* $\int \Phi_{s,t} f = \int f$ for every nonnegative $f \in L^1(\mathbb{R}^d)$;

(A6) *(cascade)* $\Phi_{t,t} = \mathrm{Id}$, and $\Phi_{s,t}\Phi_{r,s} = \Phi_{r,t}$ whenever $r \le s \le t$;

(A7) *(continuity)* for every $f \in L^1(\mathbb{R}^d)$ the map $(s,t) \mapsto \Phi_{s,t} f$ is continuous from $\{0 \le s \le t\}$ to $L^1(\mathbb{R}^d)$;

(A8) *(dilation covariance)* there is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0,\infty)$ with $D_\lambda \Phi_{s,t} = \Phi_{S_\lambda s,\, S_\lambda t} D_\lambda$ for all $0 \le s \le t$ and all $\lambda > 0$;

(ND) *(nondegeneracy)* $\Phi_{s,t} \neq \mathrm{Id}$ whenever $s < t$.

The numbering is that of the line, and Remark 3.26 compares the two at $d = 1$. Axioms (A2) and (A3) say that the measurement prefers no position and no direction. The comments on (A6), (A8) and (ND) in §2.2 apply unchanged. By (A1), (A2), (A4) and (A5) each $\Phi_{s,t}$ is convolution by a probability measure, its *kernel* (Lemma 3.3(1) below). The remaining axioms then become conditions on the kernels.

Definition 3.2 (isotropic cascade family)**.** Fix an integer $d \ge 1$. For $A \in \mathrm{GL}(d)$ write $A_*\mu := \mu \circ A^{-1}$ for the image of a measure μ on $\mathbb{R}^d$, and $\widehat{\mu}(\xi) = \int e^{i\langle \xi, x\rangle} \mu(dx)$, so that $\widehat{A_*\mu}(\xi) = \widehat{\mu}(A^{\mathsf{T}}\xi)$. An *isotropic cascade family in dimension* d is a family $(\mu_{s,t})_{0 \le s \le t}$ of probability measures on $\mathbb{R}^d$ such that

(I1) *(cascade)* $\mu_{t,t} = \delta_0$, and $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ whenever $r \le s \le t$;

(I2) *(continuity)* for every $\xi \in \mathbb{R}^d$ the map $(s,t) \mapsto \widehat{\mu}_{s,t}(\xi)$ is continuous on $\{0 \le s \le t\}$;

(I3) *(rotation invariance)* $Q_*\mu_{s,t} = \mu_{s,t}$ for every $Q \in \mathrm{O}(d)$ and every pair;

(I4) *(dilation covariance)* there is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0,\infty)$ with $(\lambda I)_*\mu_{s,t} = \mu_{S_\lambda s,\, S_\lambda t}$ for every $\lambda > 0$ and every pair;

(ND) *(nondegeneracy)* $\mu_{s,t} \neq \delta_0$ whenever $s < t$.

The two definitions correspond axiom by axiom. (I1) is the cascade (A6), (I2) the continuity (A7), (I3) the rotation invariance (A3), (I4) the dilation covariance (A8) with the same bijections S_λ, and (ND) is (ND). The next lemma says that the two definitions describe the same families. The point that needs an argument is its clause (2), that the two continuities are the same: (A7) requires convergence in $L^1(\mathbb{R}^d)$, and (I2) convergence of the transforms at each frequency. The proof is in the supplement (§S3.1).

Lemma 3.3 (operators and kernels in dimension d)**.** *Fix $d \ge 1$, and for a probability measure μ on $\mathbb{R}^d$ and $f \in L^1(\mathbb{R}^d)$ write $(\mu * f)(x) := \int f(x-y)\,\mu(dy)$.*

1. (One operator.) *Let Φ be a linear operator on $L^1(\mathbb{R}^d)$ with the four axioms of Definition 3.1 that concern one operator: (A1) Φ is bounded; (A2) $\Phi T_a = T_a \Phi$ for every $a \in \mathbb{R}^d$; (A4) $\Phi f \ge 0$ whenever $f \ge 0$; (A5) $\int \Phi f = \int f$ whenever $f \ge 0$. Then there is a unique probability measure μ on $\mathbb{R}^d$ with*

$$\Phi f = \mu * f, \qquad f \in L^1(\mathbb{R}^d).$$

   Conversely, for every probability measure μ on $\mathbb{R}^d$ the operator $f \mapsto \mu * f$ satisfies (A1), (A2), (A4) and (A5).

2. (Continuity.) *Let μ_n and μ be probability measures on $\mathbb{R}^d$. Then $\widehat{\mu}_n(\xi) \to \widehat{\mu}(\xi)$ for every $\xi \in \mathbb{R}^d$ if and only if $\mu_n * f \to \mu * f$ in $L^1(\mathbb{R}^d)$ for every $f \in L^1(\mathbb{R}^d)$.*

3. (From operators to kernels.) *Let $(\Phi_{s,t})_{0\le s\le t}$ be an isotropic cascade measurement family in dimension d (Definition 3.1), and for each pair let $\mu_{s,t}$ be the probability measure that (1) gives for $\Phi_{s,t}$. Then $(\mu_{s,t})_{0\le s\le t}$ is an isotropic cascade family in dimension d (Definition 3.2), with the bijections S_λ of (A8) in (I4).*

4. (From kernels to operators.) *Let $(\mu_{s,t})_{0\le s\le t}$ be an isotropic cascade family in dimension d. Then the operators $\Phi_{s,t}f := \mu_{s,t} * f$ on $L^1(\mathbb{R}^d)$ form an isotropic cascade measurement family in dimension d, with the bijections S_λ of (I4) in (A8), and the probability measure that (1) gives for $\Phi_{s,t}$ is $\mu_{s,t}$.*

Clause (1) is Lemma 2.2 in $\mathbb{R}^d$, without the reflection. Clauses (3) and (4) are inverse to each other: the operators that (4) builds from the kernels of (3) are the $\Phi_{s,t}$, and the last assertion of (4) says that (3) returns the kernels. So convolution is a one-to-one correspondence between the isotropic cascade families and the isotropic cascade measurement families in dimension d. From here on a family on $\mathbb{R}^d$ is given by its kernels.

3.2 Reduction to the line

This subsection classifies the isotropic families by reducing them to the line. The tool is the projection of a kernel on a line through the origin. Fix a unit vector $e \in \mathbb{R}^d$. The *marginal* of a law μ on $\mathbb{R}^d$ along e is its image under $x \mapsto \langle e, x\rangle$, a law on $\mathbb{R}$. If μ has a density p and $d \ge 2$, the marginal has the density

$$y \;\longmapsto\; \int_{e^\perp} p(ye + z)\, dz, \qquad y \in \mathbb{R},$$

the integral of p over the hyperplane through ye orthogonal to e. In the plane these are the integrals of p along the lines orthogonal to e. This is the projection of tomography: taken along every direction, the marginals are the Radon transform of p ([43], §II.1, p. 9).

Two facts make the reduction work. A rotation-invariant kernel has the same marginal along every line, because a rotation takes any line through the origin to any other and leaves the kernel unchanged. And this one marginal κ determines the kernel. The Fourier transform of a marginal is the transform of the kernel restricted to the line, $\widehat{\kappa}(\omega) = \widehat{\mu}(\omega e)$, which is the projection-slice theorem of tomography ([43], Thm. II.1.1, p. 11, for smooth rapidly decreasing densities and with another sign and normalization of the transform). The transform of a rotation-invariant kernel is radial, so its values on one line give it everywhere: $\widehat{\mu}(\xi) = \widehat{\kappa}(|\xi|)$. Lemma 3.4 states both facts.

The marginals $\kappa_{s,t}$ of an isotropic family inherit its axioms, because projection takes a convolution to a convolution and a dilation to a dilation. So they are the kernels of a family on the line in the sense of Definition 2.1 (Lemma 3.5). The classification on the line, Theorem 2.3, gives their transforms, and the identity $\widehat{\mu}_{s,t}(\xi) = \widehat{\kappa}_{s,t}(|\xi|)$ carries the answer back to $\mathbb{R}^d$ (Theorem 3.6). The subsection has three parts: the classification by projection, a test for membership in $\mathcal{A}_d$ by the polar profile (Proposition 3.11), and the Gaussian variance mixtures whose mixing law is a kernel of a time-causal family, which lie in every class (Proposition 3.12). The necessity half of the classification is proved here, and its other proofs in §A.2. The proofs of Lemma 3.5 and of Proposition 3.12 are in the supplement (§S3.2).

Lemma 3.4 (a rotation-invariant law is determined by its marginal)**.** *Let $d \ge 1$, let $e \in \mathbb{R}^d$ be a unit vector and $\pi(x) := \langle e, x\rangle$, and for a probability measure μ on $\mathbb{R}^d$ write $\pi_*\mu$ for its image on $\mathbb{R}$.*

1. *If $Q_*\mu = \mu$ for every $Q \in \mathrm{O}(d)$, then $\kappa := \pi_*\mu$ is a symmetric probability measure on $\mathbb{R}$, and*

$$\widehat{\mu}(\xi) = \widehat{\kappa}(|\xi|), \qquad \xi \in \mathbb{R}^d.$$

In particular $\widehat{\mu}$ is real and radial, and κ does not depend on e.

2. *Under the hypothesis of (1),* $\pi_*\mu = \delta_0$ *if and only if* $\mu = \delta_0$.

3. *For all probability measures* μ, μ' *on* $\mathbb{R}^d$ *and every* $\lambda > 0$, $\pi_*(\mu * \mu') = \pi_*\mu * \pi_*\mu'$, $\pi_*\big((\lambda I)_*\mu\big) = (\lambda I)_*(\pi_*\mu)$ *on* $\mathbb{R}$, *and* $\pi_*\delta_0 = \delta_0$.

Clause (2) is what carries the nondegeneracy (ND) to the line, in every dimension.

Lemma 3.5 (the marginals of an isotropic family form a family on the line)**.** *Let* $(\mu_{s,t})$ *be an isotropic cascade family in dimension* d *with dilation action* (S_λ), *let* e *and* π *be as in Lemma 3.4, and put*

$$\kappa_{s,t} := \pi_*\mu_{s,t}, \qquad \Phi_{s,t}f := \kappa_{s,t} * f \quad (f \in L^1(\mathbb{R})), \qquad 0 \le s \le t.$$

Then $\widehat{\mu}_{s,t}(\xi) = \widehat{\kappa}_{s,t}(|\xi|)$ *for every* $\xi \in \mathbb{R}^d$, *each* $\kappa_{s,t}$ *is a symmetric probability measure, and* $(\Phi_{s,t})$ *is a symmetric cascade measurement family in the sense of Definition 2.1, with the same* (S_λ).

The classification in dimension d is now the line's theorem applied to the marginals. The class $\mathcal{A}_d$ of Proposition 2.6 comes from the cascade itself: the kernel from scale zero to the scale t_1 with $\chi(t_1) = 1$ is self-decomposable, with the stages as its residuals. A gauge, self-decomposability (2.2) and the admissible exponents (2.1) are as in §2.2.

Theorem 3.6 (the isotropic classification in dimension d)**.** *Let* $d \ge 1$. *A family* $(\mu_{s,t})_{0\le s\le t}$ *of probability measures on* $\mathbb{R}^d$ *is an isotropic cascade family in dimension* d *if and only if there are a gauge* χ *and an* $F \in \mathcal{A}_d$ *with* $F \not\equiv 0$ *such that*

$$\widehat{\mu}_{s,t}(\xi) = \exp\big[-\big(F(\chi(t)|\xi|) - F(\chi(s)|\xi|)\big)\big], \qquad 0 \le s \le t,\ \xi \in \mathbb{R}^d. \tag{3.1}$$

In particular the kernels from scale zero are $\widehat{\mu}_{0,t}(\xi) = e^{-F(\chi(t)|\xi|)}$. *The pair is unique up to* $(\chi, F) \mapsto (c\,\chi,\, F(c^{-1}\,\cdot))$ *with* $c > 0$, *in the sense that if a gauge* χ' *and a real function* F' *on* $[0,\infty)$ *with* $F'(0) = 0$ *also satisfy* (3.1), *then* $\chi' = c\,\chi$ *and* $F' = F(c^{-1}\,\cdot)$ *for some* $c > 0$, *and* $c = 1$ *when* $\chi(1) = \chi'(1) = 1$.

Proof. Necessity. Let $(\mu_{s,t})$ be an isotropic cascade family and let $\kappa_{s,t}$ and $\Phi_{s,t}$ be as in Lemma 3.5. By that lemma $(\Phi_{s,t})$ is a symmetric cascade measurement family in the sense of Definition 2.1, and it is convolution by the symmetric probability measures $\kappa_{s,t}$. Theorem 2.3 gives a gauge χ, an admissible $F \not\equiv 0$ and probability measures $\nu_{s,t}$ on $\mathbb{R}$ with $\Phi_{s,t}f = \nu_{s,t} * f$ and $\widehat{\nu}_{s,t}(\omega) = \exp[-(F(\chi(t)\omega) - F(\chi(s)\omega))]$. An operator of the family has only one kernel (Lemma 2.2), so $\nu_{s,t} = \kappa_{s,t}$. With $\widehat{\mu}_{s,t}(\xi) = \widehat{\kappa}_{s,t}(|\xi|)$ from Lemma 3.5, this gives (3.1).

It remains to see $F \in \mathcal{A}_d$. F is admissible, since it has the form (2.1), and $F(0) = 0$. Put $t_1 := \chi^{-1}(1)$ and $\mu := \mu_{0,t_1}$, so that $\widehat{\mu}(\xi) = e^{-F(|\xi|)}$, because $\chi(0) = 0$. For $\lambda \in (0,1)$ put $s_\lambda := \chi^{-1}(\lambda) \in (0, t_1)$. By (3.1), $\widehat{\mu}(\lambda\xi) = e^{-F(\lambda|\xi|)}$ and $\widehat{\mu}_{s_\lambda,t_1}(\xi) = e^{-(F(|\xi|)-F(\lambda|\xi|))}$, so

$$\widehat{\mu}(\xi) = \widehat{\mu}(\lambda\xi)\,\widehat{\mu}_{s_\lambda,t_1}(\xi),$$

which is (2.2) with residual $\rho_\lambda = \mu_{s_\lambda,t_1}$. So μ is self-decomposable, $\xi \mapsto F(|\xi|)$ is its exponent, and $F \in \mathcal{A}_d$. The sufficiency and the uniqueness halves are proved in Appendix A. □

With Lemma 3.3 the theorem classifies the operators as well.

Corollary 3.7 (the isotropic classification on operators)**.** *Let* $d \ge 1$. *A family* $(\Phi_{s,t})_{0\le s\le t}$ *of linear operators on* $L^1(\mathbb{R}^d)$ *is an isotropic cascade measurement family in dimension* d *(Definition 3.1) if and only if there are a gauge* χ, *an* $F \in \mathcal{A}_d$ *with* $F \not\equiv 0$, *and probability measures* $\mu_{s,t}$ *on* $\mathbb{R}^d$ *satisfying* (3.1), *such that*

$$\Phi_{s,t}f = \mu_{s,t} * f, \qquad 0 \le s \le t,\ f \in L^1(\mathbb{R}^d).$$

The measures are then unique, and the pair is unique up to $(\chi, F) \mapsto (c\,\chi,\, F(c^{-1}\,\cdot))$ *with* $c > 0$*, in the sense that if a gauge* χ'*, a real function* F' *on* $[0,\infty)$ *with* $F'(0) = 0$*, and probability measures* $\mu'_{s,t}$ *on* $\mathbb{R}^d$ *satisfying* (3.1) *with* (χ', F') *in place of* (χ, F) *also give* $\Phi_{s,t} f = \mu'_{s,t} * f$ *for all* $0 \le s \le t$ *and all* $f \in L^1(\mathbb{R}^d)$*, then* $\mu'_{s,t} = \mu_{s,t}$ *for all* $0 \le s \le t$*, and* $\chi' = c\,\chi$ *and* $F' = F(c^{-1}\,\cdot)$ *for some* $c > 0$*, and* $c = 1$ *when* $\chi(1) = \chi'(1) = 1$*.*

The 'if' half of the theorem takes probability measures with the transforms (3.1) as given. The next lemma supplies them for every pair (χ, F), from the residuals of the law with exponent $F(|\cdot|)$. With the uniqueness clause, the pairs therefore parametrise the isotropic cascade families, up to the rescaling $(\chi, F) \mapsto (c\,\chi, F(c^{-1}\,\cdot))$.

Lemma 3.8 (every pair is realised)**.** *Let* $d \ge 1$*, let* χ *be a gauge, and let* $F \in \mathcal{A}_d$ *with* $F \not\equiv 0$*. Then there is an isotropic cascade family* $(\mu_{s,t})_{0\le s\le t}$ *in dimension* d *that satisfies* (3.1) *with the pair* (χ, F)*.*

The theorem and the lemma give the main result: a law is a kernel from scale zero to a positive scale of an isotropic family exactly when it is rotation invariant, self-decomposable and not the point mass at the origin.

Corollary 3.9 (the kernels from scale zero are the rotation-invariant self-decomposable laws other than δ_0)**.** *Let* $d \ge 1$ *and let* μ *be a probability measure on* $\mathbb{R}^d$*. Consider the condition*

(S) μ *is rotation invariant, that is* $Q_*\mu = \mu$ *for every* $Q \in \mathrm{O}(d)$*, it is self-decomposable, and* $\mu \ne \delta_0$*.*

1. (On kernels.) *(S) holds if and only if there are an isotropic cascade family* $(\mu_{s,t})_{0\le s\le t}$ *in dimension* d *and a scale* $t > 0$ *with* $\mu_{0,t} = \mu$*.*

2. (On operators.) *(S) holds if and only if there are an isotropic cascade measurement family* $(\Phi_{s,t})_{0\le s\le t}$ *in dimension* d *(Definition 3.1) and a scale* $t > 0$ *with* $\Phi_{0,t} f = \mu * f$ *for every* $f \in L^1(\mathbb{R}^d)$*.*

3. (The families with this kernel.) *Suppose (S). Then there is an* $F \in \mathcal{A}_d$ *with* $F \not\equiv 0$ *and* $\widehat{\mu}(\xi) = e^{-F(|\xi|)}$ *for every* $\xi \in \mathbb{R}^d$*, and* μ *determines* F *on* $[0,\infty)$*. Let* $t_1 > 0$*. A family* $(\mu_{s,t})_{0\le s\le t}$ *of probability measures on* $\mathbb{R}^d$ *is an isotropic cascade family in dimension* d *with* $\mu_{0,t_1} = \mu$ *if and only if it satisfies* (3.1) *with this* F *and a gauge* χ *with* $\chi(t_1) = 1$*. The gauge is then unique. For every gauge* χ *with* $\chi(t_1) = 1$ *there is exactly one such family, and its kernels from scale zero are the dilations of* μ*:* $\mu_{0,t} = (\chi(t) I)_*\mu$ *for every* $t > 0$*.*

4. (Up to the relabelling of the scales.) *Let* $t_1 > 0$*, and let* $(\mu_{s,t})_{0\le s\le t}$ *and* $(\mu'_{s,t})_{0\le s\le t}$ *be isotropic cascade families in dimension* d *with* $\mu'_{0,t_1} = \mu_{0,t_1}$*. Then there is an increasing bijection* θ *of* $[0,\infty)$ *with* $\theta(t_1) = t_1$ *and* $\mu'_{s,t} = \mu_{\theta(s),\theta(t)}$ *for all* $0 \le s \le t$*. If* χ *and* χ' *are the gauges that clause (3) gives the two families, for the law* $\mu = \mu_{0,t_1}$*, then* $\theta := \chi^{-1} \circ \chi'$ *is such a bijection.*

So the rotation-invariant self-decomposable laws on $\mathbb{R}^d$ *other than the point mass at the origin are exactly the kernels from scale zero to a positive scale of the isotropic cascade families in dimension* d*, and a family with a given kernel from scale zero to a given scale is determined by the gauge (clause (3)), that is up to the relabelling of the scales (clause (4)).*

Clause (3) says how a family is recovered from one of its kernels. The law gives the exponent, and what is left free is the gauge. By clause (4) two families with the same kernel from scale zero to a positive scale differ by a relabelling of the scales and by nothing else. The proof is in §A.2. It uses the description of the class $\mathcal{A}_d$ in Proposition 3.11(2) below.

The theorem names the class $\mathcal{A}_d$ and does not say which functions belong to it. The rest of the subsection gives the members an explicit form. On the line that form is (2.1): a Gaussian coefficient a_{V} and a nonincreasing folded profile k, with $k(x)\,dx/x$ the rate at which displacements of size about x occur. In $\mathbb{R}^d$ the same two data appear in polar coordinates. A rotation-invariant Lévy measure ν_d with a density has a radial one, $f_d(|x|)$. It is the rate of the displacements per unit volume, not the density of the kernel, which is written p from §3.3 on. The role of k is played by the *polar profile*

$$k_d(r) := |S^{d-1}|\, r^d\, f_d(r), \qquad r > 0,$$

the radial density times $|S^{d-1}|\, r^d$. The factor is chosen so that the displacements of size between r_1 and r_2 have the rate $\nu_d(\{r_1 < |x| < r_2\}) = \int_{r_1}^{r_2} k_d(r)\, dr/r$, the formula of the line. At $d = 1$ the polar profile is the folded profile k of (2.1), and the Gaussian coefficients are related by $a = 2a_{\mathrm{V}}$ (Proposition 3.11(2)). The class $\mathcal{A}_d$ then has the description of the line: its members are the exponents of the pairs (a, k_d) with k_d nonincreasing (Proposition 3.11). The next definition fixes this vocabulary. It serves §3.3 as well, where the limit $k_d(0+)$ of the polar profile at radius 0 decides how the kernel behaves at its own centre, the origin of $\mathbb{R}^d$. Hence the name *origin datum*.

Definition 3.10 (isotropic origin data in dimension d, and the polar profile)**.** Fix an integer $d \geq 1$ and write σ_{d-1} for the surface measure of the unit sphere $S^{d-1} \subset \mathbb{R}^d$ and $|S^{d-1}| = \sigma_{d-1}(S^{d-1})$ for its surface area. An *isotropic origin datum in dimension d* is a pair (a, k_d) with $a \geq 0$ and $k_d : (0, \infty) \to [0, \infty)$ nonincreasing, subject to

$$\int_0^1 k_d(r)\, r\, dr < \infty \qquad \text{and} \qquad \int_1^\infty k_d(r)\, \frac{dr}{r} < \infty.$$

Its Lévy measure is the rotation-invariant measure ν_d on $\mathbb{R}^d \setminus \{0\}$ with radial density $f_d(r) = k_d(r)\,\big(|S^{d-1}|\, r^d\big)^{-1}$, so that

$$\int_{\mathbb{R}^d} g\, d\nu_d = \frac{1}{|S^{d-1}|} \int_{S^{d-1}} \sigma_{d-1}(d\eta) \int_0^\infty g(r\eta)\, k_d(r)\, \frac{dr}{r} \tag{3.2}$$

for every nonnegative Borel g vanishing at the origin. Its *exponent* is

$$F_d(\xi) = \tfrac{1}{2} a|\xi|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle \xi, x\rangle\big)\, \nu_d(dx), \qquad \xi \in \mathbb{R}^d, \tag{3.3}$$

and its *law* is the symmetric infinitely divisible law μ on $\mathbb{R}^d$ with $\widehat{\mu} = e^{-F_d}$. We call k_d the datum's *polar profile* and $c := k_d(0+) \in [0, \infty]$ its *profile at the origin*. The datum is *subordinated* when there is a delay datum (a, k_{I}) in the sense of §2.3, with k_{I} nonincreasing, whose polar profile [16, (4.6)] is k_d:

$$k_d(r) = \frac{2}{\Gamma(d/2)} \int_0^\infty v^{d/2-1} e^{-v}\, k_{\mathrm{I}}\Big(\frac{r^2}{2v}\Big)\, dv, \qquad r > 0. \tag{3.4}$$

The polar profile is the integral over the sphere of the profile k_η of the polar criterion C2 (§2.6). The delay datum and its delay profile k_{I} are those of §2.3, and (3.4) is how a delay profile becomes a polar profile. The identity (3.2) is polar coordinates (Lemma A.1 in the appendix). The next proposition describes $\mathcal{A}_d$ in polar form, by that criterion.

Proposition 3.11 (the polar form of $\mathcal{A}_d$)**.** *Let $d \geq 1$ and $F : [0, \infty) \to [0, \infty)$.*

1. (The reading.) *$\xi \mapsto F(|\xi|)$ is the exponent of a self-decomposable law on $\mathbb{R}^d$ if and only if it is the exponent F_d of an isotropic origin datum (a, k_d) in dimension d (Definition 3.10):*

$$F(|\xi|) = \tfrac{1}{2} a|\xi|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle \xi, x\rangle\big)\, \frac{k_d(|x|)}{|S^{d-1}|\, |x|^d}\, dx, \qquad \xi \in \mathbb{R}^d.$$

The datum is then unique, k_d up to modification on a Lebesgue-null set.

2. (The class.) *If this holds, F is admissible, of the form* (2.1) *with $a_{\mathrm{V}} = \frac{1}{2}a'$ and $k = k_1$ for its datum (a', k_1) in dimension* 1. *So $\mathcal{A}_d$ is the set of F for which $F(|\cdot|)$ is the exponent of an isotropic origin datum in dimension d.*

3. (The classification in polar form.) *A family $(\mu_{s,t})_{0\le s\le t}$ of probability measures on $\mathbb{R}^d$ is an isotropic cascade family in dimension d if and only if there are a gauge χ and an isotropic origin datum (a, k_d) in dimension d, not both trivial ($a > 0$, or k_d not almost everywhere* 0*), with* (3.1) *for*

$$F(r) = \tfrac{1}{2}a\,r^2 + \int_{\mathbb{R}^d} \bigl(1 - \cos(r\langle e_1, x\rangle)\bigr)\,\frac{k_d(|x|)}{|S^{d-1}|\,|x|^d}\,dx, \qquad r \ge 0.$$

Clause (1) is the test for membership in $\mathcal{A}_d$, and the uniqueness of the datum is what lets §3.3 speak of the profile at the origin of a kernel. Clause (3) is the classification with the class written out. For example, the datum $(0, b\,\mathbf{1}_{(0,\tau)})$ has the Lévy density $b\,(|S^{d-1}|\,|x|^d)^{-1}$ on the ball of radius τ and none outside it, and its exponent is in $\mathcal{A}_d$. For $d \ge 2$ that exponent is not a member of a Cin ray (Proposition 3.22(3)), although a Cin ray has the same step as its folded profile: the polar profile is not the folded profile of the same exponent. One exponent has a datum in each dimension that admits it, with a different polar profile in each, and the folded profile belongs to its marginal.

The bridge from the time-causal theory (§2.3) carries over to $\mathbb{R}^d$. The variance mixtures of d-dimensional Gaussians against the kernels of a time-causal family form an isotropic family in every dimension. That their exponents lie in every $\mathcal{A}_d$ is Proposition 2.6(2). The next proposition adds the kernels themselves, as explicit mixtures with their gauge, and names three families by their transforms: the Matérn kernels, which are the example of §3.3, and, recorded for reference, the stable kernels and the Student-t kernels.

Proposition 3.12 (the bridge in dimension d, and three named families)**.** *Let F_{I} be causally admissible in the sense of Definition 2.4. Suppose $F_{\mathrm{I}} \not\equiv 0$. Let $(\mu^{\mathrm{I}}_{x,y})_{0\le x\le y}$ be probability measures on $[0,\infty)$ with*

$$\int_{[0,\infty)} e^{-\sigma u}\,\mu^{\mathrm{I}}_{x,y}(du) = \exp\bigl[-\bigl(F_{\mathrm{I}}(y\sigma) - F_{\mathrm{I}}(x\sigma)\bigr)\bigr], \qquad \sigma \ge 0,$$

the kernels of a time-causal family in its canonical gauge, as §2.3 takes them. Fix $d \ge 1$, let $g^{(d)}_u$ be the centred Gaussian law on $\mathbb{R}^d$ with covariance uI for $u > 0$ and $g^{(d)}_0 := \delta_0$, and put

$$\mu_{s,t} := \int_{[0,\infty)} g^{(d)}_u\;\mu^{\mathrm{I}}_{s^2,t^2}(du), \qquad 0 \le s \le t, \qquad F(r) := F_{\mathrm{I}}\bigl(\tfrac{1}{2}r^2\bigr). \tag{3.5}$$

1. (The family.) *$(\mu_{s,t})$ is an isotropic cascade family in dimension d with $S_\lambda t = \lambda t$, and $\widehat{\mu}_{s,t}(\xi) = \exp[-(F(t|\xi|) - F(s|\xi|))]$: it is the family of Theorem 3.6 with gauge $\chi = \mathrm{id}$ and exponent F, and $F \in \mathcal{A}_d$, for every $d \ge 1$.*

2. (The Matérn kernels.) *If $F_{\mathrm{I}}(\sigma) = \gamma\log(1+\sigma)$ with $\gamma > 0$, then $F(r) = \gamma\log(1 + r^2/2)$ and $\widehat{\mu}_{0,t}(\xi) = (1 + t^2|\xi|^2/2)^{-\gamma}$.*

3. (The stable kernels.) *If $F_{\mathrm{I}}(\sigma) = \sigma^\alpha$ with $0 < \alpha \le 1$, then $F(r) = 2^{-\alpha}r^{2\alpha}$ and $\widehat{\mu}_{0,t}(\xi) = \exp(-2^{-\alpha}t^{2\alpha}|\xi|^{2\alpha})$, of index $2\alpha \in (0,2]$. At $\alpha = 1$, $\mu_{s,t} = g^{(d)}_{t^2-s^2}$, the Gaussian.*

4. (The Student-t kernels.) *Let $\vartheta > 0$ and let T_1 be the scaled inverse-gamma law of shape ϑ of [16, Prop. 5.10], with density $u^{-\vartheta-1}e^{-1/(2u)}/(2^\vartheta\Gamma(\vartheta))$. For every $t > 0$ the Gaussian variance mixture $\int g^{(d)}_u\,P(t^2T_1 \in du)$ is a probability measure on $\mathbb{R}^d$ with transform*

$$\frac{2^{1-\vartheta}}{\Gamma(\vartheta)}\,(t|\xi|)^\vartheta\,K_\vartheta(t|\xi|),$$

K_ϑ being the modified Bessel function of the second kind of order ϑ. If $\mu^{\mathrm{I}}_{0,1}$ is the law of T_1, that is, if the inverse-gamma laws are the delay kernels of the family, then these mixtures are the kernels $\mu_{0,t}$ of clause (1), and their exponent is in $\mathcal{A}_d$.

The kernels $\mu^{\mathrm{I}}_{x,y}$ from which the proposition starts exist for each of the three named families. The exponents $\gamma\log(1+\sigma)$ and σ^α of clauses (2) and (3) are causally admissible, and the classification of the time-causal paper supplies a family for every causally admissible exponent ([17], version 1.1.0, Thm. 7.3 with Props. 8.8 and 8.5; at $\alpha = 1$ the exponent σ is the drift term of Definition 2.4, $b_0 = 1$ and $k_{\mathrm{I}} = 0$, and its family is that of the pure delays δ_{y-x}, Cor. 7.4 and Rem. 7.5 there). The inverse-gamma laws of clause (4) are generalized gamma convolutions (Bondesson [6], §4, pp. 51–52; Halgreen [26], pp. 13–15, for every generalized inverse Gaussian law), and a generalized gamma convolution is self-decomposable ([7], §3.1, p. 30; the class and this property are Thorin's). So they are the delay laws of a time-causal family ([17], Prop. 8.17, by its Thm. 7.3). Clause (2) is the Matérn member of §2.4 at range $1/\sqrt{2}$, carried to $\mathbb{R}^d$ with the same index.

3.3 The kernel at its centre

The classification describes the kernels of an isotropic family by their transforms. This subsection asks what a kernel from scale zero looks like at its own centre, the origin of $\mathbb{R}^d$: whether a point source is recorded as a blob of finite height or as a *spike*, a density that is unbounded at the centre. The kernels from scale zero of a family are the dilations of one law (Theorem 3.6), so a spike is there at every scale. It costs a user two things. A sampled kernel has no central value. And the peak of the record of a point source depends on the size of the cell that records it: the peak is the mass that the kernel gives the cell divided by the volume of the cell, and it grows without bound as the cell shrinks. The record of a bounded image f stays bounded by $\sup|f|$ whatever the kernel, a kernel being a probability measure.

One number decides, against the dimension. For a kernel from scale zero with no Gaussian part the number is $c = k_d(0+)$, the limit of its polar profile at radius zero (Definition 3.10). In the frequency domain c is the power at which the transform falls off (Corollary 3.14). The kernel has a bounded density exactly when $c > d$ (Theorem 3.13). Figure 1 shows one exponent in three dimensions, with $c = 3/2$: the kernel is bounded on the line and has a spike in the plane and in space.

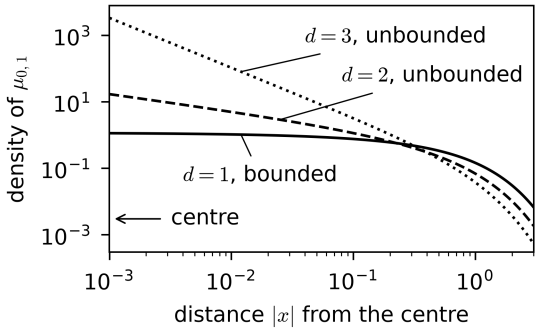


Figure 1: The radial density of the Matérn kernel $\mu_{0,1}$ of index $\gamma = 3/4$ on $\mathbb{R}^d$, $d = 1, 2, 3$, on logarithmic axes, from its closed form (3.7). Here $c = 3/2$. On the line the density is bounded and the curve levels off at the centre. In the plane and in space it is unbounded, and the curve has the slope $c - d$ there, $-\frac{1}{2}$ in the plane and $-\frac{3}{2}$ in space. Computed by a script deposited with the paper.

The theorem decides boundedness for the kernels with no Gaussian part. A kernel with a Gaussian part has a bounded smooth density in every dimension: the exponent of a datum

(a, k_d) is at least $\frac{1}{2}a|\xi|^2$, so for $a > 0$ the transform is at most $e^{-a|\xi|^2/2}$, and C1 applies with every n. In clause (4), g_t is the Gaussian density of variance t,

$$g_t(x) = (2\pi t)^{-d/2} e^{-|x|^2/2t}, \qquad x \in \mathbb{R}^d,\ t > 0,$$

which the statement takes from Lemma S.8 of the supplement. Here t is not a scale, and $\int g_t \, d\mu$ is the value at the centre of the kernel blurred by g_t.

Theorem 3.13 (the boundedness threshold in dimension d is the dimension)**.** *Let $d \geq 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0, let μ be its law, put $c := k_d(0+) \in [0, \infty]$ and suppose $c > 0$.*

1. (The profile at the origin is a decay exponent.) *For every finite $\beta < c$ there is $C_\beta < \infty$ with*
$$\left|\widehat{\mu}(\xi)\right| \leq C_\beta \, (1 + |\xi|)^{-\beta}, \qquad \xi \in \mathbb{R}^d.$$

2. (Above the threshold: bounded, and a ladder of derivatives.) *Let $N \geq 0$ be an integer. If $c > d + N$ then μ has a density of class C^N on the whole of $\mathbb{R}^d$, all of whose partial derivatives of orders $0, \ldots, N$ tend to 0 at infinity. In particular $c > d$ makes the density continuous and bounded, and $c = \infty$ makes it C^∞.*

3. (At and below the threshold, on the subordinated slice: unbounded.) *Suppose the datum is subordinated, with delay datum $(0, k_{\mathrm{I}})$ of exponent F_{I}, that W is a random delay whose law, a probability measure on $[0, \infty)$, has Laplace transform $e^{-F_{\mathrm{I}}}$, and that Z is standard normal on $\mathbb{R}^d$, independent of W. Then $c = 2\, k_{\mathrm{I}}(0+)$, $W > 0$ almost surely, μ is the law of $\sqrt{W}Z$, and μ has the radial density*
$$p(x) = \int_0^\infty (2\pi u)^{-d/2} e^{-|x|^2/2u} \, \Pr(W \in du). \tag{3.6}$$
This p is nonincreasing in $|x|$, and
$$\lim_{x \to 0} p(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{\mu}(\xi) \, d\xi = \frac{(2\pi)^{-d/2}}{\Gamma(d/2)} \int_0^\infty \sigma^{d/2-1} e^{-F_{\mathrm{I}}(\sigma)} \, d\sigma,$$
finite exactly when $c > d$. If $c \leq d$, then for every representative q of μ's density, every $\delta > 0$ and every M there is an x with $0 < |x| < \delta$ and $q(x) > M$: no representative is bounded near the origin.

4. (At and below the threshold, for every datum: unbounded.) *Suppose $c \leq d$. Then $\int_{\mathbb{R}^d} \widehat{\mu}(\xi) \, d\xi = \infty$, and there are no $\delta > 0$ and $M < \infty$ with $\mu(B) \leq M|B|$ for every Borel set $B \subseteq \{|x| < \delta\}$. In particular, for every representative q of a density of μ, every $\delta > 0$ and every M there is an x with $0 < |x| < \delta$ and $q(x) > M$. Quantitatively, with g_t the heat kernel of Lemma S.8, there is $C > 0$ with*
$$\int_{\mathbb{R}^d} g_t \, d\mu \geq C \, t^{-(d-c)/2} \quad (c < d), \qquad \int_{\mathbb{R}^d} g_t \, d\mu \geq C \, \log \frac{1}{t} \quad (c = d), \qquad 0 < t \leq 1,$$
and for every β with $0 < \beta < c$ there is C'_β with $\int g_t \, d\mu \leq C'_\beta \, t^{-(d-\beta)/2}$ for all $t > 0$. And $\log \int g_t \, d\mu / \log(1/t) \to (d - c)/2$ as $t \downarrow 0$.

5. (The dichotomy.) *The law μ has a density bounded near the origin, that is, a density q with $q(x) \leq M$ whenever $|x| < \delta$, for some $\delta > 0$ and $M < \infty$, exactly when $c > d$.*

By Theorem 3.6 and Proposition 3.11, the laws of the theorem are the kernels from scale zero with no Gaussian part, and they are all the rotation-invariant self-decomposable laws on $\mathbb{R}^d$ with no Gaussian part other than the point mass. The hypothesis $c > 0$ only excludes the point mass: k_d is nonincreasing, so c is its supremum. The result is clause (5), the threshold. Clauses (1) and (2) are its bounded side: the profile at the origin gives the decay of the transform, and the decay gives the derivatives. Clause (4) is its unbounded side, for every datum. Clause (3) is for the subordinated data, the Gaussian variance mixtures whose mixing law is a delay law of §2.3: it adds the density as an explicit mixture, its value at the centre, and c read off the delay profile, $c = 2k_1(0+)$. N continuous derivatives are enough for the derivative filters of order up to N built from the kernel to be bounded continuous functions.

The threshold is the dimension for a reason that is seen in the frequency domain. The value at the centre of a continuous density is $(2\pi)^{-d}\int_{\mathbb{R}^d}\widehat{\mu}(\xi)\,d\xi$ when the transform is integrable. For finite c and large $|\xi|$ the transform lies between a constant times $|\xi|^{-c}$ and, for every $\beta < c$, a constant times $|\xi|^{-\beta}$ (Lemma A.2(1) and clause (1)), so it is integrable exactly when $c > d$. And a law with a nonnegative transform and a density bounded near the origin has an integrable transform (Lemma S.8(2) of the supplement).

The theorem takes c from the Lévy measure in dimension d. The corollary reads it, and the Gaussian coefficient, off the exponent.

Corollary 3.14 (the exponent determines the Gaussian coefficient and the profile at the origin)**.** *Let $d \geq 1$, let (a, k_d) be an isotropic origin datum in dimension d, let F_d be its exponent, and put $c := k_d(0+) \in [0, \infty]$.*

1. (The Gaussian coefficient.) *$F_d(\xi)/|\xi|^2 \to \frac{1}{2}a$ as $|\xi| \to \infty$.*
2. (The profile at the origin.) *As $|\xi| \to \infty$,*

$$\frac{F_d(\xi) - \frac{1}{2}a|\xi|^2}{\log|\xi|} \to c,$$

   where the limit is $+\infty$ when $c = \infty$. In particular, if $a = 0$ and μ is the datum's law, then $-\log|\widehat{\mu}(\xi)|/\log|\xi| \to c$.
3. (The same numbers in every dimension.) *Let $d' \geq 1$, let $(a', k'_{d'})$ be an isotropic origin datum in dimension d' with exponent $F'_{d'}$, and suppose that the two exponents are one function of the radius: there is a function F on $[0, \infty)$ with $F_d(\xi) = F(|\xi|)$ for every $\xi \in \mathbb{R}^d$ and $F'_{d'}(\xi') = F(|\xi'|)$ for every $\xi' \in \mathbb{R}^{d'}$. Then $a' = a$ and $k'_{d'}(0+) = k_d(0+)$.*

For a datum with no Gaussian part, clause (2) is the reading of c as the power at which the transform falls off, and clause (3) removes the dimension. An exponent $F \in \mathcal{A}_D$ belongs to $\mathcal{A}_d$ for every $d \leq D$ (Proposition 2.6(1)), and in each of these dimensions $F(|\cdot|)$ is the exponent of a datum (Proposition 3.11). By clause (3) these data have one Gaussian coefficient and one c, and clause (5) of the theorem holds in each of these dimensions with that c. So a kernel that is bounded on the line can have a spike in the plane.

For the kernels that a reader knows, the number c is as follows.

- The Gaussian has a Gaussian part, and no Lévy measure. It is outside the theorem, and it is bounded and smooth.
- The stable kernels of index $p < 2$ have an exponent that is a multiple of r^p (Proposition 3.12(3)). So $F(r)/r^2 \to 0$ and $F(r)/\log r \to \infty$: by clauses (1) and (2) of the corollary they have no Gaussian part and $c = \infty$, and they are smooth in every dimension, by clause (2) of the theorem. The Poisson kernel of §2.3 is the case $p = 1$.
- The Matérn kernel of index γ has $c = 2\gamma$ (Example 3.19 below).

- On the line the member $b\,\mathrm{Cin}(\tau|\omega|)$ of a Cin ray has the folded profile $b\,\mathbf{1}_{(0,\tau)}$, so $c = b$, and the unit member sits at the threshold (Corollary 2.8(2)).

An unbounded density could be unbounded at many points. The next proposition says that it is unbounded at the centre only: the density of every kernel from scale zero, with or without a Gaussian part, is radially nonincreasing, and so finite at every positive radius.

Proposition 3.15 (the kernels from scale zero have a radially nonincreasing density)**.** *Let $d \geq 1$.*

1. *Let $(\mu_{s,t})$ be an isotropic cascade family in dimension d and let $t > 0$. Then $\mu_{0,t}$ has a density p with $p(x) = q(|x|)$ for $x \neq 0$, where $q : (0,\infty) \to [0,\infty)$ is nonincreasing.*

2. *The same holds for the law of every isotropic origin datum (a, k_d) in dimension d that is not trivial ($a > 0$, or k_d not almost everywhere 0).*

For the law at each time of a symmetric Lévy process whose Gaussian covariance is a multiple of the identity and whose Lévy measure has a radially nonincreasing density the statement is Watanabe's ([58], Proposition, p. 488; Table 3). The proof given here derives it from Wolfe's theorem C5.

The rest of the subsection gives the rate of the spike, computed from the Lévy measure. A positive function L *varies slowly* at infinity when $L(\lambda\rho)/L(\rho) \to 1$ as $\rho \to \infty$ for every $\lambda > 0$, as a constant or a logarithm does, and at 0 when the same holds as its argument decreases to 0. Two functions are *asymptotic*, written $f \sim g$, when their quotient tends to 1. The density at distance s from the centre is governed by the transform at the frequencies up to about $1/s$. For the pure power $|\xi|^{-c}$ with $c < d$ this is exact: its inverse Fourier transform is the Riesz kernel $R(d,c)\,|x|^{c-d}$, with

$$R(d,c) := \frac{\Gamma\big(\frac{d-c}{2}\big)}{2^c\,\pi^{d/2}\,\Gamma\big(\frac{c}{2}\big)}.$$

The transform of a kernel with a finite c need not be a constant times $|\xi|^{-c}$. The next proposition says what it is asymptotic to: the power $|\xi|^{-c}$ times a constant and a factor $K(1/|\xi|)$ that varies slowly, both computed from the polar profile. The lemma after it gives in closed form the one constant that depends on the dimension alone, $\varkappa_d$.

Proposition 3.16 (the exponent at infinity through the polar profile)**.** *Let $d \geq 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0 and $0 < c := k_d(0+) < \infty$, let F_d be its exponent, put $C_k := \int_1^\infty k_d(r)\,dr/r$, and let e be a unit vector of $\mathbb{R}^d$. Write $\mathrm{Cin}(X) = \int_0^X (1-\cos u)\,du/u$ and put*

$$K(x) := \exp\Big[\int_x^1 \big(c - k_d(u)\big)\,\frac{du}{u}\Big] \qquad (0 < x \leq 1).$$

1. (The factor.) *K is continuous and nonincreasing on $(0,1]$, with $1 = K(1) \leq K(x) \leq x^{-c}$. For every $a > 0$, $K(ax)/K(x) \to 1$ as $x \downarrow 0$; the quotient is defined for $0 < x \leq \min\{1, 1/a\}$. If $\int_0^1 (c - k_d(u))\,du/u < \infty$, then $K(x)$ increases, as $x \downarrow 0$, to the finite limit*

$$K(0+) := \exp\int_0^1 \big(c - k_d(u)\big)\,\frac{du}{u} \;\geq\; 1.$$

   If the integral diverges, then $K(x) \to \infty$ as $x \downarrow 0$, and $K(0+) := \infty$.

2. (Two constants.) *The limit*

$$g_\infty := \lim_{X\to\infty} \big(\mathrm{Cin}(X) - \log X\big)$$

exists, and $-3 \le g_\infty \le \frac{13}{4}$. *The function* $\eta \mapsto \log|\langle e,\eta\rangle|$ *is integrable on* S^{d-1} *for* σ_{d-1}, *and*

$$\lambda_d := \frac{1}{|S^{d-1}|}\int_{S^{d-1}} \log|\langle e,\eta\rangle|\,\sigma_{d-1}(d\eta) \in (-\infty, 0]$$

does not depend on e, *with* $\lambda_1 = 0$. *Put* $\varkappa_d := g_\infty + \lambda_d$.

3. (The exponent.) *As* $\rho \to \infty$,

$$F_d(\rho e) - \big(c\log\rho - \log K(1/\rho) + C_k\big) \to c\,\varkappa_d, \qquad \text{that is} \qquad \frac{e^{-F_d(\rho e)}}{e^{-C_k - c\varkappa_d}\,\rho^{-c}\,K(1/\rho)} \to 1.$$

Moreover, for every $\rho \ge 1$,

$$c\,(\lambda_d - 3) \;\le\; F_d(\rho e) - \big(c\log\rho - \log K(1/\rho) + C_k\big) \;\le\; \tfrac{13}{4}\,c.$$

More generally, for every $\rho > 0$,

$$c\,(\lambda_d - 3) \;\le\; F_d(\rho e) - \int_{1/\rho}^{\infty} k_d(r)\,\frac{dr}{r} \;\le\; \tfrac{13}{4}\,c,$$

and for $\rho \ge 1$ *the integral equals* $c\log\rho - \log K(1/\rho) + C_k$.

4. (When the transform is comparable to a power.) *For every* ξ *with* $|\xi| \ge 1$,

$$e^{-13c/4 - C_k}\,K(1/|\xi|) \;\le\; |\xi|^c\,e^{-F_d(\xi)} \;\le\; e^{c(3-\lambda_d) - C_k}\,K(1/|\xi|).$$

So $|\xi|^c e^{-F_d(\xi)}$ *is bounded on* $\{|\xi| \ge 1\}$ *if and only if* $\int_0^1 (c - k_d(u))\,du/u < \infty$, *and then it tends to* $e^{-C_k - c\varkappa_d}\,K(0+)$ *as* $|\xi| \to \infty$.

Lemma 3.17 (the constant of the exponent in closed form)**.** *Let* $\psi := \Gamma'/\Gamma$ *be the digamma function on* $(0,\infty)$ *and* $\gamma_{\mathrm{E}} := -\psi(1) = -\Gamma'(1)$ *Euler's constant. For an integer* $d \ge 1$ *let* g_∞, λ_d *and* $\varkappa_d = g_\infty + \lambda_d$ *be the constants of Proposition 3.16(2).*

1. $g_\infty = \gamma_{\mathrm{E}}$.
2. $\lambda_d = \frac{1}{2}\big[\psi(\frac{1}{2}) - \psi(\frac{d}{2})\big]$.
3. *Hence*

$$\varkappa_d = \gamma_{\mathrm{E}} + \tfrac{1}{2}\big[\psi(\tfrac{1}{2}) - \psi(\tfrac{d}{2})\big] = \tfrac{1}{2}\big(\gamma_{\mathrm{E}} - \psi(\tfrac{d}{2})\big) - \log 2,$$

so that $\varkappa_1 = \gamma_{\mathrm{E}}$, $\varkappa_2 = \gamma_{\mathrm{E}} - \log 2$, $\varkappa_3 = \gamma_{\mathrm{E}} - 1$, *and* $\varkappa_{d+2} = \varkappa_d - 1/d$ *for every* $d \ge 1$.

4. (The constants in closed form.) *For real numbers* $c > 0$ *and* C,

$$e^{-C - c\varkappa_d} = 2^c \exp\Big(-C - \tfrac{c}{2}\big(\gamma_{\mathrm{E}} - \psi(\tfrac{d}{2})\big)\Big).$$

Consequently, with $R(d,c)$ *the constant of Lemma S.9(1) for* $0 < c < d$,

$$R(d,c)\,e^{-C - c\varkappa_d} = \frac{\Gamma\big(\frac{d-c}{2}\big)}{\pi^{d/2}\,\Gamma\big(\frac{c}{2}\big)}\,\exp\Big(-C - \tfrac{c}{2}\big(\gamma_{\mathrm{E}} - \psi(\tfrac{d}{2})\big)\Big),$$

and

$$(2\pi)^{-d}\,|S^{d-1}|\,e^{-C - d\varkappa_d} = \frac{2}{\pi^{d/2}\,\Gamma\big(\frac{d}{2}\big)}\,\exp\Big(-C - \tfrac{d}{2}\big(\gamma_{\mathrm{E}} - \psi(\tfrac{d}{2})\big)\Big).$$

The proposition has no hypothesis on c against d, so it describes the transform above the threshold as well. Of g_∞ it uses only that the limit exists, with the bounds of clause (2), and the lemma gives its value. When $c \le d$ the density follows the transform as the Riesz kernel follows the pure power. For $c < d$ the density $q(s)$ at distance s from the centre is asymptotic, as $s \downarrow 0$, to $R(d,c)\, s^{-d}\, \widehat{\mu}(e/s)$, with e a unit vector. This step from the transform to the density is a Tauberian theorem, Lemma S.9(1) of the supplement, from which the statements here cite the constant $R(d,c)$, and §A.1 says where it is in print. At $c = d$ the transform is at the edge of integrability, and an integral of K takes the place of the power. In the statement s is a distance from the centre, not a scale.

Proposition 3.18 (the rate at the centre through the polar profile)**.** *Let $d \ge 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0, let μ be its law, and suppose that $c := k_d(0+)$ satisfies $0 < c \le d$. Let C_k, K and $\varkappa_d$ be as in Proposition 3.16, let $|S^{d-1}|$ be the surface measure of the unit sphere and V_d the volume of the unit ball, and put*

$$A := e^{-C_k - c\,\varkappa_d}.$$

The law μ has a density p with $p(x) = q(|x|)$ for $x \neq 0$, where $q : (0,\infty) \to [0,\infty)$ is nonincreasing. Let q be any such function. All limits are as $s \downarrow 0$.

1. (Below the threshold.) *Let $c < d$, let $R(d,c)$ be the constant of Lemma S.9(1), and put $Q := R(d,c)\, A$. Then*

$$\frac{q(s)}{Q\, s^{c-d}\, K(s)} \to 1 \qquad \text{and} \qquad \frac{\mu(\{|x| \le s\})}{Q\, |S^{d-1}|\, c^{-1}\, s^c\, K(s)} \to 1.$$

2. (When a pure power holds.) *Let $c < d$. If $\int_0^1 (c - k_d(u))\, du/u < \infty$, then $q(s)\, s^{d-c} \to Q\, K(0+)$, a finite positive number, and there is $C < \infty$ with $q(s) \le C\, s^{c-d}$ for every $s > 0$. If the integral diverges, then $q(s)\, s^{d-c} \to \infty$, and there is no such C.*

3. (At the threshold.) *Let $c = d$, and put*

$$\mathcal{L}(s) := \int_s^1 K(y)\, \frac{dy}{y} \quad (0 < s \le 1), \qquad B := (2\pi)^{-d}\, |S^{d-1}|\, A.$$

Then

$$\frac{q(s)}{B\, \mathcal{L}(s)} \to 1 \qquad \text{and} \qquad \frac{\mu(\{|x| \le s\})}{B\, V_d\, s^d\, \mathcal{L}(s)} \to 1.$$

Moreover $\log(1/s) \le \mathcal{L}(s) \le K(s) \log(1/s)$ for $0 < s \le 1$. If $\int_0^1 (d - k_d(u))\, du/u < \infty$, then

$$\frac{\mathcal{L}(s)}{\log(1/s)} \to K(0+) \qquad \text{and} \qquad \frac{q(s)}{\log(1/s)} \to B\, K(0+),$$

both limits finite and positive. If the integral diverges, then $\mathcal{L}(s)/\log(1/s) \to \infty$ and $q(s)/\log(1/s) \to \infty$.

Lemma 3.17(4) at $C = C_k$ gives Q and B in closed form. So the kernel at its centre is computed from its Lévy measure: the power is $c - d$, the factor $K(s)$ is one integral of the polar profile over $(s, 1)$, and the constant depends on c, on d and on the profile beyond radius 1. By clauses (2) and (3) the density is asymptotic to a constant times $|x|^{c-d}$, or times $\log(1/|x|)$ at the threshold, exactly when K is bounded. The limits for the mass of a ball give the peak of the record of a point source, the mass of the recording cell divided by its volume. For a cell of radius s round the centre it is asymptotic to $(d/c)\, Q\, s^{c-d} K(s)$ for $c < d$ and to $B\, \mathcal{L}(s)$ for $c = d$, because $|S^{d-1}| = d\, V_d$.

Example 3.19 (the Matérn kernels)**.** The Matérn kernel of index $\gamma > 0$ is the kernel from scale zero with exponent $F(r) = \gamma \log(1 + r^2/2)$ (Proposition 3.12(2)). Here $F(r) = F_{\mathrm{I}}(r^2/2)$ with $F_{\mathrm{I}}(\sigma) = \gamma \log(1 + \sigma)$, which is causally admissible with delay profile $k_{\mathrm{I}}(u) = \gamma e^{-u}$, and its delay law is the gamma law of shape γ, with density $u^{\gamma-1}e^{-u}/\Gamma(\gamma)$. So F lies in every $\mathcal{A}_d$ (Proposition 2.6(2)), its datum in each dimension is subordinated (Lemma A.6), and clause (3) of Theorem 3.13 applies with a gamma-distributed W. It gives $c = 2k_{\mathrm{I}}(0+) = 2\gamma$. So the kernel is bounded exactly when $\gamma > d/2$: for $\gamma > \frac{1}{2}$ on the line, for $\gamma > 1$ in the plane and for $\gamma > \frac{3}{2}$ in space. The mixture (3.6) is here, at scale 1 ([20], (24) and the display after it, p. 241),

$$p(x) = \frac{2}{(2\pi)^{d/2}\,\Gamma(\gamma)} \Big(\frac{|x|}{\sqrt{2}}\Big)^{\gamma-d/2} K_{\gamma-d/2}\big(\sqrt{2}\,|x|\big), \tag{3.7}$$

with K_ν the modified Bessel function of the second kind. For $\gamma > d/2$ this is the Matérn function of smoothness $\gamma - d/2$ and range $1/\sqrt{2}$ ([45], (9) of the arXiv version), so the kernel is bounded exactly where it is a Matérn covariance function. For every $\gamma > 0$ it is the Bessel kernel of potential theory, dilated by $1/\sqrt{2}$ ([2], (4,1), p. 416). The authors of the Bessel scale space (§1) note that the kernel with $2\gamma = d+1$ is continuous and not differentiable at the centre, and the kernel with $2\gamma = d+3$ twice continuously differentiable ([8], p. 87). The continuity and the two derivatives are the cases $N = 0$ and $N = 2$ of clause (2) of Theorem 3.13.

By (3.4) with the delay profile γe^{-u}, the polar profile of the kernel is

$$k_d(r) = \frac{4\gamma}{\Gamma(d/2)} \Big(\frac{r}{\sqrt{2}}\Big)^{d/2} K_{d/2}\big(\sqrt{2}\,r\big),$$

which is $2\gamma e^{-\sqrt{2}\,r}$ on the line and $2\gamma(1 + \sqrt{2}\,r)e^{-\sqrt{2}\,r}$ in space. For it the integral $\int_0^1 (c - k_d(u))\,du/u$ is finite, so the factor K is bounded, and for $\gamma < d/2$ the spike is a pure power (Proposition 3.18(2)). Its coefficient needs no integral. The transform $(1+\rho^2/2)^{-\gamma}$ is asymptotic to $2^\gamma \rho^{-2\gamma}$, and to $A\,\rho^{-c}K(1/\rho)$ by Proposition 3.16(3), so $A\,K(0+) = 2^\gamma$ in every dimension. Hence, as $x \to 0$,

$$p(x) \sim \frac{\Gamma(d/2-\gamma)}{2^\gamma \pi^{d/2}\,\Gamma(\gamma)}\,|x|^{2\gamma-d} \quad \big(\gamma < \tfrac{d}{2}\big), \qquad p(x) \sim \frac{2}{(2\pi)^{d/2}\,\Gamma(d/2)}\,\log\frac{1}{|x|} \quad \big(\gamma = \tfrac{d}{2}\big),$$

the coefficients being $Q\,K(0+)$ and $B\,K(0+)$. The second case is index 1 in the plane. Both are the behaviour of the closed form (3.7) at the centre ([2], (4,2), p. 416, in the present scaling). Table 1 has the numbers for $\gamma = 3/4$, the kernel of Figure 1.

d	$c - d$	C_k	$K(0+)$	A	Q	the density at the centre
1	$\frac{1}{2}$	0.171	4.741	0.355		bounded, with $p(0) \approx 1.180$
2	$-\frac{1}{2}$	0.359	2.023	0.831	0.277	$p(x) \sim 0.560\,\lvert x\rvert^{-1/2}$
3	$-\frac{3}{2}$	0.535	1.524	1.104	0.0701	$p(x) \sim 0.107\,\lvert x\rvert^{-3/2}$

Table 1: The Matérn kernel $\mu_{0,1}$ of index $\gamma = 3/4$, at scale 1 in the gauge $\chi(t) = t$, for which $c = 3/2$, in dimensions 1, 2 and 3: the constants of Propositions 3.16 and 3.18, and the density at the centre. In each dimension $A\,K(0+) = 2^{3/4}$. Computed by a script deposited with the paper.

The factor need not be bounded. Let $0 < c < d$, and let $k_d(u) = c\,(1 - 1/(2 + \log(1/u)))$ for $0 < u \le 1$, continued beyond radius 1 by a nonincreasing function with $C_k < \infty$. The profile approaches c like $1/\log(1/u)$, the integral diverges, and $K(x) = (1 + \frac{1}{2}\log(1/x))^c$. So $q(s) \sim Q\,s^{c-d}\,(1 + \frac{1}{2}\log(1/s))^c$: the spike exceeds every multiple of the pure power, by a power of a logarithm.

The statements are for one law, and a family has a kernel for each scale. The kernel from scale zero to scale t is the dilation of the law by $\chi(t)$, with density $\chi(t)^{-d}\,p(x/\chi(t))$. For $c<d$ this is asymptotic at the centre to $\chi(t)^{-c}\,Q\,|x|^{c-d}K(|x|)$, because K varies slowly. So the power and the factor are the same at every scale, and the constant is multiplied by $\chi(t)^{-c}$. The radius 1 in K and in C_k is the unit of length at the scale where $\chi(t)=1$.

Two bounds complement the rate, and §A.3 states them. An asymptotic equivalence gives no radius on which an inequality holds and no constant for it. For $d\ge 2$ and a finite c the kernel is bounded from below near its centre: $\mu(E)\ge c_1\int_E |x|^{c-d}\,dx$ for every Borel set E in a ball $\{|x|<\delta\}$, with δ and c_1 given in terms of d, of c and of the polar profile (Proposition A.3). For $c\le d$ and every $\beta<c$ the density is bounded from above at every radius: $q(s)\le (A_\beta/V_d)\,s^{\beta-d}$, with V_d the volume of the unit ball and a constant A_β that the supplement gives in closed form (Proposition A.4; §S3.3). For $d\ge 2$ and $c\le d$ the two bounds together give the order of the density at the centre, $\log p(x)/\log(1/|x|)\to d-c$ (its clause (3)).

The stages. The statements above are about the kernels from scale zero. A stage $\mu_{s,t}$ with $s>0$, which is what a cascade applies to a record, is a residual of such a kernel (§2.2), and need not be a dilation of it. When the kernels have no Gaussian part and c is finite, a stage does not blur everything. It returns the fraction $(\chi(s)/\chi(t))^c$ of the record at scale s unchanged, and adds a blur of that record by a density of mass $1-(\chi(s)/\chi(t))^c$. With a Gaussian part, or with $c=\infty$, as for the stable kernels, a stage is a blur by a density.

Proposition 3.20 (the stages of a cascade)**.** *Let $d\ge 1$, let $(\mu_{s,t})_{0\le s\le t}$ be an isotropic cascade family in dimension d, and let χ be a gauge and $F\in\mathcal{A}_d$ with $F\not\equiv 0$ that satisfy* (3.1) *(Theorem 3.6). Let (a,k_d) be an isotropic origin datum in dimension d with exponent $\xi\mapsto F(|\xi|)$, put $c:=k_d(0+)\in[0,\infty]$, let $0<s<t$, and put $\lambda:=\chi(s)/\chi(t)\in(0,1)$. The* Lévy measure of the stage *is the measure $\nu_{s,t}$ on $\mathbb{R}^d\setminus\{0\}$ with the density $\kappa_{s,t}(|x|)\,\big(|S^{d-1}|\,|x|^d\big)^{-1}$, where*

$$\kappa_{s,t}(r):=k_d\Big(\frac{r}{\chi(t)}\Big)-k_d\Big(\frac{r}{\chi(s)}\Big)\qquad(r>0).$$

1. (The Lévy measure of a stage.) *$\kappa_{s,t}\ge 0$, the measure $\nu_{s,t}$ is rotation invariant, and $\int_{\mathbb{R}^d}(|x|^2\wedge 1)\,\nu_{s,t}(dx)<\infty$. The transform of the stage is*

$$\widehat{\mu}_{s,t}(\xi)=\exp\Big[-\tfrac12 a\big(\chi(t)^2-\chi(s)^2\big)|\xi|^2-\int_{\mathbb{R}^d}\big(1-\cos\langle\xi,x\rangle\big)\,\nu_{s,t}(dx)\Big],\qquad \xi\in\mathbb{R}^d,\quad(3.8)$$

   and $\nu_{s,t}(\mathbb{R}^d\setminus\{0\})=c\,\log\big(\chi(t)/\chi(s)\big)$, in $[0,\infty]$. So $\nu_{s,t}$ is a finite measure exactly when $c<\infty$.

2. (No Gaussian part and a finite c: a point mass at the centre.) *Suppose that $a=0$ and that c is finite, and put $m:=c\,\log(\chi(t)/\chi(s))$. Then $c>0$, and:*

   (a) (The stage is compound Poisson.) *$\mu_{s,t}=e^{-m}\sum_{n\ge0}\nu_{s,t}^{*n}/n!$, with $\nu_{s,t}^{*0}=\delta_0$, and $e^{-m}=\lambda^c$.*

   (b) (A point mass at the centre and a smoothing part.) *The measure $\eta_{s,t}:=\lambda^c\sum_{n\ge1}\nu_{s,t}^{*n}/n!$ is absolutely continuous with respect to Lebesgue measure, of total mass $1-\lambda^c$, and*

$$\mu_{s,t}=\lambda^c\,\delta_0+\eta_{s,t}.$$

   So $\mu_{s,t}(\{0\})=(\chi(s)/\chi(t))^c\in(0,1)$, and $\mu_{s,t}$ has no density.

   (c) (The operator.) *With $p_{s,t}\ge 0$ a density of $\eta_{s,t}$, an integrable function of integral $1-\lambda^c$, the operator of the stage is*

$$\mu_{s,t}*f=\Big(\frac{\chi(s)}{\chi(t)}\Big)^c f+p_{s,t}*f,\qquad f\in L^1(\mathbb{R}^d).$$

(d) (Not self-decomposable.) *$\mu_{s,t}$ is not self-decomposable.*

3. (Infinitely divisible; the stages with a density.)

(a) $\mu_{s,t}$ is infinitely divisible and has the generating triplet $\big(a(\chi(t)^2-\chi(s)^2)\,I,\ \nu_{s,t},\ 0\big)$: its Gaussian part has the variance $a(\chi(t)^2-\chi(s)^2)$ in every direction, and its Lévy measure has the polar profile $\kappa_{s,t}$ and the mass $c\,\log(\chi(t)/\chi(s))$.

(b) If $a=0$ and $c=\infty$, then $\mu_{s,t}$ is absolutely continuous with respect to Lebesgue measure. In particular it has no point mass.

(c) If $a>0$, then $\mu_{s,t}$ has a bounded density of class C^∞, all of whose partial derivatives are bounded.

So a stage $\mu_{s,t}$ with $0<s<t$ has a point mass at the centre exactly when the datum has no Gaussian part and a finite profile at the origin, and it has a density in the two other cases.

Clauses (1) and (3) give the stage as a law: it is infinitely divisible, and its Lévy measure is the difference of two dilations of that of the kernel. The fraction of clause (2) tends to 0 as $s\downarrow 0$, and the kernels from scale zero have no point mass. The number c, the ratio $\chi(t)/\chi(s)$ and the product $a\,\chi(t)^2$ are the same for every pair (χ,F) of the family, by the uniqueness clause of Theorem 3.6. For the Matérn kernels of Example 3.19, in the gauge $\chi(t)=t$ of Proposition 3.12, the fraction is $(s/t)^{2\gamma}$. At $\gamma=1$ the stage is $\mu_{s,t}=(s/t)^2\delta_0+(1-(s/t)^2)\,\mu_{0,t}$ in every dimension, a point mass and a multiple of the kernel at scale t: its transform is $(1+s^2\sigma)/(1+t^2\sigma)=(s/t)^2+(1-(s/t)^2)/(1+t^2\sigma)$ at $\sigma=|\xi|^2/2$.

3.4 The dimension filtration

In every dimension an isotropic family is given by the same kind of object, a gauge and an exponent F, a function of one variable: the transform of its kernel from scale zero is $e^{-F(\chi(t)|\xi|)}$ (Theorem 3.6). The familiar families use one exponent in every dimension. The Gaussian has $F(r)=\frac{1}{2}r^2$, and the α-scale spaces, which are written on $\mathbb{R}^d$ [13], have a multiple of $r^{2\alpha}$ (Proposition 3.12(3)). This subsection asks whether that is general. Does the exponent of a family in dimension d give a family in dimension $d+1$ when it is read as a function of $|\xi|$? By Theorem 3.6 and Lemma 3.8, the exponents of the isotropic families in dimension d are exactly the members $F\not\equiv 0$ of the class $\mathcal{A}_d$ of Proposition 2.6. The way down is always open, since $\mathcal{A}_{d+1}\subseteq\mathcal{A}_d$. The way up is not: in every dimension some exponents are lost. The exponents admitted in every dimension are exactly those of the subordinated families of §2.3 (Proposition 2.6(3)), the Matérn and the stable kernels among them (Proposition 3.12).

The subsection has three parts. The first matches the exponents of the kernels with no Gaussian part and a finite c with probability laws on $\mathbb{R}^d$, and says which of them are admitted one dimension up. The second is a test on the Lévy measure, for every exponent, with what becomes of the simplest exponent that fails it. The third follows one family of exponents through the dimensions.

The first statement is for the kernels of §3.3 with no Gaussian part and a finite c. Their exponents are the members of $\mathcal{A}_d$ that grow like a logarithm, $F(r)/\log r\to c$ with $0<c<\infty$ (Corollary 3.14). The Matérn kernels are among them, and the Gaussian and the stable kernels are not. Let k_d be the polar profile of such a kernel. The function k_d/c decreases from 1 at radius zero to 0 at infinity, so it is, up to its values at its jumps, the tail $r\mapsto\Pr(R>r)$ of a random radius $R>0$. Let X be that radius in a direction drawn uniformly from the sphere. The transform of the law of X is a function $\phi(|\xi|)$ of the radius alone, a *radial characteristic function* on $\mathbb{R}^d$. Read as a function of a displacement, it is equally an isotropic correlation function that is valid in $\mathbb{R}^d$. The proposition says that F and the pair (c,ϕ) determine each other, and that F is admitted one dimension up exactly when the same function of the radius is still the transform of a probability law on $\mathbb{R}^{d+1}$.

In the statement $\Omega_d(t)$ is the mean of $\cos(t\langle e,\eta\rangle)$ over the unit sphere, the transform along a ray of the uniform law on the sphere: $\Omega_1 = \cos$, $\Omega_2 = J_0$, the Bessel function of order zero, and $\Omega_3(t) = \sin t/t$. A random vector X stands for its law, and $\log^+ := \max\{\log, 0\}$. Two conditions are put on the law. The first, $\mathbb{E}\log^+|X| < \infty$, a *finite logarithmic moment*, is the condition $\int_1^\infty k_d(r)\,dr/r < \infty$ of Definition 3.10, because $\mathbb{E}\log^+|X| = \int_1^\infty \Pr(|X| > r)\,dr/r$. The second, no atom at the origin, makes $k_d(0+) = c$: an atom of mass p there would give the same F with $c(1-p)$ in place of c, so the condition is what makes the pair unique.

Proposition 3.21 (the dictionary between $\mathcal{A}_d$ and the radial characteristic functions)**.** *Let $d \geq 1$. For $t \geq 0$ put*

$$\Omega_d(t) := \frac{1}{|S^{d-1}|}\int_{S^{d-1}} \cos\bigl(t\langle e,\eta\rangle\bigr)\,\sigma_{d-1}(d\eta),$$

which does not depend on the unit vector e; $\Omega_1 = \cos$. Call a random vector X in $\mathbb{R}^d$ rotation invariant *when QX has the law of X for every $Q \in \mathrm{O}(d)$. Its characteristic function is then real and radial,*

$$\mathbb{E}\,e^{i\langle\xi,X\rangle} = \phi(|\xi|), \qquad \phi(r) := \mathbb{E}\,\Omega_d\bigl(r|X|\bigr) \quad (r \geq 0),$$

the function ϕ determines the law of X, and X has the law of RU, where R has the law of $|X|$ and U is uniformly distributed on S^{d-1} and independent of R.

1. (From a law to an exponent.) *Let X be a rotation-invariant random vector in $\mathbb{R}^d$ with $\Pr(X = 0) = 0$, let ϕ be as above, and let $0 < c < \infty$. Then $\int_0^1 (1-\phi(u))\,du/u < \infty$ if and only if $\mathbb{E}\log^+|X| < \infty$. In that case the function*

$$F(r) := c\int_0^r \bigl(1-\phi(u)\bigr)\,\frac{du}{u} \qquad (r \geq 0)$$

belongs to $\mathcal{A}_d$: $\xi \mapsto F(|\xi|)$ is the exponent of the isotropic origin datum $(0, k_d)$ in dimension d with $k_d(r) = c\,\Pr(|X| > r)$. Its profile at the origin is $k_d(0+) = c$, and $F(r)/\log r \to c$ as $r \to \infty$.

2. (From an exponent to a law.) *Conversely, let $F \in \mathcal{A}_d$ and suppose that $F(r)/\log r$ has a limit $c \in (0,\infty)$ as $r \to \infty$. Then F is the function of clause (1) for this c and for a rotation-invariant X with $\Pr(X = 0) = 0$ and $\mathbb{E}\log^+|X| < \infty$, whose law is unique. Moreover F is continuously differentiable on $(0,\infty)$, with*

$$\phi(r) = 1 - \frac{r\,F'(r)}{c} \quad (r > 0), \qquad \Pr(|X| > r) = \frac{k_d(r+)}{c} \quad (r > 0)$$

for every isotropic origin datum $(0, k_d)$ in dimension d with exponent $\xi \mapsto F(|\xi|)$. So $(c,$ the law of $X) \mapsto F$ is a bijection from the pairs of clause (1) with $\mathbb{E}\log^+|X| < \infty$ onto the set of the $F \in \mathcal{A}_d$ for which $F(r)/\log r$ has a limit in $(0,\infty)$. These are the members of $\mathcal{A}_d$ with Gaussian coefficient 0 and profile at the origin in $(0,\infty)$.

3. (The next dimension.) *Let F, c, X and ϕ be as in clause (1), with $\mathbb{E}\log^+|X| < \infty$. Then $F \in \mathcal{A}_{d+1}$ if and only if $\zeta \mapsto \phi(|\zeta|)$, $\zeta \in \mathbb{R}^{d+1}$, is the characteristic function of a random vector X' in $\mathbb{R}^{d+1}$. Every such X' is rotation invariant, with $\Pr(X' = 0) = 0$ and $\mathbb{E}\log^+|X'| < \infty$, its first d coordinates have the law of X, and F is the function of clause (1) in dimension $d+1$ for the same c and for X'.*

For the Matérn exponent $F(r) = \gamma\log(1 + r^2/2)$, with $c = 2\gamma$, clause (2) gives

$$\phi(r) = 1 - \frac{rF'(r)}{2\gamma} = \Bigl(1 + \frac{r^2}{2}\Bigr)^{-1},$$

the transform of the Matérn kernel of index 1. By the same clause the polar profile of the Matérn kernel of index γ is 2γ times the tail of the radius of the Matérn kernel of index 1,

which on the line is $e^{-\sqrt{2}\,r}$. The radial functions that are characteristic functions on $\mathbb{R}^m$ form classes that decrease as m grows, the classes of Schoenberg ([52], §2.1 and Thm. 2, pp. 816–817), and they decrease strictly in every dimension ([23], p. 94). By clause (3) a function of the class of $\mathbb{R}^d$ that is not in the class of $\mathbb{R}^{d+1}$ serves as ϕ and gives an exponent in $\mathcal{A}_d$ and not in $\mathcal{A}_{d+1}$, when its law has no atom at the origin and a finite logarithmic moment.

The second statement needs no law and no characteristic function, and it holds for every exponent, with or without a Gaussian part. It follows the Lévy measure one dimension down. The *shadow* of a rotation-invariant Lévy measure on $\mathbb{R}^{d+1}$ is its image under the projection onto $\mathbb{R}^d$, restricted to $\mathbb{R}^d \setminus \{0\}$. The polar profile of the shadow is an average of the profile upstairs over the radii at least as large (Lemma A.5), and such an average is continuous. So a polar profile with a jump, which belongs to a Lévy density with a jump at that radius, is the shadow of nothing.

Proposition 3.22 (the profile one dimension down; the step datum; the Cin rays)**.** *Let $d \geq 1$.*

1. (The profile one dimension down is continuous.) *Let $F \in \mathcal{A}_{d+1}$, and let (a', k') be an isotropic origin datum in dimension $d+1$ with exponent $\zeta \mapsto F(|\zeta|)$; there is one, by Proposition 3.11(2). Then $F \in \mathcal{A}_d$, and every isotropic origin datum (a, k_d) in dimension d with exponent $\xi \mapsto F(|\xi|)$ has $a = a'$ and*

$$k_d(r) = \frac{|S^{d-1}|}{|S^d|} \int_{\mathbb{R}} k'\big(r\sqrt{1+u^2}\big)\,(1+u^2)^{-(d+1)/2}\,du \qquad \text{for every } r > 0.$$

   In particular k_d is continuous on $(0,\infty)$. So if $F \in \mathcal{A}_d$ and some isotropic origin datum in dimension d with exponent $\xi \mapsto F(|\xi|)$ has a polar profile with a discontinuity, then $F \notin \mathcal{A}_{d+1}$.

2. (Strictness at every step, with a witness.) *Let $b > 0$ and $\tau > 0$. Then $(0, b\,\mathbf{1}_{(0,\tau)})$ is an isotropic origin datum in dimension d, and its exponent is radial: it is $\xi \mapsto G(|\xi|)$ for a function G on $[0,\infty)$. This G belongs to $\mathcal{A}_d \setminus \mathcal{A}_{d+1}$. Hence $\mathcal{A}_{d+1} \subsetneq \mathcal{A}_d$ for every $d \geq 1$.*

3. (The Cin rays.) *At $d = 1$, $G(r) = b\ \mathrm{Cin}(\tau r)$. So no* Cin *ray belongs to $\mathcal{A}_2$, nor to $\mathcal{A}_D$ for any $D \geq 2$.*

Clause (1) is a test that needs one dimension only: an exponent of $\mathcal{A}_d$ whose polar profile there has a jump is admitted in no higher dimension. For example, the step $b\,\mathbf{1}_{(0,\tau)}$ of the plane has on the line the shadow $(2b/\pi)\arccos(r/\tau)$ on $(0,\tau)$, which falls continuously from b to 0: one projection removes the sharp edge. The converse of the test fails, since the family below has smooth profiles and still leaves the classes. Clause (2) makes the simplest datum a *witness* in every dimension, an exponent that is admitted there and not one dimension up. Clause (3) is about the Cin rays, the elementary exponents of §1, whose folded profile is a single step (Proposition 2.7). Every member of $\mathcal{A}_2$ is an exponent of the line, and so still a superposition of them and of a Gaussian term. What the clause excludes is a Cin ray taken alone, which is the exponent of no isotropic family in the plane or in any higher dimension.

Read in the dictionary, the first two clauses say the following (Corollary A.7, in §A.4). The law that the dictionary matches with an exponent admitted one dimension up has a radius with no atom. And the law that it matches with the step datum is the uniform law on the sphere of radius τ, whose radius is an atom, so $\zeta \mapsto \Omega_d(\tau|\zeta|)$ is not a characteristic function on $\mathbb{R}^{d+1}$: the function $\cos(\tau|\zeta|)$ is not the transform of a probability law in the plane, nor $J_0(\tau|\zeta|)$ of one in space.

The step datum is excluded from the next dimension as the exponent of a cascade. It remains the exponent of an infinitely divisible law there, and in exactly one dimension more.

Proposition 3.23 (the step exponent one and two dimensions up)**.** *Let $d \geq 1$, $b > 0$ and $\tau > 0$. For an integer $n \geq 1$ let Ω_n be the function of Proposition 3.21 in dimension n, and put*

$$G_n(\rho) := b \int_0^\tau \big(1 - \Omega_n(r\rho)\big)\, \frac{dr}{r} \qquad (\rho \geq 0),$$

so that $\xi \mapsto G_n(|\xi|)$, $\xi \in \mathbb{R}^n$, is the exponent of the step datum $(0, b\,\mathbf{1}_{(0,\tau)})$ in dimension n (Corollary A.7(2), in whose form the substitution $r = \tau u/\rho$ puts the integral). Write $G := G_d$. It is the function of Proposition 3.22(2), and by that clause with the nesting of the classes, $G \in \mathcal{A}_D$ exactly when $D \leq d$. Say that a function $\psi \geq 0$ on $\mathbb{R}^D$ is the exponent of an infinitely divisible law *when $e^{-\psi}$ is the characteristic function of an infinitely divisible probability measure on $\mathbb{R}^D$.*

1. (Exactly two dimensions more.) *Let $D \geq 1$ be an integer. The function $\zeta \mapsto G(|\zeta|)$, $\zeta \in \mathbb{R}^D$, is the exponent of an infinitely divisible law on $\mathbb{R}^D$ if and only if $D \leq d + 2$.*

2. (Dimension $d + 2$.) *For every $\rho \geq 0$,*

$$G_d(\rho) = G_{d+2}(\rho) + \frac{b}{d}\big(1 - \Omega_{d+2}(\tau\rho)\big). \tag{3.9}$$

   So the infinitely divisible law on $\mathbb{R}^{d+2}$ with exponent $\zeta \mapsto G(|\zeta|)$ is the convolution of the law of the step datum $(0, b\,\mathbf{1}_{(0,\tau)})$ of dimension $d + 2$ with the compound Poisson law $e^{-b/d} \sum_{n\geq 0} (b/d)^n \sigma_\tau^{*n}/n!$, where σ_τ is the uniform probability measure on the sphere of radius τ of $\mathbb{R}^{d+2}$: jumps at the rate b/d, uniformly distributed on that sphere. Its generating triplet is $(0,\ \nu_{d+2}^{\text{step}} + \frac{b}{d}\,\sigma_\tau,\ 0)$, with ν_{d+2}^{step} the Lévy measure of that step datum.

3. (Dimension $d + 1$.) *The infinitely divisible law on $\mathbb{R}^{d+1}$ with exponent $\zeta \mapsto G(|\zeta|)$ has the generating triplet $(0, \nu', 0)$, where ν' is the rotation-invariant measure on $\mathbb{R}^{d+1} \setminus \{0\}$ with the density $k'(|z|)\,(|S^d|\,|z|^{d+1})^{-1}$ and*

$$k'(s) := 2b\,C_{d+1}\Big[J_{d+2}\Big(\frac{\tau}{s}\Big) + \frac{1}{d}\Big(\frac{s}{\tau}\Big)^d \frac{s}{\sqrt{\tau^2 - s^2}}\Big] \quad (0 < s < \tau), \qquad k'(s) := 0 \quad (s \geq \tau),$$

   with $J_n(x) := \int_1^x v^{1-n}(v^2 - 1)^{-1/2}\, dv$ for $x \geq 1$ and $C_n = |S^{n-1}|/|S^n|$ as in Lemma A.5. The function k' is positive on $(0, \tau)$, $k'(s) \to b$ as $s \downarrow 0$, and $k'(s) \to \infty$ as $s \uparrow \tau$. For $d = 1$, $k'(s) = b\,\tau/\sqrt{\tau^2 - s^2}$, and for $d = 2$, $k'(s) = (2b/\pi)\big[\arccos(s/\tau) + s/\sqrt{\tau^2 - s^2}\big]$, on $(0, \tau)$.

In particular a Cin *ray, $G(r) = b\ \mathrm{Cin}(\tau r)$, which is the case $d = 1$ (Proposition 3.22(3)), is the exponent of an infinitely divisible law in the plane and in space, of a self-decomposable law in neither, and of no infinitely divisible law on $\mathbb{R}^4$.*

So one can still measure with the dilations of this kernel in dimensions $d + 1$ and $d + 2$, but they form no cascade: for some pairs of scales the record at the coarser one is not a convolution of the record at the finer one. Clauses (2) and (3) say what the kernel is made of. In dimension $d + 2$ it is the step kernel of that dimension with isolated displacements of the one length τ added, at the rate b/d. In dimension $d + 1$ its polar profile k' is no step: it starts from b at the centre and is unbounded at the radius τ, so it is not nonincreasing. The J_n of its formula is the integral defined in the clause, not a Bessel function.

The third statement follows one family through the dimensions, the family $F^{(w)}$, $w > 0$, of the cone paper. It has the form of the bridge, $F^{(w)}(r) = F_{\mathrm{I}}^{(w)}(r^2/2)$ for a delay datum (§2.3), and its delay profile is a power plus a bump of height w on $[1, 2]$. In dimension d its polar profile is the sum of a multiple of $1/r$, which decreases, and of w times the polar profile of the bump, which rises before it falls (its closed form is (S.18) of the supplement). The sum is nonincreasing

exactly when $w\,z^m\big(e^{-z}-2^{-d/2}e^{-z/2}\big) \le \frac{1}{2}\Gamma(m)$ for every $z>0$, with $m=(d+1)/2$, that is, when the bump is no higher than the decrease of the power allows (Lemma S.10 of the supplement). The definition names the largest such height w_d, through the supremum of the left side. These are the *critical values* of the family.

Definition 3.24 (the separating family and its critical values)**.** Let $\mathcal{S}$, $\mathcal{A}_d$ and $\mathcal{A}_\infty$ be the subordinated slice and the classes of Proposition 2.6, sets of functions on $[0,\infty)$. For $w>0$ let $D^{(w)}$ be the delay datum (§2.3) with drift 0 and profile $k_{\mathrm{I}}^{(w)}(u)=u^{-1/2}+w\,\mathbf{1}_{[1,2]}(u)$, let $F_{\mathrm{I}}^{(w)}$ be its exponent, put $F^{(w)}(\omega):=F_{\mathrm{I}}^{(w)}(\omega^2/2)$ for $\omega\ge 0$, and write $k_d^{(w)}$ for its polar profile (3.4) in dimension d. Finally put

$$M_d:=\sup_{z>0} z^m\big(e^{-z}-2^{-d/2}e^{-z/2}\big), \qquad w_d:=\frac{\Gamma(m)}{2M_d}, \qquad m:=\frac{d+1}{2}. \tag{3.10}$$

The theorem says in which dimensions each member of the family is admitted.

Theorem 3.25 (the separating family and its critical values)**.** *Let $F^{(w)}$, $w>0$, be the separating family of Definition 3.24 and $w_1,w_2,\dots$ its critical values.*

1. (Membership.) *For every integer $D\ge 1$ and every $w>0$, $F^{(w)}\in\mathcal{A}_D$ if and only if $w\le w_D$.*
2. (The critical values decrease strictly.) $w_1>w_2>w_3>\cdots>0$.
3. (Their limit.) $\sqrt{D}\,w_D\to\sqrt{\pi}$ *as* $D\to\infty$.
4. (The dimensions in which a member is admitted.) *Let $w>0$. If $w\in(w_{d+1},w_d]$ for an integer $d\ge 1$, then $F^{(w)}\in\mathcal{A}_D$ for $D=1,\dots,d$ and for no $D>d$. If $w>w_1$, then $F^{(w)}$ belongs to no $\mathcal{A}_D$. Every $w\in(0,w_1]$ lies in exactly one of the intervals $(w_{d+1},w_d]$. Each of these intervals is nonempty, so $\mathcal{A}_d\setminus\mathcal{A}_{d+1}$ contains a member of the family, and $\mathcal{A}_{d+1}\subsetneq\mathcal{A}_d$, for every integer $d\ge 1$.*

So no member is admitted in every dimension. Figure 2 shows the member $w=4$. Here $w_3<4\le w_2$, so $F^{(4)}$ is admitted on the line and in the plane and not in space, as the cone paper proves ([16, Lem. 4.12(3)]). dimensions 1 and 2, and in dimension 3 it rises on an interval, so it is not the polar profile of a datum there. The critical values are not given in closed form: M_d is the maximum of a function whose stationary point solves a transcendental equation. Those of the figure are computed numerically from (3.10), and the first three are the values that the cone paper reports, 12.61, 5.24 and 3.21 ([16, Rem. 4.10], where w is written c). Clause (3) gives the order of the decrease, $d^{-1/2}$, and not the critical values in the dimensions of imaging: w_2 and w_3 are about 4.2 and 3.1 times $\sqrt{\pi/d}$. There the computed values are the guide. Two bounds hold in closed form (Lemma S.11 of the supplement): with $m=(d+1)/2$,

$$\frac{e^{-1/2}}{2\sqrt{m}} \;\le\; w_d \quad (d\ge 1), \qquad w_d \;\le\; \frac{\sqrt{\pi/2}+\sqrt{\pi}+4}{2\,(1-\varepsilon_d)\sqrt{m}} \quad (d\ge 3), \qquad \varepsilon_d:=2^{-d/2}e^{m/2}.$$

In this family each dimension tolerates a lower bump.

As laws the members of the family are ordinary kernels in every dimension. For every $w>0$ and every D the function $e^{-F^{(w)}(|\xi|)}$ is the transform of a Gaussian variance mixture on $\mathbb{R}^D$, a rotation-invariant infinitely divisible law (Proposition A.8(2)). In the dimensions that do not admit $F^{(w)}$ that law lacks self-decomposability alone, the property that gives a cascade its stages (its clause (3)). So the family separates a kernel in every dimension from a cascade in every dimension. Its delay profile is not monotone, and its delay exponent $F_{\mathrm{I}}^{(w)}$ is not causally admissible: otherwise $F^{(w)}$ would be a member of $\mathcal{S}$, and so of every $\mathcal{A}_D$ (Proposition 2.6(2)).

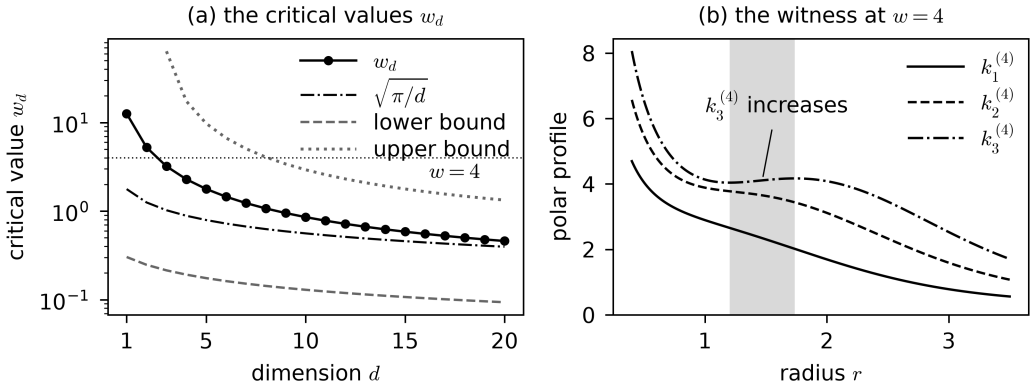


Figure 2: (a) The critical values w_d of Definition 3.24 for $d = 1, \dots, 20$, computed numerically, on a logarithmic scale, with the lower and the upper bound displayed in the text, the upper one for $d \geq 3$, and the asymptote $\sqrt{\pi/d}$ of Theorem 3.25(3). The horizontal line $w = 4$ crosses the sequence between $d = 2$ and $d = 3$. (b) The polar profiles $k_d^{(4)}$ of $F^{(4)}$ (Definition 3.24) in dimensions 1, 2 and 3: nonincreasing in dimensions 1 and 2, increasing on the marked interval, about $(1.20, 1.74)$, in dimension 3. Computed by a script deposited with the paper.

Remark 3.26 (relation to the line). At $d = 1$ the definitions, the classification and the threshold of this section are those of the line. $\mathrm{O}(1) = \{\pm 1\}$, so Definition 3.1 is Definition 2.1, Corollary 3.7 is Theorem 2.3, and $\mathcal{A}_1$ is the admissible cone. The polar profile k_1 is the folded profile of (2.1), so $k_1(0+)$ is the constant c of Corollary 2.8, whose two regimes and threshold are Theorem 3.13(2), (4) and (5). What of the line theory is not carried to $\mathbb{R}^d$ is said in §4.

4 Conclusion

The hemigroup axioms define, in every dimension, the scale spaces whose kernels from scale zero are the dilations of a rotation-invariant self-decomposable law, and every such law other than the point mass at the origin is such a kernel (Corollary 3.9). §1 says what follows for the kernel at its centre and for the dimension.

The paper has four limitations. It gives no generator of the scale evolution: the evolution equation that the cone paper has on the line [16, Prop. 6.2] is not carried to $\mathbb{R}^d$. It proves nothing in $\mathbb{R}^d$ on locality or on the extrema of the smoothed signal. Its statements on the centre are about the kernels from scale zero: of a stage $\mu_{s,t}$ with $s > 0$ and no Gaussian part it gives the point mass and its size (Proposition 3.20), and it does not examine the smoothing part at its centre. And the dictionary covers the exponents of logarithmic growth only: for the kernels with a Gaussian part or with $c = \infty$ the paper matches no law on $\mathbb{R}^d$ with the exponent.

4.1 Open problems

The derivatives. Let $d + N - 1 < c < d + N$ with an integer $N \geq 1$. The density then has $N - 1$ continuous derivatives (Theorem 3.13(2)), and the theorem says nothing of the next one. Is the radial derivative of order N of the density asymptotic at the centre to a constant times $|x|^{c-d-N}$ times the slowly varying factor K of Proposition 3.18, as the density itself is for $c < d$? And what takes the place of the logarithm at $c = d + N$? On the line both are answered by Sato and Yamazato, for every finite c ([48], Thms. 1.6 and 1.7).

The stages. Let the kernels have no Gaussian part. Is the absolutely continuous part of a stage $\mu_{s,t}$ with $s > 0$ bounded near the centre when c is finite (Proposition 3.20(2)), and what number decides it? And is the density of a stage bounded when $c = \infty$?

A Notation, sources, and the proofs given in print

This appendix has the notation (Table 2), the sources of the statements (§A.1), the proofs of the classification (§A.2), five statements that the main text names and does not state (§A.3), and the proofs of the statements that are proved in print only (§A.4). The proofs that Lean checked are in the supplement, except those of the classification and that of Proposition 3.22, which is in §A.4.

symbol	role
ξ, ω	frequency on $\mathbb{R}^d$, on the line
$s \le t$, χ, S_λ	scales; a gauge; the action of a dilation on the scales
λ, ρ_λ; λ_d	a dilation factor; for $\lambda \in (0,1)$ in (2.2), the contraction ratio, and ρ_λ the residual; the number $2^{-d/2}$ in Lemma S.10, and the mean of $\log\lvert\langle e,\eta\rangle\rvert$ over the sphere in Proposition 3.16 (in C2, λ is a measure on the sphere)
$\Phi_{s,t}$; T_a, R_Q, D_λ	the operators of a family; on $L^1(\mathbb{R}^d)$, the translation by a, the action of $Q \in \mathrm{O}(d)$ and the dilation by λ (Definition 3.1; on the line τ_a, R and D_λ)
$\mu_{s,t}$; $\nu_{s,t}$, $\kappa_{s,t}$	the kernels of a family on $\mathbb{R}^d$; kernels on the line
x; $\mu^{\mathrm{I}}_{x,y}$	a point of $\mathbb{R}^d$ or of $\mathbb{R}$; in $\mu^{\mathrm{I}}_{x,y}$, $x \le y$ are scales of a time-causal family
u_K, Λ	in C5, the uniform law on a convex set K, and the mixing measure
F, F_d, F_{I}	an exponent on the line (2.1); the exponent of an isotropic origin datum on $\mathbb{R}^d$; a causally admissible exponent
a_{V}, a, b_0	the Gaussian coefficient on the line; that of an isotropic origin datum, $a_{\mathrm{V}} = \frac{1}{2}a$ (in Lemma A.2(2), a dilation factor); the drift of a delay datum
k, k_d, k_{I}	the folded profile; the polar profile, the radial Lévy density times $\lvert S^{d-1}\rvert\, r^d$; the delay profile
c	the profile at the origin, $k_d(0+)$, and $k(0+)$ on the line; in the uniqueness clause of Theorem 3.6, the ratio of two gauges
ν_d, f_d; p, q	the Lévy measure of an isotropic origin datum, and its radial density; the density of a kernel and its radial form, $p(x) = q(\lvert x\rvert)$
K, C_k, $\varkappa_d$	the slowly varying factor of the rate, the integral of the polar profile beyond radius 1, and a constant of the dimension (Proposition 3.16); K_ν is a Bessel function, and in Lemma S.11 K is a number
A, Q, B; $R(d,c)$	the constants of Proposition 3.18, and that of Lemma S.9. Elsewhere A_β is the constant of Proposition A.4, Q an orthogonal matrix (Definition 3.1, Proposition 3.21), B a Borel set (Theorem 3.13(4) and C2), R_Q the action of Q, and R a random radius (Proposition 3.21)
L; $\mathcal{L}$	the slowly varying factor of a transform, $\widehat{\mu} = \lvert\xi\rvert^{-c} L(\lvert\xi\rvert)$; the function $\int_s^1 K(y)\,dy/y$ of Proposition 3.18(3)
e	a unit vector of $\mathbb{R}^d$, as in $\widehat{\mu}(e/s)$; in e^{-C_k} and the like, the base of the exponential
Ω_d, ϕ, X	in §3.4, the characteristic function of the uniform law on the unit sphere along a ray; that of a rotation-invariant random vector X (in §2.4 ϕ_1 is a density, and in Proposition 3.16 X is a real variable)
$\mathcal{A}_d$, $\mathcal{A}_\infty$, $\mathcal{S}$	the classes of Proposition 2.6; their intersection; the subordinated slice
b Cin$(\tau\lvert\omega\rvert)$; τ_a	a member of a Cin ray, of folded profile $b\,\mathbf{1}_{(0,\tau)}$; the translation by a
γ, θ; α; ϑ	the Matérn index and range; the stable kernels, of index 2α; the shape of the Student-t kernels
W, Z; T_1	a random delay and a standard normal vector; the scaled inverse-gamma delay
w, w_d, M_d, m	the parameter of the separating family, its critical values, $m = (d+1)/2$ (Definition 3.24)
τ	the cut-off of a Cin ray and of a step datum; in §A.1, the time of a Lévy process, the kernel being its law at $\tau = 1$; a constant of Proposition A.3
s	in Proposition 3.18 and in Lemma S.9 of the supplement, a distance from the centre of the kernel, not a scale

g_t, $g_u^{(d)}$	the heat kernel at variance t (here t is not a scale); the centred Gaussian law of covariance uI

Table 2: Notation of §3 and Appendix A.

A.1 Sources

Table 3 says where the main statements of §3 are in print, statement by statement, with the case that each source covers. Several of the sources are written for a Lévy process (§2.3): with a law of exponent F_d they study the laws with the transforms $e^{-\tau F_d}$, $\tau > 0$, the laws of the process at the times τ. The time is not the scale of this paper. The time multiplies the exponent, and the scale rescales the frequency. A kernel is the law at time 1, and the law at time τ is a kernel with τc in place of c.

statement	where it is in print, and for which case
Thm. 3.6 and Cor. 3.9, the classification	Read on laws: a process with independent increments on $\mathbb{R}^d$ that is self-similar up to a change of scale and a deterministic shift has self-decomposable laws, and every self-decomposable law but a point mass is the law at time 1 of one ([47], Thm. 16.1, p. 99, due to Sato [46]). On groups, a law is self-decomposable exactly when it is the limit of the kernels $\mu_{0,t}$ of a continuous hemigroup that a contracting one-parameter group of automorphisms maps into itself ([27], Thm. 2.14.21, p. 383).
Thm. 3.13, the threshold	On the line: §2.4. For the laws of a Lévy process on $\mathbb{R}^d$: bounded when $c > d$, unbounded when $c \leq d/2$ ([33], p. 126, and the proof of Lemma 2.6, p. 132). The threshold with its boundary case $c = d$, for two families: the Matérn kernels (the sources quoted in Example 3.19, where two of the kernels above the threshold are given with their derivatives) and the laws with transform $(1 + \lvert\xi\rvert^{\alpha})^{-\tau}$ ([24], Example 3).
Cor. 3.14, c read off the exponent	Clause (1), for every infinitely divisible law: [47], proof of Thm. 8.1(ii), p. 40. Clause (2) for a finite c: on the line, an asymptotic form of the transform with its constant ([48], Lemma 5.1, p. 298); in $\mathbb{R}^d$ it follows from an equivalence for the Lévy processes whose laws are rotation invariant with radially nonincreasing densities ([24], Thm. 3.5) by de Haan's theorem on functions whose increments are governed by a slowly varying function ([3], Thm. 3.7.3, pp. 162–163).
Prop. 3.15, the shape	For the law at each time of a symmetric Lévy process whose Gaussian covariance is a multiple of the identity and whose Lévy measure has a radially nonincreasing density: [58], Proposition, p. 488. That a self-decomposable law on $\mathbb{R}^d$ which is not degenerate is absolutely continuous: [47], Thm. 27.13, p. 181, proved by Sato in 1982 (p. 196).
Lemma S.9 of the supplement, the rate through the transform: clause (1) for $c < d$, quoted before Proposition 3.18, and clause (2) for $c = d$	Clause (1), for a rotation-invariant probability law on $\mathbb{R}^d$, $d \geq 1$, whose transform is $\lvert\xi\rvert^{-c}$ times a slowly varying factor, $0 < c < d$: the mass of a ball ([36], Thm. 1.4.3, p. 65, where the law is the spectral measure of a random field and the transform its covariance function), and the density when it is a nonincreasing function of the radius (the same book, Thm. 1.4.4, pp. 66–67). The book gives the two theorems for a transform asymptotic to a power to Leonenko and Olenko, and their proofs with a slowly varying factor to Olenko (p. 351). For a power the mass of a ball is [37], Corollary, p. 1542, with a constant that is not given in closed form. The survey [38] collects both theorems, as its Thms. 3 and 4, with related ones.
Lemma S.9, through Hankel transforms	For $c > (d-3)/2$ the mass of a ball in clause (1) also follows from the Abel–Tauber theorem for Hankel transforms, in the direction from a function that is ultimately decreasing to zero to its transform. For a bounded function, which is the case $d \geq 3$: [53], Thm. 1, p. 169, with the remark on p. 171, and, as an equivalence for a function decreasing on the half-line, [54], Thm. 17(a), p. 613, with pp. 617–618 for the Hankel kernel. With local integrability in place of boundedness, and so for every d: [4], Thm. 1.2. The mass of a ball in clause (2) follows from the boundary case of that theorem ([53], Thm. 2, p. 169, and [54], Thm. 18(a), pp. 613–614, for a bounded function; [4], Thm. 1.1 with (1.4)).

statement	where it is in print, and for which case
Lemma S.9, on the line, and the method of its proof	On the line: the theorem of Sato and Yamazato quoted in §2.4. The method of proof, an average of the transform against a positive kernel read as a Laplace transform of the mass of balls: with the Poisson kernel, for the spectral measure of a random field, [36], Lemma 1.4.11, p. 72, and the proof on p. 104; with the Gaussian kernel, in $\mathbb{R}^d$, for the measure of the expected time that a Lévy process spends in a set, [11], Thm. 8 with its proof, and Thm. 9. The constant of the mass of a ball in clause (1), $R(d,c)\,|S^{d-1}|/c$, is the constant of these two theorems at the index c.
Prop. 3.18, the rate through the profile	On the line: the same theorem of Sato and Yamazato, with their lemma cited at Corollary 3.14 above. In $\mathbb{R}^d$, for the Lévy processes whose laws have radially nonincreasing densities and as the time τ tends to 0: the density at x is asymptotic to τ times the Lévy density at x times the transform of the law at the frequencies of length $1/|x|$ ([24], Thm. 3.9). The constant $R(d,\tau c)$ of the proposition for the law at time τ agrees with it in that limit.
Props. A.3 and A.4, the two bounds; Lemma A.2	For the Lévy processes whose laws have radially nonincreasing densities, a lower bound and the comparison of a radially nonincreasing density with its average, on which the upper bound rests ([24], Prop. 5.3 and (5.4)). The lemma on the line: [57], Prop. 1, p. 58, and the same lemma of Sato and Yamazato. The two-sided estimates of [5] for such processes assume an exponent whose growth lies between two powers of positive order ((19), p. 11, of the arXiv version), which a finite c excludes.
Prop. 3.20, the stages	On the line, for the Matérn kernel of index 1: the stage between two scales of ratio r as a mixture in which no displacement has the weight r^2 ([16, Prop. 7.2], with (7.2)).
Prop. 3.21, the dictionary	Clauses (1) and (2) are a case of the representation of a self-decomposable law as the limit law of the integrals $\int_0^t e^{-s}\,dY(s)$ of a Lévy process Y ([31], Thms. 3.2 and 4.1, pp. 252 and 254; [50], Thm. 4.1, p. 86): the case in which Y moves by isolated jumps, at the rate c and with the law of X. The representation is also in [49], Thms. 5.1–5.3, pp. 546–547 (the law of a stationary process of Ornstein–Uhlenbeck type), in [28], Thms. 2.1 and 3.1, pp. 388 and 390 (operator-self-decomposable laws on a Banach space), and, on the line, in [62], Thm. 1, p. 306, and Cor. 3, p. 307 (the limit in law of $e^{-\gamma t}\int_0^t e^{\gamma u}\,dB(u)$, $\gamma>0$). In radial form: [29], Cor. 1.7, pp. 1056–1057, restated as [30], Prop. 3.9.6, pp. 180–182. The classes of the radial characteristic functions on $\mathbb{R}^m$ decrease in m, with the mixtures of Gaussians as their intersection ([52], §2.1 and Thm. 2, pp. 816–817), and strictly ([23], p. 94).
Prop. 3.22 and Cor. A.7, the test on the Lévy measure	For probability laws, the radius of a projection of a rotation-invariant law ([10], Lemma 2, p. 371, and Cor. 2, p. 372) and the density of the projection ([18], Thm. 2.10, p. 36). See also [12]: Prop. 11 with (3.8), p. 1821, for the radius law of a radial characteristic function in a lower dimension, the law of its radius times an independent variable with values in $(0,1)$. The formula of Lemma A.5 is (3.8) with the Lévy measure in place of the law. And Lemma 12, p. 1822: Ω_d is a characteristic function on $\mathbb{R}^d$ and not on $\mathbb{R}^{d+1}$, which is the last assertion of Corollary A.7(2). The representation of a self-decomposable law of which the Cin rays are the elementary terms on the line: [31], Thm. 7.2, p. 259.
Thm. 3.25, the separating family	The family, the members $w=8$ and $w=4$ as witnesses in dimensions 1 and 2, and the first critical values, computed numerically: [16, Lem. 4.12 and Rem. 4.10].

Table 3: Where the main statements of §3 are in print, and for which case. Three sources are cited in the numbering of the version that was read, which is not the journal's: the Hokkaido preprint for [4], and the arXiv versions for [11] and [38]. The numbers of [24] are those of the published article, and the pages of [37] those of the English translation.

A.2 The proofs of the classification

The necessity half of Theorem 3.6 is proved in §3.2. This subsection has the other proofs of the classification: those of Lemma 3.4, of the sufficiency and uniqueness halves of the theorem, of Corollary 3.7, of Lemma 3.8, of Proposition 3.11 and of Corollary 3.9. The proofs of Lemmas 3.3 and 3.5, which carry arguments of the line paper from the line to $\mathbb{R}^d$, are in the supplement.

Proof of Lemma 3.4. (1). $\pi_*\mu$ is a probability measure because π is measurable, and for $\omega \in \mathbb{R}$

$$\widehat{\kappa}(\omega) = \int_{\mathbb{R}} e^{i\omega y}\, \kappa(dy) = \int e^{i\omega\langle e,x\rangle}\, \mu(dx) = \widehat{\mu}(\omega e).$$

A rotation-invariant law has a radial transform: if $|\eta| = |\eta'|$, let Q be the identity when $\eta = \eta'$ and otherwise the reflection in the hyperplane orthogonal to $\eta - \eta'$, an element of $\mathrm{O}(d)$ with $Q^{\mathsf{T}} = Q$ and $Q\eta = \eta'$; then $\widehat{\mu}(\eta) = \widehat{Q_*\mu}(\eta) = \widehat{\mu}(Q^{\mathsf{T}}\eta) = \widehat{\mu}(\eta')$. Since $|\,|\xi|e\,| = |\xi|$, this gives $\widehat{\mu}(\xi) = \widehat{\mu}(|\xi|e) = \widehat{\kappa}(|\xi|)$ for every $\xi \in \mathbb{R}^d$. Since $-I \in \mathrm{O}(d)$ and $\pi(-x) = -\pi(x)$, the image of κ under $\omega \mapsto -\omega$ is $\pi_*((-I)_*\mu) = \kappa$, so κ is symmetric and $\widehat{\kappa}$ is real; hence so is $\widehat{\mu}$. For another unit vector e' with marginal κ', $|\omega e'| = |\omega e|$ gives $\widehat{\kappa'}(\omega) = \widehat{\mu}(\omega e') = \widehat{\mu}(\omega e) = \widehat{\kappa}(\omega)$ for every $\omega \in \mathbb{R}$, so $\kappa' = \kappa$ by uniqueness of the transform on $\mathbb{R}$.

(2). If $\mu = \delta_0$ then $\pi_*\mu = \delta_0$. Conversely, if $\kappa = \delta_0$ then $\widehat{\kappa} \equiv 1$, so $\widehat{\mu} \equiv 1 = \widehat{\delta_0}$ by (1), and $\mu = \delta_0$ by uniqueness of the transform on $\mathbb{R}^d$.

(3). $\mu * \mu'$ is the image of $\mu \otimes \mu'$ under $(x, y) \mapsto x + y$, and $\pi(x + y) = \pi(x) + \pi(y)$, so $\pi_*(\mu * \mu')$ is the image of $\pi_*\mu \otimes \pi_*\mu'$ under addition on $\mathbb{R}$, which is $\pi_*\mu * \pi_*\mu'$. The other two identities are $\pi(\lambda x) = \lambda\pi(x)$ and $\pi(0) = 0$. □

Proof of Theorem 3.6, sufficiency and uniqueness. Sufficiency. Let χ be a gauge and $F \in \mathcal{A}_d$ with $F \not\equiv 0$, and let $(\mu_{s,t})$ be a family of probability measures on $\mathbb{R}^d$ satisfying (3.1). First, F is continuous on $[0, \infty)$ with $F(0) = 0$: on $0 \le r \le \Omega$ the integrand of (2.1) is at most $\max\{2, \Omega^2/2\}\,(1 \wedge x^2)\,k(x)/x$, which is integrable by the two conditions on k, so dominated convergence applies. The axioms are read off the transform, a probability measure on $\mathbb{R}^d$ being determined by its transform. (I1): the exponents of $\mu_{s,t}$ and $\mu_{r,s}$ add up to that of $\mu_{r,t}$, and the exponent of $\mu_{t,t}$ is 0. (I2): F is continuous, and so is χ, an increasing bijection of $[0, \infty)$. (I3): $\widehat{Q_*\mu_{s,t}}(\xi) = \widehat{\mu}_{s,t}(Q^{\mathsf{T}}\xi)$ and $|Q^{\mathsf{T}}\xi| = |\xi|$. (I4): with $S_\lambda := \chi^{-1}(\lambda\,\chi(\cdot))$, an increasing bijection of $[0, \infty)$, $\widehat{(\lambda I)_*\mu_{s,t}}(\xi) = \widehat{\mu}_{s,t}(\lambda\xi) = \widehat{\mu}_{S_\lambda s, S_\lambda t}(\xi)$. (ND): let $s < t$ and suppose $\mu_{s,t} = \delta_0$, so that $F(\chi(t)r) = F(\chi(s)r)$ for every $r \ge 0$. If $s = 0$ this says $F(\chi(t)r) = 0$ for every r, and $\chi(t) > 0$ gives $F \equiv 0$. If $s > 0$ it says $F(r) = F(\varkappa r)$ for every r, with $\varkappa := \chi(s)/\chi(t) \in (0, 1)$. Iterating, $F(r) = F(\varkappa^n r) \to F(0) = 0$, so again $F \equiv 0$ (§2.5). Both contradict $F \not\equiv 0$.

Uniqueness. Let (χ', F') also satisfy (3.1). Both exponents are real, so taking logarithms of the kernels from scale zero, $F(\chi(t)r) = F'(\chi'(t)r)$ for all $t, r \ge 0$, since $\chi(0) = \chi'(0) = 0$ and $F(0) = F'(0) = 0$. At $t' := \chi'^{-1}(1)$ this reads $F'(r) = F(c^{-1}r)$ with $c^{-1} := \chi(t') > 0$. Hence $F(\chi(t)r) = F(c^{-1}\chi'(t)r)$ for all t and r. If $\chi(t) \ne c^{-1}\chi'(t)$ for some $t > 0$, then $F(r) = F(\varkappa r)$ for every r with $\varkappa := c^{-1}\chi'(t)/\chi(t) \ne 1$, and $F \equiv 0$ as in the previous paragraph (with $\varkappa^{-1}$ in place of $\varkappa$ if $\varkappa > 1$), a contradiction. So $\chi' = c\,\chi$. When $\chi(1) = \chi'(1) = 1$ this gives $c = 1$. □

Proof of Corollary 3.7. Necessity. Let $(\Phi_{s,t})$ be an isotropic cascade measurement family in dimension d. By Lemma 3.3(1) each $\Phi_{s,t}$ is convolution by a probability measure $\mu_{s,t}$ on $\mathbb{R}^d$, and by Lemma 3.3(3) the family $(\mu_{s,t})$ is an isotropic cascade family in dimension d. The necessity direction of Theorem 3.6 gives a gauge χ and an $F \in \mathcal{A}_d$ with $F \not\equiv 0$ satisfying (3.1).

Sufficiency. Let χ, F and $(\mu_{s,t})$ be as in the statement. By the sufficiency direction of Theorem 3.6 the family $(\mu_{s,t})$ is an isotropic cascade family in dimension d. By Lemma 3.3(4) the operators $f \mapsto \mu_{s,t} * f$, which are the $\Phi_{s,t}$, form an isotropic cascade measurement family in dimension d.

Uniqueness. Let χ', F' and $(\mu'_{s,t})$ be as in the last sentence of the statement. For $0 \le s \le t$ the probability measures $\mu_{s,t}$ and $\mu'_{s,t}$ give the same operator $\Phi_{s,t}$, so they are equal, by the uniqueness clause of Lemma 3.3(1). Hence (χ', F') satisfies (3.1) for the family $(\mu_{s,t})$, and the uniqueness clause of Theorem 3.6 gives c. □

Proof of Lemma 3.8. Let μ be a self-decomposable probability measure on $\mathbb{R}^d$ with $\widehat{\mu}(\xi) = e^{-F(|\xi|)}$, which the definition of $\mathcal{A}_d$ provides. For $c \in (0, 1)$ let ρ_c be a residual of μ. As

$\widehat{\mu}(c\,\xi) = e^{-F(c|\xi|)} \neq 0$, (2.2) forces $\widehat{\rho}_c(\xi) = e^{-(F(|\xi|)-F(c|\xi|))}$; put $\rho_0 := \mu$ and $\rho_1 := \delta_0$, so that this formula holds for every $c \in [0,1]$, as $F(0) = 0$. For $0 \le s \le t$ put

$$\mu_{s,t} := \big(\chi(t)I\big)_*\,\rho_{\chi(s)/\chi(t)},$$

with $\chi(s)/\chi(t) := 0$ when $\chi(t) = 0$, that is, when $s = t = 0$; then $\mu_{0,0}$ is the image of μ under the zero map, δ_0. Since $0 \le \chi(s) \le \chi(t)$, the index lies in $[0,1]$, and

$$\widehat{\mu}_{s,t}(\xi) = \widehat{\rho}_{\chi(s)/\chi(t)}\big(\chi(t)\xi\big) = \exp\big[-\big(F(\chi(t)|\xi|) - F(\chi(s)|\xi|)\big)\big],$$

because $\chi(s)/\chi(t)\cdot\chi(t)|\xi| = \chi(s)|\xi|$, both sides being 0 when $\chi(t) = 0$. Each $\mu_{s,t}$ is a probability measure, the image of one, so by the sufficiency direction of Theorem 3.6 $(\mu_{s,t})$ is an isotropic cascade family, and it satisfies (3.1) with (χ, F) by construction. □

The proofs of this paper integrate against the Lévy measure of an isotropic origin datum in polar coordinates. The identity (3.2) of Definition 3.10 is stated once, as a lemma.

Lemma A.1 (the polar decomposition of an isotropic origin datum's Lévy measure)**.** *Let $d \ge 1$ and let (a, k_d) be an isotropic origin datum in dimension d with Lévy measure ν_d. Then* (3.2) *holds for every nonnegative Borel g on $\mathbb{R}^d$ vanishing at the origin, σ_{d-1} being the surface measure of S^{d-1} and $|S^{d-1}|$ its surface area.*

Proof. By definition ν_d has the density $f_d(|x|) = k_d(|x|)(|S^{d-1}||x|^d)^{-1}$ on $\mathbb{R}^d \setminus \{0\}$. In polar coordinates $x = r\eta$, $dx = r^{d-1}dr\,\sigma_{d-1}(d\eta)$, so for $g \ge 0$ Borel and vanishing at the origin $\int g\,d\nu_d = \int_{S^{d-1}} \sigma_{d-1}(d\eta) \int_0^\infty g(r\eta)\,k_d(r)(|S^{d-1}|r^d)^{-1}r^{d-1}\,dr$, which is (3.2). □

Proof of Proposition 3.11. (1), forward. Let μ be self-decomposable on $\mathbb{R}^d$ with $\widehat{\mu}(\xi) = e^{-F(|\xi|)}$. By C4 it is infinitely divisible, and by C3(i) it has a generating triplet (A, ν, γ). For $Q \in \mathrm{O}(d)$, $\widehat{Q_*\mu}(\xi) = \widehat{\mu}(Q^{\mathsf{T}}\xi) = \widehat{\mu}(\xi)$, so $Q_*\mu = \mu$. The unit ball is Q-invariant, so substituting $x = Q^{\mathsf{T}}y$ in the representation shows that $Q_*\mu$ has triplet $(QAQ^{\mathsf{T}}, Q_*\nu, Q\gamma)$. By uniqueness, C3(ii), $QAQ^{\mathsf{T}} = A$ and $Q_*\nu = \nu$ for every Q. For unit vectors e, e' choose Q with $Qe = e'$. Then $\langle e', Ae'\rangle = \langle e, Q^{\mathsf{T}}AQ\,e\rangle = \langle e, Ae\rangle$, so $\langle \xi, A\xi\rangle = a|\xi|^2$ with $a := \langle e, Ae\rangle \ge 0$. The real part of $e^{i\theta} - 1 - i\theta\mathbf{1}_D$ is $\cos\theta - 1$, so taking moduli in the representation,

$$F(|\xi|) = -\log\big|\widehat{\mu}(\xi)\big| = \tfrac{1}{2}a|\xi|^2 + \int_{\mathbb{R}^d}\big(1 - \cos\langle\xi, x\rangle\big)\,\nu(dx).$$

Since μ is self-decomposable, C2 gives $\nu(B) = \int_S \lambda(d\eta)\int_0^\infty \mathbf{1}_B(r\eta)\,k_\eta(r)\,dr/r$ with λ a finite measure on $S = S^{d-1}$ and $k_\eta(r)$ measurable in η and nonincreasing in r. A nonincreasing k_η differs from its right limit $k_\eta(r+) = \sup\{k_\eta(q) : q \in \mathbb{Q},\ q > r\}$ at countably many r only, a Lebesgue-null set. Replacing $k_\eta(r)$ by $k_\eta(r+)$ therefore changes no integral in the display, and makes the kernel jointly measurable in (η, r), as a countable supremum of measurable functions. Taking $B = \{x : |x| \in C\}$ for Borel $C \subset (0,\infty)$ and exchanging the integrals, the image of ν under $x \mapsto |x|$ is

$$\nu\big(\{x : |x| \in C\}\big) = \int_C \bar{k}(r)\,\frac{dr}{r}, \qquad \bar{k}(r) := \int_S k_\eta(r+)\,\lambda(d\eta),$$

and $\bar{k}$ is nonincreasing with values in $[0,\infty]$. Integrating $|x|^2 \wedge 1$ against ν through this image gives $\int_0^\infty (r^2\wedge 1)\bar{k}(r)\,dr/r = \int(|x|^2\wedge 1)\,\nu(dx) < \infty$, which contains the pair of conditions of Definition 3.10. They force $\bar{k} < \infty$ on $(0,\infty)$, because a nonincreasing function infinite at some r_0 is infinite on $(0, r_0]$. So $(a, \bar{k})$ is an isotropic origin datum. Let ν_d be its Lévy measure, the right side of (3.2). By Lemma A.1 the image of ν_d under $x \mapsto |x|$ is $\bar{k}(r)\,dr/r$ on $(0,\infty)$, the same as that of ν.

It remains to see that $\nu_d = \nu$. Both are invariant under every $Q \in \mathrm{O}(d)$ (ν by the first paragraph, ν_d by the invariance of σ_{d-1}), both vanish at the origin, and both are finite on each set $U_\varepsilon = \{x : |x| > \varepsilon\}$, $\varepsilon > 0$. A finite measure m invariant under $\mathrm{O}(d)$ has a transform that depends only on its radial image: $\widehat{m}(Q\xi) = \widehat{m}(\xi)$ for every Q, a reflection carries ξ to $|\xi|\theta$ for any unit θ, and averaging over θ against $\sigma_{d-1}/|S^{d-1}|$ and exchanging the integrals gives

$$\widehat{m}(\xi) = \frac{1}{|S^{d-1}|} \int_{[0,\infty)} \Phi(|\xi|, r)\, (|\cdot|_* m)(dr), \qquad \Phi(s,r) := \int_S e^{isr\langle e_1, \theta\rangle}\, \sigma_{d-1}(d\theta),$$

the inner sphere integral $\int_S e^{is\langle x,\theta\rangle} \sigma_{d-1}(d\theta)$ depending on x only through $|x|$, again by the invariance of σ_{d-1}. Applied to the restrictions of ν and ν_d to U_ε, which are invariant and have the same radial image, this gives equal transforms, so the restrictions are equal. Letting $\varepsilon \downarrow 0$ along $1/(n+1)$ gives $\nu = \nu_d$ off the origin, hence everywhere. The display for F is therefore $F(|\xi|) = \frac{1}{2}a|\xi|^2 + \int (1 - \cos\langle \xi, x\rangle)\, \nu_d(dx)$, the exponent of the datum $(a, \bar{k})$.

(1), converse. Let (a, k_d) be a datum with $F_d = F(|\cdot|)$. Its Lévy measure ν_d satisfies $\int (|x|^2 \wedge 1)\, d\nu_d < \infty$ by the two conditions, so by C3(iii) there is an infinitely divisible μ with triplet $(aI, \nu_d, 0)$. As ν_d is symmetric, the odd part of the integrand integrates to 0 and $\widehat{\mu} = e^{-F_d}$: μ is the datum's law. Its Lévy measure has the form of C2 with $\lambda = \sigma_{d-1}$ and $k_\eta = k_d/|S^{d-1}|$ for every η, nonincreasing in r. So μ is self-decomposable by C2.

(1), uniqueness. Two data with the same exponent have the same law, hence by C3(ii) the same Gaussian matrix aI and the same Lévy measure. The images of the two Lévy measures under $x \mapsto |x|$ are then equal; by Lemma A.1 they are $k_d(r)\, dr/r$ and $k'_d(r)\, dr/r$ on $(0,\infty)$, so $k_d = k'_d$ almost everywhere.

(2). Let μ be the law of clause (1), e_1 the first basis vector and $\pi(x) := \langle e_1, x\rangle$. Then $\pi_*\mu$ is self-decomposable on $\mathbb{R}$. If $\rho_\varkappa$ is a residual of μ for the ratio $\varkappa \in (0,1)$, then $\pi_*\rho_\varkappa$ is one of $\pi_*\mu$:

$$\widehat{\pi_*\mu}(\omega) = \widehat{\mu}(\omega e_1) = \widehat{\mu}(\varkappa\omega e_1)\, \widehat{\rho}_\varkappa(\omega e_1) = \widehat{\pi_*\mu}(\varkappa\omega)\, \widehat{\pi_*\rho_\varkappa}(\omega).$$

Its exponent is $F(|\omega|)$. Clause (1) at $d = 1$ gives a datum (a', k_1) in dimension 1 with, as $|S^0| = 2$ and the integrand is even,

$$F(|\omega|) = \tfrac{1}{2}a'\omega^2 + \int_{\mathbb{R}} \big(1 - \cos\omega x\big) \frac{k_1(|x|)}{2|x|}\, dx = \tfrac{1}{2}a'\omega^2 + \int_0^\infty \big(1 - \cos\omega x\big)\, k_1(x)\, \frac{dx}{x},$$

which is (2.1) with $a_{\mathrm{V}} = \frac{1}{2}a'$ and $k = k_1$. The two integrability conditions are the same pair. With the definition of $\mathcal{A}_d$ in Proposition 2.6 this is the description of the class.

(3). By Theorem 3.6 and clause (2), the family is isotropic exactly when (3.1) holds for a gauge and an $F \not\equiv 0$ with $F(|\cdot|)$ the exponent F_d of a datum. The displayed $F(r)$ is $F_d(re_1)$, and F_d is radial. Finally $F \equiv 0$ exactly when the datum is trivial. If $a = 0$ and $k_d = 0$ almost everywhere then $F_d \equiv 0$. Conversely $F_d \equiv 0$ makes the law δ_0, whose triplet is $(0,0,0)$, so $a = 0$ and $k_d = 0$ almost everywhere by the uniqueness of clause (1). □

Proof of Corollary 3.9. A kernel from scale zero satisfies (S). Let $(\mu_{s,t})$ be an isotropic cascade family in dimension d and let $t > 0$. Then $\mu_{0,t}$ is rotation invariant by (I3), and $\mu_{0,t} \neq \delta_0$ by (ND). By Theorem 3.6 there are a gauge χ and an $F \in \mathcal{A}_d$ with (3.1), so that $\widehat{\mu}_{0,t}(\xi) = e^{-F(\chi(t)|\xi|)}$. By the definition of $\mathcal{A}_d$ (Proposition 2.6) there is a self-decomposable probability measure μ' on $\mathbb{R}^d$ with $\widehat{\mu'}(\xi) = e^{-F(|\xi|)}$. Put $\lambda := \chi(t) > 0$. Then $\widehat{(\lambda I)_*\mu'}(\xi) = \widehat{\mu'}(\lambda\xi) = e^{-F(\lambda|\xi|)} = \widehat{\mu}_{0,t}(\xi)$, so $\mu_{0,t} = (\lambda I)_*\mu'$, a probability measure on $\mathbb{R}^d$ being determined by its transform. The image of a self-decomposable law under λI is self-decomposable: if ρ_c is a residual of μ' for the ratio $c \in (0,1)$, then $\widehat{\mu'}(\lambda\xi) = \widehat{\mu'}(c\lambda\xi)\, \widehat{\rho}_c(\lambda\xi)$ is (2.2) for $(\lambda I)_*\mu'$ with the residual $(\lambda I)_*\rho_c$. So $\mu_{0,t}$ satisfies (S).

The exponent of a law with (S). Let μ satisfy (S). By Lemma 3.4(1) its transform is real and radial. It has no zero: μ is infinitely divisible by C4, so by C3(i) it has a generating

triplet, and its transform is the exponential of a complex-valued function. A real continuous function on $\mathbb{R}^d$ that has no zero and takes the value 1 at the origin is positive. Fix a unit vector e and put $F(r) := -\log \widehat{\mu}(re)$ for $r \geq 0$. Then $F(0) = 0$, $F \geq 0$ because $|\widehat{\mu}| \leq 1$, and $\widehat{\mu}(\xi) = \widehat{\mu}(|\xi|e) = e^{-F(|\xi|)}$ for every ξ, the transform being radial. So $\xi \mapsto F(|\xi|)$ is the exponent of the self-decomposable law μ, and F is admissible by Proposition 3.11(2). Hence $F \in \mathcal{A}_d$. If F vanished on $[0, \infty)$ the transform of μ would be 1 and μ would be δ_0, so $F \not\equiv 0$. A function F' with $\widehat{\mu}(\xi) = e^{-F'(|\xi|)}$ for every ξ has $F'(r) = -\log \widehat{\mu}(re) = F(r)$ for every $r \geq 0$.

Clause (3), the equivalence. Let μ satisfy (S), let F be its exponent, let $t_1 > 0$, and let $(\mu_{s,t})$ be a family of probability measures on $\mathbb{R}^d$. Suppose first that it satisfies (3.1) with F and a gauge χ with $\chi(t_1) = 1$. By the sufficiency direction of Theorem 3.6, F being in $\mathcal{A}_d$ and not identically 0, it is an isotropic cascade family, and $\widehat{\mu}_{0,t_1}(\xi) = e^{-F(\chi(t_1)|\xi|)} = e^{-F(|\xi|)} = \widehat{\mu}(\xi)$, because $\chi(0) = 0$ and $F(0) = 0$. So $\mu_{0,t_1} = \mu$. Conversely let $(\mu_{s,t})$ be an isotropic cascade family with $\mu_{0,t_1} = \mu$. By the necessity direction of Theorem 3.6 there are a gauge χ' and an $F' \in \mathcal{A}_d$ with (3.1). Put $c := \chi'(t_1) > 0$. The kernel from scale zero to scale t_1 gives $e^{-F'(c|\xi|)} = \widehat{\mu}(\xi) = e^{-F(|\xi|)}$, that is $F'(cr) = F(r)$ for every $r \geq 0$. Then $\chi := c^{-1}\chi'$ is a gauge with $\chi(t_1) = 1$, and $F'(\chi'(t)|\xi|) = F'(c\,\chi(t)|\xi|) = F(\chi(t)|\xi|)$ for every $t \geq 0$. So the family satisfies (3.1) with (χ, F).

Clause (3), the rest. Let two gauges χ and χ'' with $\chi(t_1) = \chi''(t_1) = 1$ both satisfy (3.1) with F for one family. By the uniqueness clause of Theorem 3.6, applied with $F' = F$, which vanishes at 0, there is $c > 0$ with $\chi'' = c\,\chi$, and at t_1 this reads $c = 1$. Let χ be any gauge with $\chi(t_1) = 1$. By Lemma 3.8 there is a family of probability measures that satisfies (3.1) with (χ, F), and by the equivalence it is an isotropic cascade family with $\mu_{0,t_1} = \mu$. It is the only one, because (3.1) gives the transform of every $\mu_{s,t}$. For $t > 0$, $\widehat{(\chi(t)I)_*\mu}(\xi) = \widehat{\mu}(\chi(t)\xi) = e^{-F(\chi(t)|\xi|)} = \widehat{\mu}_{0,t}(\xi)$, so $\mu_{0,t} = (\chi(t)I)_*\mu$.

Clause (1). One direction is the first paragraph. For the other let μ satisfy (S). Clause (3) with $t_1 = 1$ and the gauge $\chi = \mathrm{id}$ gives an isotropic cascade family with $\mu_{0,1} = \mu$.

Clause (2). Let $(\Phi_{s,t})$ be an isotropic cascade measurement family in dimension d and let $t > 0$ with $\Phi_{0,t} f = \mu * f$ for every f. By Lemma 3.3(1) each $\Phi_{s,t}$ is convolution by a probability measure $\mu_{s,t}$, which it determines, and by Lemma 3.3(3) the family $(\mu_{s,t})$ is an isotropic cascade family in dimension d. As $\mu_{0,t}$ and μ give the same operator, $\mu_{0,t} = \mu$, and (S) holds by clause (1). Conversely let μ satisfy (S). By clause (1) there are an isotropic cascade family $(\mu_{s,t})$ and a $t > 0$ with $\mu_{0,t} = \mu$. By Lemma 3.3(4) the operators $\Phi_{s,t} f := \mu_{s,t} * f$ form an isotropic cascade measurement family in dimension d, and $\Phi_{0,t} f = \mu * f$.

Clause (4). Let the two families be as in the clause and put $\mu := \mu_{0,t_1}$. By the first paragraph μ satisfies (S). Let F be its exponent, and let χ and χ' be the gauges of the two families by clause (3), so that $\chi(t_1) = \chi'(t_1) = 1$. Then $\theta := \chi^{-1} \circ \chi'$ is an increasing bijection of $[0, \infty)$ with $\theta(t_1) = t_1$ and $\chi \circ \theta = \chi'$. Let $0 \leq s \leq t$, so that $0 \leq \theta(s) \leq \theta(t)$. By (3.1) the transforms of $\mu'_{s,t}$ and of $\mu_{\theta(s),\theta(t)}$ are both $\exp[-(F(\chi'(t)|\xi|) - F(\chi'(s)|\xi|))]$, so $\mu'_{s,t} = \mu_{\theta(s),\theta(t)}$. □

A.3 Statements named in the main text

The five statements of this subsection are named in §3 or used in the proofs of §A.4. Lean checked them, and their proofs are in the supplement. The first bounds the transform of a kernel with a finite c from below.

Lemma A.2 (a profile bounded at the origin gives a logarithmic exponent)**.** *Let $d \geq 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0 and $c = k_d(0+) < \infty$, let F_d be its exponent, and put $C_k := \int_1^\infty k_d(r)\, dr/r$.*

1. (The exponent grows at most like $c \log|\xi|$.) *For every ξ with $|\xi| \geq 1$,*

$$F_d(\xi) \leq c \log|\xi| + \tfrac{13}{4}\, c + 2C_k, \qquad \text{so} \qquad e^{-F_d(\xi)} \geq e^{-13c/4 - 2C_k}\, |\xi|^{-c}.$$

2. (The exponent is logarithmic.) *For every $a > 0$, $F_d(a\xi) - F_d(\xi) \to c \log a$ as $|\xi| \to \infty$. Equivalently, for a unit vector e the function $g(u) := F_d(\sqrt{u}\, e)$, $u \geq 0$, satisfies $g(au) - g(u) \to \frac{c}{2} \log a$ as $u \to \infty$, for every $a > 0$.*

Clause (1) is the lower bound on the transform quoted in §3.3, and it is used in Theorem 3.13(4) and in Corollary 3.14(2). For a finite c, clause (2) says more than that corollary: $e^{-F_d(\xi)} = |\xi|^{-c} L(|\xi|)$ with L slowly varying, and Proposition 3.16 gives L through the polar profile.

Two bounds complement the rate of §3.3. Each is for one kernel, with its constants. The first bounds the spike from below, for $d \geq 2$ and a finite c, and is stated for the kernel as a measure before it is stated for its density.

Proposition A.3 (a pointwise lower bound at the origin)**.** *Let $d \geq 2$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0, let μ be its law, and suppose that $c := k_d(0+)$ is finite, $0 < c < \infty$. Then there are $\delta > 0$ and $c_1 > 0$ with*

$$\mu(E) \geq c_1 \int_E |x|^{c-d}\, dx \qquad \text{for every Borel set } E \subseteq \{|x| < \delta\}. \tag{A.1}$$

In particular:

1. *the density q of the absolutely continuous part of μ satisfies $q(x) \geq c_1 |x|^{c-d}$ for Lebesgue-almost every x with $|x| < \delta$; so does every density of μ when μ has one, and every representative of q is at least $c_1|x|^{c-d}$ almost everywhere near the origin, which tends to ∞ as $x \to 0$ when $c < d$;*
2. *$\mu(\{|x| < r\}) \geq c_1 |S^{d-1}|\, c^{-1}\, r^c$ for $0 < r \leq \delta$.*

The constants can be taken as follows. Let $\varepsilon_0 \in (0, 1]$ satisfy $k_d(\varepsilon_0) \geq c/2$, put $C_k := \int_1^\infty k_d(r)\, dr/r$, let V_d be the volume of the unit ball, and put

$$\tau := \min\Big\{\tfrac{1}{2},\ \big(\tfrac{3}{4d^2 c}\big)^{1/2}\Big\}, \quad \kappa_0 := \Big(\frac{c}{2|S^{d-1}|}\Big)^2 \frac{V_d}{8^d}, \quad \delta := \frac{\varepsilon_0}{2}, \quad c_1 := \frac{\kappa_0}{4}\, \tau^c e^{-C_k} \min\{2^{c-d}, 4^{c-d}\}.$$

For $c \leq d$ the power cannot be improved. A bound $c_1|x|^{\beta-d}$ with $\beta < c$ would give $\int g_t\, d\mu \geq C\, t^{-(d-\beta)/2}$, against the upper bound of Theorem 3.13(4) at a power between β and c. For $c > d$ the bound tends to 0 and says little, the density being continuous and positive at the centre. The proof uses $d \geq 2$ at one step, the choice of a nonzero vector orthogonal to a given point. On the line, where that step fails, the rate gives the bound near the centre for $c < 1$, without a radius.

The second proposition bounds the spike from above, when $c \leq d$, at every power below c. The upper bound of clause (4) of Theorem 3.13 on the blurred kernel gives a bound on the mass of a ball, and a radially nonincreasing density is at most its average over a ball.

Proposition A.4 (an upper bound at the origin, and the order of the density there)**.** *Let $d \geq 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0, let μ be its law, and suppose that $c := k_d(0+)$ satisfies $0 < c \leq d$. Let V_d be the volume of the unit ball of $\mathbb{R}^d$.*

1. (The mass of a ball.) *For every β with $0 < \beta < c$ there is $A_\beta < \infty$ with*

$$\mu(\{|x| < r\}) \leq A_\beta\, r^\beta \qquad \text{for every } r > 0.$$

One may take $A_\beta = (2\pi)^{d/2} e^{1/2}\, C'_\beta$, for every constant C'_β with which the upper bound of Theorem 3.13(4) holds.

2. (The density.) *The law μ has a density p with $p(x) = q(|x|)$ for $x \neq 0$, where $q : (0,\infty) \to [0,\infty)$ is nonincreasing. For every such q, every $0 < \beta < c$, and $A_\beta = (2\pi)^{d/2} e^{1/2}\, C'_\beta$ with C'_β as in clause (1),*

$$q(r) \le \frac{A_\beta}{V_d}\, r^{\beta - d} \qquad \text{for every } r > 0.$$

3. (The order at the origin, for $d \ge 2$.) *Let $d \ge 2$, let δ and c_1 be the constants displayed at the end of Proposition A.3, and let q be as in clause (2). Then $q(r) \ge c_1\, r^{c-d}$ for every r with $0 < r < \delta$, with no exceptional set. With $p(x) = q(|x|)$,*

$$\frac{\log p(x)}{\log(1/|x|)} \to d - c \quad (x \to 0), \qquad \frac{\log \mu(\{|x| < r\})}{\log r} \to c \quad (r \downarrow 0).$$

With the lower bound, clause (3) gives the logarithmic order of the density for $d \ge 2$. At $c = d$ its first limit is 0, and how the density grows there is clause (3) of Proposition 3.18. the supplement gives the constant A_β in closed form, after the proof.

Proposition 3.22 rests on the next lemma: the part of a rotation-invariant Lévy measure on $\mathbb{R}^{d+1}$ that a projection sees in $\mathbb{R}^d$, its *shadow*, has a polar profile in closed form, and that profile is continuous.

Lemma A.5 (the polar profile of a shadow)**.** *Let $d \ge 1$, let (a, k') be an isotropic origin datum in dimension $d+1$, with Lévy measure ν' and exponent F', and let $\Pi(x, y) := x$ be the projection of $\mathbb{R}^{d+1} = \mathbb{R}^d \times \mathbb{R}$ on the first d coordinates. Put*

$$C_d := \frac{|S^{d-1}|}{|S^d|}, \qquad \bar{k}(r) := C_d \int_{\mathbb{R}} k'\big(r\sqrt{1+u^2}\big)\,(1+u^2)^{-(d+1)/2}\, du \quad (r > 0).$$

1. *$C_d \int_{\mathbb{R}} (1+u^2)^{-(d+1)/2}\, du = 1$. So $\bar{k}(r)$ is a mean of the values of k' on $[r,\infty)$, and $0 \le \bar{k}(r) \le k'(r)$.*

2. *$\bar{k}$ is nonincreasing and continuous on $(0,\infty)$, and $\bar{k}(0+) = k'(0+)$ in $[0,\infty]$.*

3. *$(a, \bar{k})$ is an isotropic origin datum in dimension d. Its Lévy measure $\bar{\nu}$ is the image of ν' under Π, restricted to $\mathbb{R}^d \setminus \{0\}$: for every nonnegative Borel g on $\mathbb{R}^d$ vanishing at the origin, $\int g(\Pi z)\, \nu'(dz) = \int g\, d\bar{\nu}$. The restriction removes nothing: the image of ν' gives the origin of $\mathbb{R}^d$ the mass $\nu'(\{z \neq 0 : \Pi z = 0\}) = 0$. The exponent of $(a, \bar{k})$ is $\xi \mapsto F'\big((\xi, 0)\big)$, $\xi \in \mathbb{R}^d$.*

In particular, if $F'(\zeta) = F(|\zeta|)$ for a function F on $[0,\infty)$, then $(a, \bar{k})$ is an isotropic origin datum in dimension d with exponent $\xi \mapsto F(|\xi|)$.

Applied n times, the lemma gives the shadow in dimension d of a datum in dimension $d+n$. The last lemma says that the bridge formula and the polar profile (3.4) hold for a delay datum whose delay profile is not monotone, as that of the separating family is not.

Lemma A.6 (the polar profile of a delay datum, without monotonicity)**.** *Let $d \ge 1$ and let (b_0, k_{I}) be a delay datum in the sense of §2.3, with exponent F_{I}; its delay profile k_{I} is measurable and not necessarily monotone. Let k_d be its polar profile* (3.4)*. Then k_d is finite and measurable on $(0,\infty)$, with*

$$\int_0^1 k_d(r)\, r\, dr < \infty \qquad \text{and} \qquad \int_1^\infty k_d(r)\, \frac{dr}{r} < \infty,$$

and

$$F_{\mathrm{I}}\big(\tfrac{1}{2}|\xi|^2\big) = \tfrac{1}{2} b_0 |\xi|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle \xi, x\rangle\big)\, \frac{k_d(|x|)}{|S^{d-1}|\,|x|^d}\, dx, \qquad \xi \in \mathbb{R}^d.$$

In particular, if k_d is nonincreasing, then (b_0, k_d) is an isotropic origin datum in dimension d whose exponent is $\xi \mapsto F_{\mathrm{I}}(\frac{1}{2}|\xi|^2)$.

The last clause of the lemma is also what makes the datum of the Matérn kernel in Example 3.19 a subordinated one.

A.4 The statements proved in print

This subsection has the proofs of the statements that are not machine-checked (§1.1). Three of them are stated in the main text, Lemma 3.17 and Propositions 3.21 and 3.23, and two are stated here, Corollary A.7 and Proposition A.8. The proof of Proposition 3.22, which Lean checked, is given with them, because the corollary reads that proposition in the dictionary.

Proof of Lemma 3.17. Three facts about Γ. For $a > 0$, $\Gamma(a) = \int_0^\infty t^{a-1}e^{-t}\,dt$. On a compact interval $[a_0, a_1] \subset (0,\infty)$ the derivative of the integrand in a, $t^{a-1}e^{-t}\log t$, is bounded in absolute value by $(t^{a_0-1} + t^{a_1-1})\,e^{-t}\,|\log t|$, which is integrable on $(0,\infty)$. So one may differentiate under the integral sign:

$$\Gamma'(a) = \int_0^\infty t^{a-1}e^{-t}\log t\,dt,$$

the integral converging absolutely. Consequently, if a random variable T has the gamma law of shape a, with density $t^{a-1}e^{-t}/\Gamma(a)$ on $(0,\infty)$, then $\log T$ is integrable and $\mathbb{E}\log T = \psi(a)$. At $a = 1$ this reads $\gamma_{\mathrm{E}} = -\int_0^\infty e^{-v}\log v\,dv$. Next, $\Gamma(a+1) = a\,\Gamma(a)$ gives $\psi(a+1) = \psi(a) + 1/a$. Last, the duplication formula $\Gamma(a)\,\Gamma(a+\frac12) = 2^{1-2a}\sqrt{\pi}\,\Gamma(2a)$ gives, on taking logarithms and differentiating, $\psi(a) + \psi(a+\frac12) = 2\psi(2a) - 2\log 2$. At $a = \frac12$,

$$\psi(\tfrac12) = \psi(1) - 2\log 2 = -\gamma_{\mathrm{E}} - 2\log 2.$$

Clause (1). Let $G(X) = \mathrm{Cin}(X) - \log^+ X$ be as in the proof of Proposition 3.16, so that $-3 \le G \le \frac{13}{4}$ and $G(X) \to g_\infty$ as $X \to \infty$. For $0 < \varepsilon < 1$ put

$$J(\varepsilon) := \int_0^\infty \bigl(1 - \cos u\bigr)\,e^{-\varepsilon u}\,\frac{du}{u}.$$

(a) $(1-\cos u)/u = \int_0^1 \sin(tu)\,dt$, and $\int_0^\infty \sin(tu)\,e^{-\varepsilon u}\,du = t/(\varepsilon^2+t^2)$. The function $|\sin(tu)|\,e^{-\varepsilon u}$ is integrable on $(0,1)\times(0,\infty)$, so Fubini's theorem gives

$$J(\varepsilon) = \int_0^1 \frac{t\,dt}{\varepsilon^2 + t^2} = \tfrac12\log\bigl(1+\varepsilon^{-2}\bigr) = \log\frac{1}{\varepsilon} + \tfrac12\log\bigl(1+\varepsilon^2\bigr).$$

(b) $e^{-\varepsilon u} = \int_u^\infty \varepsilon e^{-\varepsilon x}\,dx$. The integrand $(1-\cos u)\,u^{-1}\,\varepsilon e^{-\varepsilon x}\,\mathbf{1}\{u < x\}$ is nonnegative, so Tonelli's theorem gives

$$J(\varepsilon) = \int_0^\infty \varepsilon e^{-\varepsilon x}\ \mathrm{Cin}(x)\,dx = \int_0^\infty e^{-v}\ \mathrm{Cin}(v/\varepsilon)\,dv.$$

(c) For $v > 0$,

$$\mathrm{Cin}(v/\varepsilon) - \log\frac{1}{\varepsilon} = G(v/\varepsilon) + \Bigl[\log^+\frac{v}{\varepsilon} - \log\frac{1}{\varepsilon}\Bigr].$$

The bracket equals $\log v$ when $v \ge \varepsilon$. When $v < \varepsilon$ it equals $\log\varepsilon$, which lies between $\log v$ and 0. So the bracket is bounded by $|\log v|$ in absolute value and tends to $\log v$ as $\varepsilon \downarrow 0$. As $|G| \le \frac{13}{4}$ and $G(v/\varepsilon) \to g_\infty$, dominated convergence with the integrable function $e^{-v}(\frac{13}{4} + |\log v|)$ gives, by (b) and $\int_0^\infty e^{-v}dv = 1$,

$$J(\varepsilon) - \log\frac{1}{\varepsilon} \to g_\infty + \int_0^\infty e^{-v}\log v\,dv = g_\infty - \gamma_{\mathrm{E}} \qquad (\varepsilon \downarrow 0).$$

(d) By (a) the left side is $\frac12\log(1+\varepsilon^2)$, which tends to 0. Hence $g_\infty = \gamma_{\mathrm{E}}$.

Clause (2). Let Z be a standard Gaussian vector of $\mathbb{R}^d$, with density $(2\pi)^{-d/2}e^{-|x|^2/2}$, and e a unit vector. In polar coordinates, for a Borel $h \ge 0$ on S^{d-1} and a Borel $f \ge 0$ on $(0,\infty)$,

$$\mathbb{E}\bigl[h(Z/|Z|)\,f(|Z|)\bigr] = \int_{S^{d-1}} h\,d\sigma_{d-1}\int_0^\infty f(r)\,(2\pi)^{-d/2}e^{-r^2/2}\,r^{d-1}\,dr.$$

With $h = 1$ and the substitution $t = r^2/2$, using $|S^{d-1}| = 2\pi^{d/2}/\Gamma(d/2)$, this shows that $|Z|^2/2$ has the gamma law of shape $d/2$. With $f = 1$ it shows that $Z/|Z|$ has the law $\sigma_{d-1}/|S^{d-1}|$. The coordinate $\langle e, Z\rangle$ is a standard Gaussian variable, so $\langle e, Z\rangle^2/2$ has the gamma law of shape $\frac{1}{2}$, by the case $d = 1$. By the first fact, $\log|Z|$ and $\log|\langle e, Z\rangle|$ are integrable, with

$$\mathbb{E}\log|Z| = \tfrac{1}{2}\big[\log 2 + \psi(\tfrac{d}{2})\big], \qquad \mathbb{E}\log|\langle e, Z\rangle| = \tfrac{1}{2}\big[\log 2 + \psi(\tfrac{1}{2})\big].$$

Since $\langle e, Z\rangle = |Z|\,\langle e, Z/|Z|\rangle$,

$$\lambda_d = \mathbb{E}\log\big|\langle e, Z/|Z|\rangle\big| = \mathbb{E}\log|\langle e, Z\rangle| - \mathbb{E}\log|Z| = \tfrac{1}{2}\big[\psi(\tfrac{1}{2}) - \psi(\tfrac{d}{2})\big].$$

Clause (3). The first expression is clauses (1) and (2). The second follows from $\psi(\frac{1}{2}) = -\gamma_{\mathrm{E}} - 2\log 2$. At $d = 1$ the first expression is γ_{E}. At $d = 2$ the second is $\frac{1}{2}(\gamma_{\mathrm{E}} - \psi(1)) - \log 2 = \gamma_{\mathrm{E}} - \log 2$. By $\psi(a+1) = \psi(a) + 1/a$ at $a = d/2$, $\varkappa_{d+2} = \varkappa_d - \frac{1}{2}\cdot\frac{2}{d} = \varkappa_d - 1/d$, and so $\varkappa_3 = \varkappa_1 - 1$.

Clause (4). By the second expression of clause (3), $-c\varkappa_d = c\log 2 - \frac{c}{2}(\gamma_{\mathrm{E}} - \psi(\frac{d}{2}))$, which is the first display. The second follows from $R(d,c) = \Gamma(\frac{d-c}{2})/(2^c\pi^{d/2}\Gamma(\frac{c}{2}))$. For the third take $c = d$ in the first display and use $|S^{d-1}| = 2\pi^{d/2}/\Gamma(\frac{d}{2})$: $(2\pi)^{-d}\cdot 2\pi^{d/2}\,\Gamma(\frac{d}{2})^{-1}\cdot 2^d = 2/(\pi^{d/2}\Gamma(\frac{d}{2}))$. □

Proof of Proposition 3.21. The function Ω_d. For unit vectors e, e' take $Q \in \mathrm{O}(d)$ with $Qe = e'$. Then $\langle e', \eta\rangle = \langle e, Q^{\mathsf{T}}\eta\rangle$, and σ_{d-1} is invariant under Q^{T}, so the integral defining $\Omega_d(t)$ is the same for e' as for e. At $d = 1$ the sphere is $\{\pm 1\}$ and $\Omega_1(t) = \cos t$. Ω_d is continuous, by dominated convergence, $|\Omega_d| \le 1$ and $\Omega_d(0) = 1$. For $x \in \mathbb{R}^d$ and $t \ge 0$,

$$\frac{1}{|S^{d-1}|}\int_{S^{d-1}} e^{it\langle x,\theta\rangle}\,\sigma_{d-1}(d\theta) = \Omega_d\big(t|x|\big). \tag{A.2}$$

Indeed the imaginary part vanishes by the symmetry $\theta \mapsto -\theta$ of σ_{d-1}. The real part is the definition of $\Omega_d(t|x|)$ with $e = x/|x|$ when $x \ne 0$, and both sides are 1 when $x = 0$.

The preamble. Let X be rotation invariant. As $-I \in \mathrm{O}(d)$, $-X$ has the law of X, so the characteristic function is real: $\mathbb{E}\,e^{i\langle\xi,X\rangle} = \mathbb{E}\cos\langle\xi, X\rangle$. Let $\xi \in \mathbb{R}^d$ and let θ be a unit vector. There is $Q \in \mathrm{O}(d)$ with $Q\xi = |\xi|\theta$, and $Q^{\mathsf{T}}X$ has the law of X, so

$$\mathbb{E}\cos\langle\xi, X\rangle = \mathbb{E}\cos\langle\xi, Q^{\mathsf{T}}X\rangle = \mathbb{E}\cos\big(|\xi|\langle\theta, X\rangle\big).$$

The left side does not depend on θ. Averaging over θ against $\sigma_{d-1}/|S^{d-1}|$, exchanging the two integrals of a bounded continuous function, and using (A.2) gives $\mathbb{E}\,e^{i\langle\xi,X\rangle} = \mathbb{E}\,\Omega_d(|\xi||X|) = \phi(|\xi|)$. By the uniqueness theorem for characteristic functions, ϕ determines the law of X. Let R have the law of $|X|$ and let U be uniform on S^{d-1}, that is of law $\sigma_{d-1}/|S^{d-1}|$, and independent of R. By (A.2) and Fubini's theorem, $\mathbb{E}\,e^{i\langle\xi,RU\rangle} = \mathbb{E}\,\Omega_d(R|\xi|) = \phi(|\xi|)$. So RU has the law of X, by the same theorem.

Step A: the identity. Let R be a random variable with values in $(0,\infty)$, let $0 < c < \infty$, and put $k(r) := c\,\Pr(R > r)$ for $r > 0$ and $\phi(v) := \mathbb{E}\,\Omega_d(vR)$ for $v \ge 0$. Then for every $\rho > 0$,

$$\int_0^\infty \big(1 - \Omega_d(\rho r)\big)\,k(r)\,\frac{dr}{r} = c\int_0^\rho \big(1 - \phi(v)\big)\,\frac{dv}{v}, \tag{A.3}$$

both sides in $[0,\infty]$. Indeed, as $1 - \Omega_d \ge 0$, Tonelli's theorem applies twice, with the substitution $r = vR/\rho$ in between:

$$\begin{aligned}\int_0^\infty \big(1 - \Omega_d(\rho r)\big)\,c\,\Pr(R > r)\,\frac{dr}{r} &= c\,\mathbb{E}\int_0^R \big(1 - \Omega_d(\rho r)\big)\frac{dr}{r} = c\,\mathbb{E}\int_0^\rho \big(1 - \Omega_d(vR)\big)\frac{dv}{v}\\ &= c\int_0^\rho \big(1 - \phi(v)\big)\frac{dv}{v}.\end{aligned}$$

Moreover, if ν is the measure on $\mathbb{R}^d \setminus \{0\}$ with the density $k(|x|)\,(|S^{d-1}|\,|x|^d)^{-1}$, then for $\xi \neq 0$, by polar coordinates, Tonelli's theorem and (A.2),

$$\int_{\mathbb{R}^d} \bigl(1 - \cos\langle \xi, x\rangle\bigr)\,\nu(dx) = \int_0^\infty \bigl(1 - \Omega_d(|\xi| r)\bigr)\,k(r)\,\frac{dr}{r}. \tag{A.4}$$

The same holds for every nonnegative Borel k in place of this one.

Step B: the two integrability conditions. With R, c and k as in Step A, Tonelli's theorem gives

$$\int_1^\infty k(r)\,\frac{dr}{r} = c\,\mathbb{E}\int_1^{\max\{R,1\}} \frac{dr}{r} = c\,\mathbb{E}\log^+ R,$$

and $\int_0^1 k(r)\,r\,dr \le c/2$ because $k \le c$.

Step C: the logarithmic moment read on ϕ. Put $H_d(s) := \int_0^s (1 - \Omega_d(v))\,dv/v$ for $s \ge 0$. It is nonnegative and nondecreasing. The pair $(0, \mathbf{1}_{(0,1)})$ is an isotropic origin datum in dimension d, with profile at the origin 1 and with $\int_1^\infty \mathbf{1}_{(0,1)}(r)\,dr/r = 0$. By (A.4) and (A.3) with $R \equiv 1$ and $c = 1$, its exponent at a point of norm s is $H_d(s)$. So Lemma A.2(1) gives $H_d(s) \le \log s + \frac{13}{4}$ for $s \ge 1$, and hence $H_d(s) \le \log^+ s + \frac{13}{4}$ for every $s \ge 0$, H_d being nondecreasing. And Corollary 3.14(2) gives $H_d(s)/\log s \to 1$ as $s \to \infty$, so there is $s_0 > 1$ with $H_d(s) \ge \frac{1}{2}\log s$ for $s \ge s_0$. Now let R and ϕ be as in Step A. By Tonelli's theorem and the substitution $v \mapsto vR$,

$$\int_0^1 \bigl(1 - \phi(v)\bigr)\frac{dv}{v} = \mathbb{E}\int_0^1 \bigl(1 - \Omega_d(vR)\bigr)\frac{dv}{v} = \mathbb{E}\,H_d(R).$$

By the two bounds on H_d, $\frac{1}{2}\,\mathbb{E}\log^+ R - \frac{1}{2}\log s_0 \le \mathbb{E}\,H_d(R) \le \mathbb{E}\log^+ R + \frac{13}{4}$. So $\int_0^1 (1-\phi(v))\,dv/v$ is finite if and only if $\mathbb{E}\log^+ R$ is.

Clause (1). Let X be as in the clause and $R := |X|$, which is positive almost surely. By the preamble $\phi(v) = \mathbb{E}\,\Omega_d(vR)$, and the equivalence is Step C. Suppose $\mathbb{E}\log^+|X| < \infty$ and put $k_d(r) := c\,\Pr(|X| > r)$. It is nonincreasing with values in $[0, c]$, and $k_d(0+) = c\,\Pr(|X| > 0) = c$. By Step B it satisfies the two conditions of Definition 3.10. So $(0, k_d)$ is an isotropic origin datum in dimension d. By (A.4) and (A.3) its exponent at $\xi \neq 0$ is $c\int_0^{|\xi|}(1 - \phi(v))\,dv/v = F(|\xi|)$, and both vanish at $\xi = 0$. By Proposition 3.11(2), $F \in \mathcal{A}_d$. By Corollary 3.14(2) for this datum, read along a ray, $F(r)/\log r \to c$.

Clause (2), existence. Let $F \in \mathcal{A}_d$ with $F(r)/\log r \to c \in (0, \infty)$. By Proposition 3.11(2), $\xi \mapsto F(|\xi|)$ is the exponent of an isotropic origin datum (a, k_d) in dimension d. By Corollary 3.14(1), $F(r)/r^2 \to \frac{1}{2}a$; as $F(r)/\log r$ has a finite limit, $F(r)/r^2 \to 0$, so $a = 0$. Then Corollary 3.14(2) gives $F(r)/\log r \to k_d(0+)$, so $k_d(0+) = c$. As in the proof of Lemma A.2, $k_d(r) \to 0$ as $r \to \infty$. Put $G(r) := k_d(r+)/c$ for $r > 0$. It is nonincreasing and right-continuous, it tends to 1 as $r \downarrow 0$ and to 0 as $r \to \infty$, and $c\,G = k_d$ off the countable set of the discontinuities of k_d. So there is a random variable R with values in $(0, \infty)$ and $\Pr(R > r) = G(r)$ for every $r > 0$. Let U be uniform on S^{d-1} and independent of R, and $X := RU$. Then X is rotation invariant, $|X| = R$, $\Pr(X = 0) = 0$, and by Step B and $c\,G = k_d$ almost everywhere, $c\,\mathbb{E}\log^+|X| = \int_1^\infty k_d(r+)\,dr/r = \int_1^\infty k_d(r)\,dr/r < \infty$. Its function ϕ is that of Step A. Let $\rho > 0$ and let e be a unit vector. By (A.4) for k_d, the equality $k_d = c\,\Pr(R > \cdot)$ almost everywhere, and (A.3),

$$F(\rho) = \int_0^\infty \bigl(1 - \Omega_d(\rho r)\bigr)\,k_d(r)\,\frac{dr}{r} = c\int_0^\rho \bigl(1 - \phi(v)\bigr)\frac{dv}{v}.$$

So F is the function of clause (1) for c and X. As ϕ is continuous, the integrand $(1 - \phi(v))/v$ is continuous on $(0, \infty)$, so F is continuously differentiable there, with $rF'(r) = c\,(1 - \phi(r))$. The datum $(0, k_d)$ was any datum with exponent $F(|\cdot|)$, and $\Pr(|X| > r) = k_d(r+)/c$ by construction.

Clause (2), uniqueness. Suppose F is also the function of clause (1) for c_1 and X_1, with $\mathbb{E}\log^+|X_1| < \infty$ and radial function ϕ_1. By clause (1), $F(r)/\log r \to c_1$, so $c_1 = c$. Both

integrands being continuous, differentiation gives $c\,(1-\phi_1(r)) = rF'(r) = c\,(1-\phi(r))$ for every $r>0$, and $\phi_1(0)=\phi(0)=1$. So $\phi_1=\phi$, and X_1 has the law of X by the preamble. In particular $\Pr(|X|>r) = k_d(r+)/c$ holds for the one law and every datum, since the construction above can start from any datum. *The bijection.* By clause (1) the map sends a pair $(c, \text{law of } X)$ into the set named; by the existence part it is onto, and by the uniqueness part it is one to one. The set is that of the members of $\mathcal{A}_d$ with Gaussian coefficient 0 and profile at the origin in $(0,\infty)$: one inclusion was shown at the start of the existence part, and the other is Corollary 3.14(2).

Clause (3), if. Let X' be a random vector in $\mathbb{R}^{d+1}$ with characteristic function $\zeta \mapsto \phi(|\zeta|)$. For $Q \in \mathrm{O}(d+1)$ the characteristic function of QX' is $\zeta \mapsto \phi(|Q^{\mathsf{T}}\zeta|) = \phi(|\zeta|)$, so QX' has the law of X': X' is rotation invariant. Let Π be the projection on the first d coordinates. The characteristic function of $\Pi X'$ at $\xi \in \mathbb{R}^d$ is that of X' at $(\xi,0)$, that is $\phi(|\xi|)$. So $\Pi X'$ has the law of X, and $\Pr(X'=0) \le \Pr(\Pi X'=0) = \Pr(X=0) = 0$. By the preamble in dimension $d+1$, the radial function of X' is ϕ. As $\int_0^1 (1-\phi(u))\,du/u = F(1)/c < \infty$, clause (1) in dimension $d+1$ gives $\mathbb{E}\log^+|X'| < \infty$ and $F \in \mathcal{A}_{d+1}$, with F the function of that clause for c and X'.

Clause (3), only if. Let $F \in \mathcal{A}_{d+1}$. By clause (1) in dimension d, $F(r)/\log r \to c$. So clause (2) applies in dimension $d+1$: there is a rotation-invariant X' in $\mathbb{R}^{d+1}$ with $\Pr(X'=0)=0$ and $\mathbb{E}\log^+|X'| < \infty$ whose radial function is $r \mapsto 1 - rF'(r)/c$ on $(0,\infty)$. By clause (2) in dimension d this function is ϕ. Both radial functions equal 1 at 0. So the characteristic function of X' is $\zeta \mapsto \phi(|\zeta|)$. □

Proof of Proposition 3.22. Clause (1). By Lemma A.5, with $\bar{k}$ the function displayed in the clause, $(a', \bar{k})$ is an isotropic origin datum in dimension d with exponent $\xi \mapsto F(|\xi|)$. So $F \in \mathcal{A}_d$, by Proposition 3.11(2). Let (a, k_d) be any isotropic origin datum in dimension d with that exponent. By the uniqueness in Proposition 3.11(1), $a = a'$ and $k_d = \bar{k}$ almost everywhere on $(0,\infty)$. Let $r>0$, and choose points $r_n \downarrow r$ and $s_n \uparrow r$ at which $k_d = \bar{k}$. As k_d is nonincreasing, $\bar{k}(r_n) = k_d(r_n) \le k_d(r) \le k_d(s_n) = \bar{k}(s_n)$, and both ends tend to $\bar{k}(r)$, because $\bar{k}$ is continuous (Lemma A.5(2)). So $k_d(r) = \bar{k}(r)$ for every $r>0$, and k_d is continuous. The next sentence of the clause is the contrapositive.

Clause (2). The function $k := b\,\mathbf{1}_{(0,\tau)}$ is nonnegative and nonincreasing on $(0,\infty)$, with $\int_0^1 k(r)\,r\,dr \le b/2$ and $\int_1^\infty k(r)\,dr/r = b\log^+\tau$. So $(0,k)$ is an isotropic origin datum in dimension d (Definition 3.10). Its exponent F_d is finite and nonnegative, as that of every datum, and radial: for $Q \in \mathrm{O}(d)$ the substitution $x = Qy$ in (3.3) gives $F_d(Q\xi) = F_d(\xi)$, because Q preserves Lebesgue measure and the radial density of the Lévy measure. So $F_d(\xi) = G(|\xi|)$ with $G(r) := F_d(re_1)$ for $r \ge 0$, and $G \in \mathcal{A}_d$ by Proposition 3.11(2). The profile k has the left limit b and the right limit 0 at τ. By clause (1), $G \notin \mathcal{A}_{d+1}$. So $\mathcal{A}_{d+1} \ne \mathcal{A}_d$. The inclusion $\mathcal{A}_{d+1} \subseteq \mathcal{A}_d$ follows from clause (1), since for every $F \in \mathcal{A}_{d+1}$, $\zeta \mapsto F(|\zeta|)$ is the exponent of an isotropic origin datum in dimension $d+1$, by Proposition 3.11(2).

Clause (3). At $d=1$ the sphere is $S^0 = \{\pm 1\}$, with σ_0 the counting measure. By (3.2) the exponent of the datum $(0, b\,\mathbf{1}_{(0,\tau)})$ at $\omega \in \mathbb{R}$ is $b\int_0^\tau (1-\cos\omega r)\,dr/r$, which is $b\,\mathrm{Cin}(\tau|\omega|)$: for $\omega \ne 0$ by the substitution $v = |\omega| r$, and both vanish at $\omega = 0$. So $G(r) = b\,\mathrm{Cin}(\tau r)$. This is the Cin ray of Proposition 2.7, the admissible exponent with Gaussian coefficient 0 and profile $b\,\mathbf{1}_{(0,\tau)}$. By clause (2) it is not in $\mathcal{A}_2$. By clause (1), with the datum that Proposition 3.11(2) supplies, $\mathcal{A}_{D+1} \subseteq \mathcal{A}_D$ for every $D \ge 1$, so $\mathcal{A}_D \subseteq \mathcal{A}_2$ for $D \ge 2$, and the ray is in none of them. □

The corollary reads the first two clauses of Proposition 3.22 in the dictionary of Proposition 3.21.

Corollary A.7 (the test and the step datum, read in the dictionary)**.** *Let $d \ge 1$, and let Ω_d be the function of Proposition 3.21.*

1. (The jump size has no atom.) *In the terms of Proposition 3.21: if F and X are as in its clause (1), with $\mathbb{E}\log^+|X| < \infty$, and $F \in \mathcal{A}_{d+1}$, then $\Pr(|X| = r) = 0$ for every $r > 0$.*

2. (The step datum.) *Let $b > 0$ and $\tau > 0$, and put*

$$G(r) := b \int_0^r \big(1 - \Omega_d(\tau u)\big)\, \frac{du}{u} \qquad (r \geq 0).$$

Then $\xi \mapsto G(|\xi|)$ is the exponent of the datum $(0, b\,\mathbf{1}_{(0,\tau)})$ in dimension d, so that G is the function of Proposition 3.22(2), and in Proposition 3.21 it is the function of $c = b$ and of X uniformly distributed on the sphere of radius τ. Hence $\zeta \mapsto \Omega_d(\tau|\zeta|)$ is not a characteristic function on $\mathbb{R}^{d+1}$.

Both clauses are read off the proposition through clause (1) of the dictionary.

Proof of Corollary A.7. Clause (1). Let F and X be as in Proposition 3.21(1), with $\mathbb{E}\log^+|X| < \infty$, and $F \in \mathcal{A}_{d+1}$. By that clause $(0, c\Pr(|X| > \cdot))$ is a datum in dimension d with exponent $F(|\cdot|)$, so its profile is continuous, by Proposition 3.22(1), and $\Pr(|X| = r) = \Pr(|X| \geq r) - \Pr(|X| > r)$ is $1/c$ times the jump of that profile at r, which is 0.

Clause (2). Let U be uniformly distributed on S^{d-1} and $X := \tau U$. It is rotation invariant, $\Pr(X = 0) = 0$, $\mathbb{E}\log^+|X| = \log^+\tau < \infty$, and its radial function is $\phi(r) = \Omega_d(\tau r)$. By Proposition 3.21(1) with $c = b$, the function G belongs to $\mathcal{A}_d$, and $\xi \mapsto G(|\xi|)$ is the exponent of the datum $(0, k_d)$ with $k_d(r) = b\Pr(\tau > r) = b\,\mathbf{1}_{(0,\tau)}(r)$. So G is the function of Proposition 3.22(2), and $G \notin \mathcal{A}_{d+1}$ by that clause. By Proposition 3.21(3), $G \notin \mathcal{A}_{d+1}$ says that $\zeta \mapsto \phi(|\zeta|) = \Omega_d(\tau|\zeta|)$ is not a characteristic function on $\mathbb{R}^{d+1}$. □

Proof of Proposition 3.23. Four facts. (a) For $n \geq 1$ and $\zeta \in \mathbb{R}^n$, (A.2) with $t = \tau$ and $x = \zeta$ says that $\zeta \mapsto \Omega_n(\tau|\zeta|)$ is the characteristic function of the uniform probability measure on the sphere of radius τ of $\mathbb{R}^n$, the image of $\sigma_{n-1}/|S^{n-1}|$ under $\theta \mapsto \tau\theta$. (b) Ω_n is continuous and $0 \leq 1 - \Omega_n(t) \leq t^2/2$, because $1 - \cos u \leq u^2/2$ and $|\langle e, \eta\rangle| \leq 1$ on the sphere. (c) Let $(0, k)$ be an isotropic origin datum in dimension n. By (3.2) with $g(x) = 1 - \cos\langle\xi, x\rangle$, by Tonelli's theorem for a nonnegative integrand, and by the definition of Ω_n with $e = \xi/|\xi|$, its exponent at $\xi \neq 0$ is

$$\frac{1}{|S^{n-1}|}\int_{S^{n-1}} \sigma_{n-1}(d\eta) \int_0^\infty \big(1 - \cos(r\langle\xi, \eta\rangle)\big)\, k(r)\, \frac{dr}{r} = \int_0^\infty \big(1 - \Omega_n(r|\xi|)\big)\, k(r)\, \frac{dr}{r},$$

and both sides vanish at $\xi = 0$. For $k = b\,\mathbf{1}_{(0,\tau)}$ this is $G_n(|\xi|)$, which is finite by (b). (d) Let ν be a symmetric Lévy measure on $\mathbb{R}^n$ and $v \geq 0$. The infinitely divisible law with triplet $(vI, \nu, 0)$ that C3(iii) supplies has the transform $\exp[-\frac{1}{2}v|\zeta|^2 - \int(1 - \cos\langle\zeta, z\rangle)\,\nu(dz)]$. Indeed the integrand $e^{i\langle\zeta,z\rangle} - 1 - i\langle\zeta, z\rangle\mathbf{1}_D(z)$ of the representation has the real part $\cos\langle\zeta, z\rangle - 1$, and its imaginary part is odd in z and integrable against ν, being at most $|\zeta|^2|z|^2/2$ in modulus on D (as $|\sin u - u| \leq u^2/2$ for every real u) and at most 1 off D; so the integral of the imaginary part against the symmetric measure ν is 0.

The shadow of the step two dimensions down. The step datum $(0, b\,\mathbf{1}_{(0,\tau)})$ of dimension $d+2$ has the exponent $\zeta \mapsto G_{d+2}(|\zeta|)$, by (c). By Lemma A.5 with $d+1$ in place of d, its shadow $(0, k_1)$ is an isotropic origin datum in dimension $d+1$ with exponent $\zeta \mapsto G_{d+2}(|\zeta|)$, $\zeta \in \mathbb{R}^{d+1}$, and

$$k_1(s) = C_{d+1}\int_{\mathbb{R}} b\,\mathbf{1}_{\{s\sqrt{1+w^2}<\tau\}}\,(1+w^2)^{-(d+2)/2}\,dw = 2b\,C_{d+1}\,J_{d+2}\Big(\frac{\tau}{s}\Big) \quad (0 < s < \tau),$$

by the substitution $v = \sqrt{1+w^2}$ on $w > 0$, for which $dw = v\,(v^2-1)^{-1/2}dv$; and $k_1(s) = 0$ for $s \geq \tau$. By the same lemma, the shadow $(0, k_2)$ of $(0, k_1)$ is an isotropic origin datum in dimension d with exponent $\xi \mapsto G_{d+2}(|\xi|)$, $\xi \in \mathbb{R}^d$, and

$$k_2(r) = C_d\,C_{d+1}\,b\int_{\mathbb{R}}\int_{\mathbb{R}} \mathbf{1}_{\{r\sqrt{(1+u^2)(1+w^2)}<\tau\}}\,(1+w^2)^{-(d+2)/2}\,(1+u^2)^{-(d+1)/2}\,dw\,du.$$

Substitute $w = y/\sqrt{1+u^2}$ in the inner integral. Then $(1+u^2)(1+w^2) = 1+u^2+y^2$ and $(1+w^2)^{-(d+2)/2}(1+u^2)^{-(d+1)/2}\,dw = (1+u^2+y^2)^{-(d+2)/2}\,dy$, so with $q=(u,y)$

$$k_2(r) = C_d\,C_{d+1}\,b\int_{\mathbb{R}^2} \mathbf{1}_{\{r\sqrt{1+|q|^2}<\tau\}}\,(1+|q|^2)^{-(d+2)/2}\,dq.$$

The same computation with the indicator replaced by 1, and the normalisation of Lemma A.5(1) in the dimensions d and $d+1$, give $C_d\,C_{d+1}\int_{\mathbb{R}^2}(1+|q|^2)^{-(d+2)/2}\,dq = 1$. In polar coordinates of the plane, with $v=\sqrt{1+|q|^2}$, the integral of $\mathbf{1}_{\{v<x\}}(1+|q|^2)^{-(d+2)/2}$ is $2\pi\int_1^x v^{-d-1}\,dv = (2\pi/d)(1-x^{-d})$ for $x\ge 1$, and 0 for $x<1$. At $x=\infty$ this gives

$$2\pi\,C_d\,C_{d+1} = d, \tag{A.5}$$

and at $x=\tau/r$ it gives

$$k_2(r) = b\Big(1-\Big(\frac{r}{\tau}\Big)^d\Big)\quad(0<r<\tau),\qquad k_2(r)=0\quad(r\ge\tau).$$

Clause (2), the identity. Fix $\rho\ge 0$ and put $h(r) := 1-\Omega_d(r\rho)$ and $h_2(r) := 1-\Omega_{d+2}(r\rho)$. By (c) for the datum $(0,k_2)$, whose exponent is $G_{d+2}(|\cdot|)$, and by the definition of G_{d+2},

$$\int_0^\tau h_2(r)\,\frac{dr}{r} = \frac{G_{d+2}(\rho)}{b} = \int_0^\tau h(r)\,\frac{dr}{r} - \tau^{-d}\int_0^\tau h(r)\,r^{d-1}\,dr. \tag{A.6}$$

This holds for every $\tau>0$, the parameter τ having been arbitrary. By (b) the functions h and h_2 are continuous with $0\le h(r), h_2(r)\le r^2\rho^2/2$, so the three integrals are continuously differentiable functions of $\tau>0$. Differentiating (A.6) in τ,

$$\frac{h_2(\tau)}{\tau} = \frac{h(\tau)}{\tau} + d\,\tau^{-d-1}\int_0^\tau h(r)\,r^{d-1}\,dr - \frac{h(\tau)}{\tau},$$

that is

$$\tau^{-d}\int_0^\tau h(r)\,r^{d-1}\,dr = \frac{1}{d}\big(1-\Omega_{d+2}(\tau\rho)\big).$$

By the definition of G_d and (A.6), $G_d(\rho) - G_{d+2}(\rho) = b\,\tau^{-d}\int_0^\tau h(r)\,r^{d-1}\,dr$, which is (3.9).

Clause (2), the law. Let μ^{step} be the law of the step datum of dimension $d+2$, the infinitely divisible law with triplet $(0,\nu^{\mathrm{step}}_{d+2},0)$ that C3(iii) supplies; its transform is $e^{-G_{d+2}(|\zeta|)}$, by (c) and (d). Let $\pi := e^{-b/d}\sum_{n\ge0}(b/d)^n\sigma_\tau^{*n}/n!$. It is a probability measure, and by (a) its transform is $e^{-b/d}\sum_n\big((b/d)\,\Omega_{d+2}(\tau|\zeta|)\big)^n/n! = \exp\big[-\frac{b}{d}(1-\Omega_{d+2}(\tau|\zeta|))\big]$. It is infinitely divisible: for $m\ge1$ the law defined in the same way with $b/(dm)$ in place of b/d has the transform $\widehat{\pi}^{1/m}$, so its m-fold convolution is π. The convolution of two infinitely divisible laws is infinitely divisible, the convolution of their roots being a root. So $\mu^{\mathrm{step}}*\pi$ is infinitely divisible, and by (3.9) its transform is $e^{-G_d(|\zeta|)}$. Put $\nu^{(d+2)} := \nu^{\mathrm{step}}_{d+2} + \frac{b}{d}\,\sigma_\tau$. It is the sum of a Lévy measure and a finite measure that gives the origin no mass, so it is a Lévy measure, and it is symmetric. By (a) and (c), $\int(1-\cos\langle\zeta,z\rangle)\,\nu^{(d+2)}(dz) = G_{d+2}(|\zeta|)+\frac{b}{d}(1-\Omega_{d+2}(\tau|\zeta|)) = G_d(|\zeta|)$. The infinitely divisible law with triplet $(0,\nu^{(d+2)},0)$, which C3(iii) supplies, therefore has the transform $e^{-G_d(|\zeta|)}$, by (d). So it is $\mu^{\mathrm{step}}*\pi$, and the triplet of a law is unique (C3(ii)).

Clause (3). Write $k' = k_1 + k_2'$ on $(0,\infty)$, with k_1 as above and

$$k_2'(s) := (2b\,C_{d+1}/d)\,(s/\tau)^d\,s\,(\tau^2-s^2)^{-1/2}\quad(0<s<\tau),\qquad k_2'(s) := 0\quad(s\ge\tau).$$

Both are nonnegative and Borel. The identity (S.17) holds for every nonnegative Borel profile h, monotone or not. For $h=k_1$ its function $\bar h$ is k_2. For $h=k_2'$ and $0<r<\tau$ put $A := \sqrt{\tau^2-r^2}$.

The integrand of $\bar h(r)$ vanishes unless $|u| < A/r$, and there $k_2'(r\sqrt{1+u^2})\,(1+u^2)^{-(d+1)/2} = (2b\,C_{d+1}/d)\,r^{d+1}\tau^{-d}\,(A^2-r^2u^2)^{-1/2}$. Since $\int_{-A/r}^{A/r}(A^2-r^2u^2)^{-1/2}\,du = \pi/r$, and by (A.5),

$$\bar h(r) = C_d\cdot\frac{2b\,C_{d+1}}{d}\cdot\frac{r^{d+1}}{\tau^d}\cdot\frac{\pi}{r} = b\Big(\frac{r}{\tau}\Big)^d \quad (0<r<\tau), \qquad \bar h(r)=0 \quad (r\ge\tau).$$

Adding the two, (S.17) for $h=k'$ has $\bar h = k_2 + b\,(r/\tau)^d\mathbf{1}_{(0,\tau)} = b\,\mathbf{1}_{(0,\tau)}$: for every nonnegative Borel g on $\mathbb{R}^d$ vanishing at the origin,

$$\int_{\mathbb{R}^{d+1}} g(\Pi z)\,\nu'(dz) = \int_{\mathbb{R}^d} g\,d\nu_d^{\rm step}, \tag{A.7}$$

with Π the projection on the first d coordinates and $\nu_d^{\rm step}$ the Lévy measure of the step datum of dimension d. The measure ν' is rotation invariant, having a radial density, and gives the origin no mass. It is a Lévy measure: in polar coordinates of $\mathbb{R}^{d+1}$, $\int(|z|^2\wedge 1)\,\nu'(dz) \le \int_0^\tau k'(s)\,s\,ds$, and k' is bounded on $(0,\tau/2]$ and at most a constant times $(\tau-s)^{-1/2}$ on $[\tau/2,\tau)$. Put $\psi(\zeta) := \int(1-\cos\langle\zeta,z\rangle)\,\nu'(dz)$ for $\zeta\in\mathbb{R}^{d+1}$, which is finite. It depends on $|\zeta|$ only, by the invariance of ν'. For $\xi\in\mathbb{R}^d$, (A.7) with $g(x)=1-\cos\langle\xi,x\rangle$ gives $\psi((\xi,0)) = G_d(|\xi|)$, by (c). So $\psi(\zeta)=G_d(|\zeta|)$ for every $\zeta\in\mathbb{R}^{d+1}$. The infinitely divisible law with triplet $(0,\nu',0)$, which C3(iii) supplies, has the transform $e^{-\psi}$, by (d), and its triplet is unique (C3(ii)). The properties of k': both terms are positive on $(0,\tau)$, since $J_{d+2}(x)>0$ for $x>1$. As $s\downarrow 0$ the second term tends to 0, and $2C_{d+1}J_{d+2}(\tau/s)\to 2C_{d+1}J_{d+2}(\infty) = C_{d+1}\int_{\mathbb{R}}(1+w^2)^{-(d+2)/2}\,dw = 1$, by Lemma A.5(1) in dimension $d+1$. As $s\uparrow\tau$ the first term tends to 0 and the second to ∞. The closed forms. By Lemma A.5(1), $C_2 = (\int_{\mathbb{R}}(1+u^2)^{-3/2}du)^{-1} = \frac12$ and $C_3 = (\int_{\mathbb{R}}(1+u^2)^{-2}du)^{-1} = 2/\pi$. Differentiation shows $J_3(x) = \sqrt{x^2-1}/x$ and $J_4(x) = \frac12\big[\sqrt{x^2-1}/x^2 + \arccos(1/x)\big]$, both sides vanishing at $x=1$. For $d=1$ this gives $k'(s) = b\big[\sqrt{\tau^2-s^2}/\tau + s^2/(\tau\sqrt{\tau^2-s^2})\big] = b\,\tau/\sqrt{\tau^2-s^2}$, and for $d=2$, $k'(s) = (2b/\pi)\big[s\sqrt{\tau^2-s^2}/\tau^2 + \arccos(s/\tau) + s^3/(\tau^2\sqrt{\tau^2-s^2})\big] = (2b/\pi)\big[\arccos(s/\tau) + s/\sqrt{\tau^2-s^2}\big]$.

Clause (1), "if". Let $D\le d+2$, let μ be the law of clause (2), and let P be the projection of $\mathbb{R}^{d+2}$ on its first D coordinates. The image $P_*\mu$ has the transform $\zeta\mapsto\widehat\mu((\zeta,0)) = e^{-G_d(|\zeta|)}$, $\zeta\in\mathbb{R}^D$. It is infinitely divisible: the image of a convolution under a linear map is the convolution of the images, so the image of an m-th convolution root of μ is one of $P_*\mu$.

Clause (1), "only if". Let $D\ge d+3$ and suppose that an infinitely divisible law on $\mathbb{R}^D$ has the transform $e^{-G_d(|\zeta|)}$. Its image under the projection on the first $d+3$ coordinates is infinitely divisible with the transform $e^{-G_d(|\zeta|)}$ on $\mathbb{R}^{d+3}$, as in the previous paragraph. So let $D=d+3$, and let μ' be such a law. By C3(i) it has a triplet (A,ν',γ). Its transform is invariant under $\mathrm{O}(D)$, so, as in the first paragraph of the proof of Proposition 3.11 from its third sentence on, which reads of the law only that it is infinitely divisible with a radial transform, C3(ii) gives $\langle\zeta,A\zeta\rangle = \alpha|\zeta|^2$ with $\alpha\ge 0$ and $Q_*\nu' = \nu'$ for every $Q\in\mathrm{O}(D)$, and

$$G_d(|\zeta|) = \tfrac12\alpha|\zeta|^2 + \int_{\mathbb{R}^{d+3}}\big(1-\cos\langle\zeta,z\rangle\big)\,\nu'(dz), \qquad \zeta\in\mathbb{R}^{d+3}.$$

Only the modulus of the representation is used for this display: μ' is not assumed symmetric, and $-\log|\widehat{\mu'}(\zeta)| = G_d(|\zeta|)$ because e^{-G_d} is real and positive. Let Π be the projection of $\mathbb{R}^{d+3}$ on its first $d+2$ coordinates, and let $\bar\nu$ be the image of ν' under Π, restricted to $\mathbb{R}^{d+2}\setminus\{0\}$. It is symmetric, and it is a Lévy measure, because $|\Pi z|\le|z|$ gives $\int(|y|^2\wedge1)\,\bar\nu(dy)\le\int(|z|^2\wedge1)\,\nu'(dz)<\infty$. For $\eta\in\mathbb{R}^{d+2}$, $\langle(\eta,0),z\rangle = \langle\eta,\Pi z\rangle$, and the integrand vanishes where $\Pi z=0$, so $G_d(|\eta|) = \frac12\alpha|\eta|^2 + \int(1-\cos\langle\eta,y\rangle)\,\bar\nu(dy)$. By C3(iii) there is an infinitely divisible law on $\mathbb{R}^{d+2}$ with triplet $(\alpha I,\bar\nu,0)$, and by (d) its transform is $e^{-G_d(|\eta|)}$. The law of clause (2) has the same transform and the triplet $(0,\nu^{(d+2)},0)$. The two laws are equal, and by C3(ii) $\alpha=0$ and $\bar\nu=\nu^{(d+2)}$. Let $E := \{y\in\mathbb{R}^{d+2}:|y|=\tau\}$. Then $\bar\nu(E) = \nu^{(d+2)}(E) \ge \frac{b}{d}\,\sigma_\tau(E) = \frac{b}{d} > 0$.

On the other hand $\bar\nu(E) = \nu'(\Pi^{-1}E)$, and $\Pi^{-1}E = \{z \in \mathbb{R}^{d+3} : |\Pi z| = \tau\}$ lies in $\{|z| \ge \tau\}$. Let m be the restriction of ν' to $\{|z| \ge \tau\}$. It is a finite measure, because $\int (|z|^2 \wedge 1)\, d\nu' < \infty$, it is invariant under $\mathrm{O}(d+3)$, and $m(\Pi^{-1}E) = \bar\nu(E) > 0$, so $m \ne 0$. Let Z be a random vector in $\mathbb{R}^{d+3}$ with the law $m/m(\mathbb{R}^{d+3})$. It is rotation invariant, so by the preamble of Proposition 3.21 it has the law of RU, with R of the law of $|Z|$, so that $R \ge \tau$, and U uniformly distributed on S^{d+2} and independent of R. Hence

$$\frac{m(\Pi^{-1}E)}{m(\mathbb{R}^{d+3})} = \Pr\big(R\,|\Pi U| = \tau\big) = \int \Pr\big(|\Pi U| = \tau/r\big)\ \Pr(R \in dr).$$

For every $w \in [0,1]$, $\Pr(|\Pi U| = w) = 0$. Indeed $|\Pi u|^2 = 1 - u_{d+3}^2$ on the sphere, so the event is $\{U \in K\}$ with $K := \{z \in \mathbb{R}^{d+3} : z_{d+3}^2 = (1-w^2)|z|^2\}$, a cone. The cone is Lebesgue-null: if $w > 0$ it is the set $\{w^2 z_{d+3}^2 = (1-w^2)|\Pi z|^2\}$, which meets every line parallel to the last axis in at most two points, and if $w = 0$ it is the last axis. In polar coordinates the Lebesgue measure of $K \cap \{1 < |z| < 2\}$ is $\sigma_{d+2}(K \cap S^{d+2}) \int_1^2 r^{d+2}\, dr$, so $\sigma_{d+2}(K \cap S^{d+2}) = 0$. Therefore $m(\Pi^{-1}E) = 0$, a contradiction.

The last sentence. For $d = 1$, $G(r) = b\,\mathrm{Cin}(\tau r)$ and the ray is in no $\mathcal{A}_D$ with $D \ge 2$, by Proposition 3.22(3). The rest is clause (1) with $d + 2 = 3$. □

The last proposition says what the members of the separating family of Definition 3.24 are as laws.

Proposition A.8 (the separating family as laws)**.** *Let $w > 0$, and let $F_{\mathrm{I}}^{(w)}$, $F^{(w)}$ and the critical values w_D be as in Definition 3.24.*

1. (Closed form.) *For $\sigma \ge 0$ and $r \ge 0$,*

$$F_{\mathrm{I}}^{(w)}(\sigma) = 2\sqrt{\pi\sigma} + w\int_1^2 \big(1 - e^{-\sigma u}\big)\frac{du}{u}, \qquad F^{(w)}(r) = \sqrt{2\pi}\, r + w\int_1^2 \big(1 - e^{-ur^2/2}\big)\frac{du}{u}.$$

2. (A Gaussian variance mixture in every dimension.) *Let Z_0 be a standard Gaussian real variable, N a Poisson variable of mean $w\log 2$, and $U_1, U_2, \dots$ variables with the density $(u\log 2)^{-1}$ on $[1,2]$, all independent, and put*

$$W := \frac{2\pi}{Z_0^2} + \sum_{j=1}^{N} U_j.$$

Then $W > 0$ almost surely and $\mathbb{E}\, e^{-\sigma W} = e^{-F_{\mathrm{I}}^{(w)}(\sigma)}$ for every $\sigma \ge 0$. For every integer $D \ge 1$, if Z is a standard Gaussian vector of $\mathbb{R}^D$ independent of W, then $\sqrt{W}\,Z$ has the characteristic function $\xi \mapsto e^{-F^{(w)}(|\xi|)}$ on $\mathbb{R}^D$. Its law $\mu_D^{(w)}$ is rotation invariant and infinitely divisible. It is the convolution of the law with characteristic function $e^{-\sqrt{2\pi}|\xi|}$, a rotation-invariant Cauchy law, and of a compound Poisson law, with characteristic function $\exp\big[-w\int_1^2 (1 - e^{-u|\xi|^2/2})\, du/u\big]$.

3. (Self-decomposable up to the critical dimension.) *For every integer $D \ge 1$, $\mu_D^{(w)}$ is self-decomposable if and only if $F^{(w)} \in \mathcal{A}_D$, that is, by Theorem 3.25(1), if and only if $w \le w_D$. Consequently, if $w \in (w_{d+1}, w_d\,]$ for an integer $d \ge 1$, then $\mu_D^{(w)}$ is self-decomposable exactly for $D \le d$, and if $w > w_1$ it is self-decomposable in no dimension.*

In clause (2) the two summands of W are a one-sided stable variable of index $\frac12$ and a compound Poisson sum.

Proof of Proposition A.8. Clause (1). By Definition 3.24 the delay datum $D^{(w)}$ has drift 0 and profile $k_{\mathrm{I}}^{(w)}(u) = u^{-1/2} + w\,\mathbf{1}_{[1,2]}(u)$, so $F_{\mathrm{I}}^{(w)}(\sigma) = \int_0^\infty (1-e^{-\sigma u})\,u^{-3/2}\,du + w\int_1^2 (1-e^{-\sigma u})\,du/u$. Integrating by parts, for $\sigma > 0$,

$$\begin{aligned}\int_0^\infty \bigl(1-e^{-\sigma u}\bigr)\,u^{-3/2}\,du &= \Bigl[-2u^{-1/2}\bigl(1-e^{-\sigma u}\bigr)\Bigr]_0^\infty + 2\sigma\int_0^\infty u^{-1/2}e^{-\sigma u}\,du\\ &= 2\sigma\cdot\Gamma(\tfrac12)\,\sigma^{-1/2} = 2\sqrt{\pi\sigma},\end{aligned}$$

the bracket vanishing at both ends. At $\sigma = 0$ both sides are 0. The second formula is the first at $\sigma = r^2/2$.

Clause (2), a Gaussian integral. For $\beta \ge 0$,

$$\int_0^\infty e^{-x^2-\beta^2/x^2}\,dx = \frac{\sqrt{\pi}}{2}\,e^{-2\beta}. \tag{A.8}$$

For $\beta = 0$ this is the Gaussian integral. Let $\beta > 0$. Then $x^2 + \beta^2/x^2 = (x-\beta/x)^2 + 2\beta$. Put $J := \int_0^\infty e^{-(x-\beta/x)^2}dx$. The substitution $x = \beta/y$ gives $J = \int_0^\infty e^{-(y-\beta/y)^2}\,\beta y^{-2}\,dy$. Adding the two expressions, $2J = \int_0^\infty e^{-(x-\beta/x)^2}(1+\beta/x^2)\,dx = \int_{\mathbb{R}} e^{-t^2}dt = \sqrt{\pi}$, by the substitution $t = x - \beta/x$, an increasing bijection of $(0,\infty)$ onto $\mathbb{R}$ with $dt = (1+\beta/x^2)\,dx$.

Clause (2), the Laplace transform of W. Let $\varkappa > 0$ and $\sigma \ge 0$. With $z = \sqrt{2}\,x$ and (A.8) at $\beta^2 = \sigma\varkappa/2$,

$$\mathbb{E}\,e^{-\sigma\varkappa/Z_0^2} = \frac{2}{\sqrt{2\pi}}\int_0^\infty e^{-z^2/2-\sigma\varkappa/z^2}\,dz = \frac{2}{\sqrt{\pi}}\int_0^\infty e^{-x^2-\sigma\varkappa/(2x^2)}\,dx = e^{-\sqrt{2\sigma\varkappa}}.$$

At $\varkappa = 2\pi$ this is $e^{-2\sqrt{\pi\sigma}}$. For the sum, conditioning on N,

$$\begin{aligned}\mathbb{E}\,e^{-\sigma\sum_{j\le N}U_j} &= \sum_{n\ge0} e^{-w\log 2}\frac{(w\log 2)^n}{n!}\bigl(\mathbb{E}\,e^{-\sigma U_1}\bigr)^n = \exp\bigl[-w\log 2\,\bigl(1-\mathbb{E}\,e^{-\sigma U_1}\bigr)\bigr]\\ &= \exp\Bigl[-w\int_1^2\bigl(1-e^{-\sigma u}\bigr)\frac{du}{u}\Bigr],\end{aligned}$$

because $\log 2\,(1-\mathbb{E}\,e^{-\sigma U_1}) = \int_1^2(1-e^{-\sigma u})\,du/u$. By independence and clause (1), $\mathbb{E}\,e^{-\sigma W} = e^{-F_{\mathrm{I}}^{(w)}(\sigma)}$. And $W \ge 2\pi/Z_0^2 > 0$ almost surely.

Clause (2), the mixture. Let $D \ge 1$ and let Z be a standard Gaussian vector of $\mathbb{R}^D$ independent of W. Conditioning on W, and using that $\sqrt{u}\,Z$ has the characteristic function $e^{-u|\xi|^2/2}$,

$$\mathbb{E}\,e^{i\langle\xi,\sqrt{W}Z\rangle} = \mathbb{E}\,e^{-W|\xi|^2/2} = e^{-F_{\mathrm{I}}^{(w)}(|\xi|^2/2)} = e^{-F^{(w)}(|\xi|)}.$$

The characteristic function is radial, so the law $\mu_D^{(w)}$ of $\sqrt{W}\,Z$ is invariant under $\mathrm{O}(D)$, by the uniqueness theorem for characteristic functions. By clause (1) it factors as $e^{-\sqrt{2\pi}|\xi|}$ times $\exp[-w\int_1^2(1-e^{-u|\xi|^2/2})\,du/u]$. The same computation with W replaced by each of its two summands shows that the first factor is the characteristic function of $\sqrt{2\pi}\,Z/|Z_0|$ and the second that of $\sqrt{\sum_{j\le N}U_j}\,Z$. The second is also the characteristic function of the compound Poisson sum $\sum_{j\le N}\sqrt{U_j}\,Z_j$, with $Z_1, Z_2, \ldots$ independent standard Gaussian vectors independent of the rest, by the computation made above for the Laplace transform. So $\mu_D^{(w)}$ is the convolution of the two laws.

Clause (2), infinite divisibility. Let $n \ge 1$, let N_n be a Poisson variable of mean $(w\log 2)/n$, independent of the rest, and put $W_n := 2\pi/(n^2Z_0^2) + \sum_{j\le N_n}U_j$. The two computations above, at $\varkappa = 2\pi/n^2$ and with w/n in place of w, give $\mathbb{E}\,e^{-\sigma W_n} = e^{-F_{\mathrm{I}}^{(w)}(\sigma)/n}$. So the law μ_n of $\sqrt{W_n}\,Z$ has the characteristic function $e^{-F^{(w)}(|\xi|)/n}$, and its n-fold convolution has the characteristic function of $\mu_D^{(w)}$. Hence $\mu_D^{(w)} = \mu_n^{*n}$ for every n: it is infinitely divisible.

Clause (3), the reading in laws. $F^{(w)} \in \mathcal{A}_D$ means that $F^{(w)}$ is admissible and that $\xi \mapsto F^{(w)}(|\xi|)$ is the exponent of a self-decomposable law on $\mathbb{R}^D$. A law with the characteristic function $e^{-F^{(w)}(|\xi|)}$ is $\mu_D^{(w)}$, by clause (2) and the uniqueness theorem. And if $\mu_D^{(w)}$ is self-decomposable, $F^{(w)}$ is admissible, by Proposition 3.11(2). So $F^{(w)} \in \mathcal{A}_D$ exactly when $\mu_D^{(w)}$ is self-decomposable.

Clause (3), the criterion and the windows. By Theorem 3.25(1), $F^{(w)} \in \mathcal{A}_D$ exactly when $w \le w_D$. By its clause (4) this holds exactly for $D \le d$ when $w \in (w_{d+1}, w_d]$, and for no D when $w > w_1$. With the reading in laws these are the assertions of the clause. □

B Verification

This appendix says where the formal development behind §1.1 is, in which versions, under which names it holds the cited facts, and which cited facts each statement rests on. the supplement has the declarations statement by statement, with Lean's report (§S4). A *declaration* is a named definition or proved statement of the formal development. An *axiom*, in Lean's report, is a declaration taken without proof, and the report on a declaration lists every axiom that its proof uses. The seven cited facts of this paper are nine such axioms, C3 being three. The guard of the repository admits twelve, because the formal development also states the three cited facts of the companion paper.

Where the formal development is. The formal development, its ledger of the cited facts with the page-verified citation of each, and the blueprint, the sources in which the statements and their proofs were developed, are in the repository accompanying this paper (see the code availability statement at the end). They are the directories `Formalization/` and `blueprint/` of that repository. The repository is `https://github.com/danielfagerstrom/spatial-hemigroup-scale-space-isotropic`, and this version of the paper is its tag `v0.1`. It is a public verification export: it is taken from the development repository, which is private, and it is deposited with this version of the paper under the DOI printed on its first page. It holds the formal development, the ledger, the blueprint sources of the two chapters behind this paper, and the scripts of the figures and of Table 1, each at the path that this paper or the supplement prints.

Versions and commands. The versions are fixed in the repository: Lean 4 `v4.31.0`, Mathlib `v4.31.0`, and the shared library `ScaleSpaceCore` at its tag `v0.4.0`. The build configuration also names the formal development of the two earlier papers, from the public export of the cone paper at its tag `v0.1`, and no file of this development uses it. `lake build` in `Formalization/` reproduces the verification: it checks every proof. On request Lean reports, for a declaration, the assumptions it depends on, which it calls its axioms. The criterion for this paper is that every axiom reported for a declaration of the list of §S4 is one of Lean's three logical axioms, `propext`, `Classical.choice` and `Quot.sound`, or one of the nine names below.

The names of the cited facts. The nine names are the seven cited facts of this paper, C3 being three. Table 4 says which of them each statement rests on. They are in the namespace `AffineHemigroup`:

C1	`density_of_integrable_charFun`	C3(iii)	`levy_khintchine_converse`
C2	`sato_polar_criterion`	C4	`selfDecomposable_infinitelyDivisible`
C3(i)	`levy_khintchine_exists`	C5	`wolfe_symmetric_unimodal`
C3(ii)	`levy_khintchine_unique`	C6	`schoenberg_radial_bernstein`
		C7	`karamata_tauberian_zero`

statement	cited facts its proof rests on
Thm. 3.6 and Cor. 3.7, the classification	C3(ii) and (iii), on the line
Cor. 3.9, the main result as one statement	C2, C3, C4
Prop. 3.11, the polar form of $\mathcal{A}_d$	C2, C3, C4
Thm. 3.13, the threshold	C1, in clauses (2) and (5)
Prop. 3.15, the radially nonincreasing density	C5; C3(ii) and (iii) in clause (1); C2 and C3(iii) in clause (2)
Prop. A.3, the lower bound	C3(iii)
Prop. A.4, the upper bound	none for the bounds of clauses (1) and (2); C3(iii) for clause (3); C2, C3(iii) and C5 for the existence of the density
Lemma S.9 of the supplement, the rate through the transform	C7 for the limits; C2, C3(iii) and C5 for the existence of the density
Prop. 3.18, the rate through the profile	C7 for the limits; C2, C3(iii) and C5 for the existence of the density
Prop. 3.20, the stages	C3(iii) in clause (3), items (a) and (b); C1 in its item (c); none in clauses (1) and (2)
Prop. 3.21, the dictionary, Prop. 3.22, Cor. A.7 and Prop. 3.23	C2, C3, C4
Thm. 3.25, the separating family	C2, C3, C4, in clauses (1) and (4)
Prop. A.8, the separating family as laws	C2, C3, C4, in clause (3)
Lemma S.13 of the supplement, the top of the filtration, $\mathcal{A}_\infty = \mathcal{S}$	C2, C3, C4, C6
every other statement	none

Table 4: The statements of the paper and of the supplement and the cited facts that their proofs rest on. For a statement with a formal counterpart these are the assumptions that Lean reports for its declarations, and for the statements of the second list of §1.1 the cited facts that the printed proof uses. "None" means none of C1 to C7: the statement follows from the definitions and the logical foundations of the proof assistant alone, or, where there is no formal counterpart, its printed proof cites no fact.

The statements restated from the two earlier papers. The printed proofs cite the statements restated in §2, and the formal proofs assume none of them. So in two places the formal proof derives from the cited facts what the printed proof cites from an earlier paper: the uniqueness of the Lévy measure in the classification on the line, from C3, and the identification $\mathcal{A}_\infty = \mathcal{S}$, from C6 with C2, C3 and C4. In their own papers the restated statements stand as follows. Of those of the line paper, the passage from operators to kernels, the classification and the two facts used in proofs are verified formally there, the classification on two cited facts, and that the admissible exponents are those of the symmetric self-decomposable laws is cited from the literature ([15], §1.1). Of those of the cone paper, the bridge on exponents, the extreme rays and the threshold on the line are verified formally there, on cited facts that it lists, and the dimension filtration is proved in its text ([16], §1.1).

Author contribution and use of AI

This article was written in close collaboration with generative AI, beyond editing assistance, at every stage.

The programme, the question, the decisions of scope and the organization of the material are mine; the prose, statements, proofs and code, the formal development in Lean that machine-checks the statements listed in §1.1 among them, are the agents' work, mostly unattended, and its central threshold began as an agent's conjecture.

Agents with no part in the writing reviewed it blind throughout, a system of another vendor

among them; each finding was checked before use (the trust base, §1.1). Beyond me, no human has reviewed the paper, and the formal statements in Lean have been reviewed by AI agents only, under review protocols that I set up.

The division of labour, with its derivation and dates, is in `notes/PROCESS.md`; the agents' words outnumber mine by between twenty-two and forty-two to one (computed 2026-10-13, a lower bound) over the module's first sixteen archived days.

Responsibility, errors included, is mine alone.

Declarations

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Data and code availability. No data were generated or analysed. The scripts that draw Figure 1 from the closed form of the Matérn kernel's density and Figure 2 from the closed forms of the polar profiles of the separating family and from its critical values, computed numerically, the script that computes Table 1, the LaTeX sources in which the statements and their proofs were developed, and the formal development in Lean 4 that machine-checks formal counterparts of the statements listed as such in §1.1, with its ledger of the cited facts (Appendix B), are in the repository accompanying this paper, `https://github.com/danielfagerstrom/spatial-hemigroup-scale-space-isotropic`, at its tag `v0.1`. It is the public verification export of this version, taken from the development repository, which is private, and it is deposited with this version of the paper under the DOI printed on its first page.

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